Scheduling, Preemption and Eviction

In Kubernetes, scheduling refers to making sure that Pods are matched to Nodes so that the kubelet can run them. Preemption is the process of terminating Pods with lower Priority so that Pods with higher Priority can schedule on Nodes. Eviction is the process of terminating one or more Pods on Nodes.

Scheduling

- Kubernetes Scheduler
- Assigning Pods to Nodes
- Pod Overhead
- Pod Topology Spread Constraints
- Taints and Tolerations
- Scheduling Framework
- Dynamic Resource Allocation
- Scheduler Performance Tuning
- Resource Bin Packing for Extended Resources
- Pod Scheduling Readiness

- PodGroup Scheduling
- Gang Scheduling
- Topology-aware Scheduling
- Workload-Aware preemption
- Descheduler
- Node Declared Features

Pod Disruption

Pod disruption is the process by which Pods on Nodes are terminated either voluntarily or involuntarily.

Voluntary disruptions are started intentionally by application owners or cluster administrators. Involuntary disruptions are unintentional and can be triggered by unavoidable issues like Nodes running out of resources, or by accidental deletions.

- Pod Priority and Preemption
- Node-pressure Eviction
- API-initiated Eviction

1 - Kubernetes Scheduler

In Kubernetes, *scheduling* refers to making sure that Pods are matched to Nodes so that Kubelet can run them.

Scheduling overview

A scheduler watches for newly created Pods that have no Node assigned. For every Pod that the scheduler discovers, the scheduler becomes responsible for finding the best Node for that Pod to run on. The scheduler reaches this placement decision taking into account the scheduling principles described below.

If you want to understand why Pods are placed onto a particular Node, or if you're planning to implement a custom scheduler yourself, this page will help you learn about scheduling.

kube-scheduler

kube-scheduler is the default scheduler for Kubernetes and runs as part of the control plane. kube-scheduler is designed so that, if you want and need to, you can write your own scheduling component and use that instead.

Kube-scheduler selects an optimal node to run newly created or not yet scheduled (unscheduled) pods. Since containers in pods - and pods themselves - can have different requirements, the scheduler filters out any nodes that don't meet a Pod's specific scheduling needs. Alternatively, the API lets you specify a node for a Pod when you create it, but this is unusual and is only done in special cases.

In a cluster, Nodes that meet the scheduling requirements for a Pod are called *feasible* nodes. If none of the nodes are suitable, the pod remains unscheduled until the scheduler is able to place it.

The scheduler finds feasible Nodes for a Pod and then runs a set of functions to score the feasible Nodes and picks a Node with the highest score among the feasible ones to run the Pod. The scheduler then notifies the API server about this decision in a process called *binding*.

Factors that need to be taken into account for scheduling decisions include individual and collective resource requirements, hardware / software / policy constraints, affinity and anti-affinity specifications, data locality, inter-workload interference, and so on.

Node selection in kube-scheduler

kube-scheduler selects a node for the pod in a 2-step operation:

1. Filtering
2. Scoring

The *filtering* step finds the set of Nodes where it's feasible to schedule the Pod. For example, the PodFitsResources filter checks whether a candidate Node has enough available resources to meet a Pod's specific resource requests. After this step, the node list contains any suitable Nodes; often, there will be more than one. If the list is empty, that Pod isn't (yet) schedulable.

In the *scoring* step, the scheduler ranks the remaining nodes to choose the most suitable Pod placement. The scheduler assigns a score to each Node that survived filtering, basing this score on the active scoring rules.

Finally, kube-scheduler assigns the Pod to the Node with the highest ranking. If there is more than one node with equal scores, kube-scheduler selects one of these at random.

There are two supported ways to configure the filtering and scoring behavior of the scheduler:

1. Scheduling Policies allow you to configure *Predicates* for filtering and *Priorities* for scoring.
2. Scheduling Profiles allow you to configure Plugins that implement different scheduling stages, including: `QueueSort`, `Filter`, `Score`, `Bind`, `Reserve`, `Permit`, and others. You can also configure the kube-scheduler to run different profiles.

What's next

- Read about scheduler performance tuning
- Read about Pod topology spread constraints
- Read the reference documentation for kube-scheduler
- Read the kube-scheduler config (v1) reference
- Learn about configuring multiple schedulers
- Learn about topology management policies
- Learn about Pod Overhead
- Learn about scheduling of Pods that use volumes in:
  - Volume Topology Support
  - Storage Capacity Tracking
  - Node-specific Volume Limits

2 - Topology-Aware Workload Scheduling

ⓘ **FEATURE STATE:** Kubernetes v1.36 [alpha](disabled by default)

Topology-Aware Scheduling (TAS) is a placement scheduling algorithm that allows to find the optimal placement for the considered PodGroup, guaranteeing that all pods will be collocated within the same topology domain. Users can accomodate TAS to their specific needs by changing TAS plugins configuration.

Scheduling framework: TAS plugins configuration

The scheduler includes new and extended in-tree plugins that implement the TAS extension points:

- `TopologyPlacement` : Implements the `PlacementGeneratePlugin` interface. It generates candidate placements by grouping nodes based on the distinct values of the requested topology `key` (defined in the PodGroup).
- `NodeResourcesFit` : Extended to implement the `PlacementScorePlugin` interface. Following similar logic to standard pod bin-packing, it scores placements based on the allocation ratio across all nodes within the placement. It uses the `MostAllocated` strategy to maximize resource utilization within a placement, and it inherits resource weights from the standard pod-by-pod plugin settings.
- `PodGroupPodsCount` : Implements the `PlacementScorePlugin` interface. It scores candidate placements based on the total number of pods in the PodGroup that you can successfully schedule.

Customizing plugin weights and bin-packing resource weights

By default, the `NodeResourcesFit` and `PodGroupPodsCount` plugins are configured with equal weights (both default to 1) to maintain a good balance between bin-packing logic and scheduling as many pods as possible.

You can adjust these weights, or the resource weights in the bin-packing strategy in your KubeSchedulerConfiguration. Here is an example snippet showing how to change

the weights for both plugins, and how to override the `NodeResourcesFit` resource weights. The latter change will apply both to pod-by-pod and placement scoring algorithms:

```yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
  - schedulerName: default-scheduler
    plugins:
      placementScore:
        enabled:
          # 1) Change the default weights of the placement score plugins
          - name: NodeResourcesFit
            weight: 2
          - name: PodGroupPodsCount
            weight: 5
    pluginConfig:
      - name: NodeResourcesFit
        args:
          # 2) Changing the scoring resource weights for both pod-by-pod and placer
          # algorithms
          scoringStrategy:
            # The type will only be considered in pod-by-pod scheduling. Placement
            # uses MostAllocated strategy
            type: LeastAllocated
            # Resource weights will be used in both pod-by-pod and placement scori
            resources:
              - name: cpu
                weight: 2
              - name: memory
                weight: 3
```

What's next

- Learn more about Topology-aware scheduling API.
- Read about pod group scheduling.
- Read about pod group policies.

3 - Assigning Pods to Nodes

You can constrain a Pod so that it is *restricted* to run on particular node(s), or to *prefer* to run on particular nodes. There are several ways to do this and the recommended approaches all use label selectors to facilitate the selection. Often, you do not need to set any such constraints; the scheduler will automatically do a reasonable placement (for example, spreading your Pods across nodes so as not place Pods on a node with insufficient free resources). However, there are some circumstances where you may want to control which node the Pod deploys to, for example, to ensure that a Pod ends up on a node with an SSD attached to it, or to co-locate Pods from two different services that communicate a lot into the same availability zone.

You can use any of the following methods to choose where Kubernetes schedules specific Pods:

- nodeSelector field matching against node labels
- Affinity and anti-affinity
- nodeName field
- Pod topology spread constraints

Node labels

Like many other Kubernetes objects, nodes have labels. You can attach labels manually. Kubernetes also populates a standard set of labels on all nodes in a cluster.

> **Note:**
> The value of these labels is cloud provider specific and is not guaranteed to be reliable. For example, the value of `kubernetes.io/hostname` may be the same as the node name in some environments and a different value in other environments.

Node isolation/restriction

Adding labels to nodes allows you to target Pods for scheduling on specific nodes or groups of nodes. You can use this functionality to ensure that specific Pods only run on nodes with certain isolation, security, or regulatory properties.

If you use labels for node isolation, choose label keys that the kubelet cannot modify. This prevents a compromised node from setting those labels on itself so that the scheduler schedules workloads onto the compromised node.

The `NodeRestriction` admission plugin prevents the kubelet from setting or modifying labels with a `node-restriction.kubernetes.io/` prefix.

To make use of that label prefix for node isolation:

1. Ensure you are using the Node authorizer and have *enabled* the `NodeRestriction` admission plugin.
2. Add labels with the `node-restriction.kubernetes.io/` prefix to your nodes, and use those labels in your node selectors. For example, `example.com.node-restriction.kubernetes.io/fips=true` or `example.com.node-restriction.kubernetes.io/pci-dss=true`.

nodeSelector

`nodeSelector` is the simplest recommended form of node selection constraint. You can add the `nodeSelector` field to your Pod specification and specify the node labels you want the target node to have. Kubernetes only schedules the Pod onto nodes that have each of the labels you specify.

See Assign Pods to Nodes for more information.

Affinity and anti-affinity

`nodeSelector` is the simplest way to constrain Pods to nodes with specific labels. Affinity and anti-affinity expand the types of constraints you can define. Some of the benefits of affinity and anti-affinity include:

- The affinity/anti-affinity language is more expressive. `nodeSelector` only selects nodes with all the specified labels. Affinity/anti-affinity gives you more control over the selection logic.
- You can indicate that a rule is *soft* or *preferred*, so that the scheduler still schedules the Pod even if it can't find a matching node.
- You can constrain a Pod using labels on other Pods running on the node (or other topological domain), instead of just node labels, which allows you to define rules for which Pods can be co-located on a node.

The affinity feature consists of two types of affinity:

- *Node affinity* functions like the `nodeSelector` field but is more expressive and allows you to specify soft rules.
- *Inter-pod affinity/anti-affinity* allows you to constrain Pods against labels on other Pods.

Node affinity

Node affinity is conceptually similar to `nodeSelector`, allowing you to constrain which nodes your Pod can be scheduled on based on node labels. There are two types of node affinity:

- `requiredDuringSchedulingIgnoredDuringExecution`: The scheduler can't schedule the Pod unless the rule is met. This functions like `nodeSelector`, but with a more expressive syntax.
- `preferredDuringSchedulingIgnoredDuringExecution`: The scheduler tries to find a node that meets the rule. If a matching node is not available, the scheduler still schedules the Pod.

> **Note:**
> In the preceding types, `IgnoredDuringExecution` means that if the node labels change after Kubernetes schedules the Pod, the Pod continues to run.

You can specify node affinities using the `.spec.affinity.nodeAffinity` field in your Pod spec.

For example, consider the following Pod spec:

pods/pod-with-node-affinity.yaml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: with-node-affinity
spec:
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
        - matchExpressions:
          - key: topology.kubernetes.io/zone
            operator: In
            values:
            - antarctica-east1
            - antarctica-west1
      preferredDuringSchedulingIgnoredDuringExecution:
      - weight: 1
        preference:
          matchExpressions:
```

```
          - key: another-node-label-key
            operator: In
            values:
            - another-node-label-value
  containers:
  - name: with-node-affinity
    image: registry.k8s.io/pause:3.8
```

In this example, the following rules apply:

- The node *must* have a label with the key `topology.kubernetes.io/zone` and the value of that label *must* be either `antarctica-east1` or `antarctica-west1`.
- The node *preferably* has a label with the key `another-node-label-key` and the value `another-node-label-value`.

You can use the `operator` field to specify a logical operator for Kubernetes to use when interpreting the rules. You can use `In`, `NotIn`, `Exists`, `DoesNotExist`, `Gt` and `Lt`.

Read Operators to learn more about how these work.

`NotIn` and `DoesNotExist` allow you to define node anti-affinity behavior. Alternatively, you can use node taints to repel Pods from specific nodes.

> **Note:**
> If you specify both `nodeSelector` and `nodeAffinity`, *both* must be satisfied for the Pod to be scheduled onto a node.
>
> If you specify multiple terms in `nodeSelectorTerms` associated with `nodeAffinity` types, then the Pod can be scheduled onto a node if one of the specified terms can be satisfied (terms are ORed).
>
> If you specify multiple expressions in a single `matchExpressions` field associated with a term in `nodeSelectorTerms`, then the Pod can be scheduled onto a node only if all the expressions are satisfied (expressions are ANDed).

See Assign Pods to Nodes using Node Affinity for more information.

Node affinity weight

You can specify a `weight` between 1 and 100 for each instance of the `preferredDuringSchedulingIgnoredDuringExecution` affinity type. When the scheduler finds nodes that meet all the other scheduling requirements of the Pod, the scheduler

iterates through every preferred rule that the node satisfies and adds the value of the `weight` for that expression to a sum.

The final sum is added to the score of other priority functions for the node. Nodes with the highest total score are prioritized when the scheduler makes a scheduling decision for the Pod.

For example, consider the following Pod spec:

pods/pod-with-affinity-preferred-weight.yaml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: with-affinity-preferred-weight
spec:
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
        - matchExpressions:
          - key: kubernetes.io/os
            operator: In
            values:
            - linux
      preferredDuringSchedulingIgnoredDuringExecution:
      - weight: 1
        preference:
          matchExpressions:
          - key: label-1
            operator: In
            values:
            - key-1
      - weight: 50
        preference:
          matchExpressions:
          - key: label-2
            operator: In
            values:
            - key-2
  containers:
  - name: with-node-affinity
    image: registry.k8s.io/pause:3.8
```

If there are two possible nodes that match the `preferredDuringSchedulingIgnoredDuringExecution` rule, one with the `label-1:key-1` label and another with the `label-2:key-2` label, the scheduler considers the `weight` of each node and adds the weight to the other scores for that node, and schedules the Pod onto the node with the highest final score.

> **Note:**
> If you want Kubernetes to successfully schedule the Pods in this example, you must have existing nodes with the `kubernetes.io/os=linux` label.

Node affinity per scheduling profile

FEATURE STATE: Kubernetes v1.20 [beta]

When configuring multiple scheduling profiles, you can associate a profile with a node affinity, which is useful if a profile only applies to a specific set of nodes. To do so, add an `addedAffinity` to the `args` field of the `NodeAffinity` plugin in the scheduler configuration. For example:

```yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration

profiles:
  - schedulerName: default-scheduler
  - schedulerName: foo-scheduler
    pluginConfig:
      - name: NodeAffinity
        args:
          addedAffinity:
            requiredDuringSchedulingIgnoredDuringExecution:
              nodeSelectorTerms:
              - matchExpressions:
                - key: scheduler-profile
                  operator: In
                  values:
                  - foo
```

The `addedAffinity` is applied to all Pods that set `.spec.schedulerName` to `foo-scheduler`, in addition to the NodeAffinity specified in the PodSpec. That is, in order to match the

Pod, nodes need to satisfy `addedAffinity` and the Pod's `.spec.NodeAffinity`.

Since the `addedAffinity` is not visible to end users, its behavior might be unexpected to them. Use node labels that have a clear correlation to the scheduler profile name.

> **Note:**
> The DaemonSet controller, which creates Pods for DaemonSets, does not support scheduling profiles. When the DaemonSet controller creates Pods, the default Kubernetes scheduler places those Pods and honors any `nodeAffinity` rules in the DaemonSet controller.

Inter-pod affinity and anti-affinity

Inter-pod affinity and anti-affinity allow you to constrain which nodes your Pods can be scheduled on based on the labels of Pods already running on that node, instead of the node labels.

Types of Inter-pod Affinity and Anti-affinity

Inter-pod affinity and anti-affinity take the form "this Pod should (or, in the case of anti-affinity, should not) run in an X if that X is already running one or more Pods that meet rule Y", where X is a topology domain like node, rack, cloud provider zone or region, or similar and Y is the rule Kubernetes tries to satisfy.

You express these rules (Y) as label selectors with an optional associated list of namespaces. Pods are namespaced objects in Kubernetes, so Pod labels also implicitly have namespaces. Any label selectors for Pod labels should specify the namespaces in which Kubernetes should look for those labels.

You express the topology domain (X) using a `topologyKey`, which is the key for the node label that the system uses to denote the domain. For examples, see Well-Known Labels, Annotations and Taints.

> **Note:**
> Inter-pod affinity and anti-affinity require substantial amounts of processing which can slow down scheduling in large clusters significantly. We do not recommend using them in clusters larger than several hundred nodes.

> **Note:**

> Pod anti-affinity requires nodes to be consistently labeled, in other words, every node in the cluster must have an appropriate label matching `topologyKey`. If some or all nodes are missing the specified `topologyKey` label, it can lead to unintended behavior.

Similar to node affinity are two types of Pod affinity and anti-affinity as follows:

- `requiredDuringSchedulingIgnoredDuringExecution`
- `preferredDuringSchedulingIgnoredDuringExecution`

For example, you could use `requiredDuringSchedulingIgnoredDuringExecution` affinity to tell the scheduler to co-locate Pods of two services in the same cloud provider zone because they communicate with each other a lot. Similarly, you could use `preferredDuringSchedulingIgnoredDuringExecution` anti-affinity to spread Pods from a service across multiple cloud provider zones.

To use inter-pod affinity, use the `affinity.podAffinity` field in the Pod spec. For inter-pod anti-affinity, use the `affinity.podAntiAffinity` field in the Pod spec.

Scheduling Behavior

When scheduling a new Pod, the Kubernetes scheduler evaluates the Pod's affinity/anti-affinity rules in the context of the current cluster state:

1. Hard Constraints (Node Filtering):
    - `podAffinity.requiredDuringSchedulingIgnoredDuringExecution` and `podAntiAffinity.requiredDuringSchedulingIgnoredDuringExecution`：
        - The scheduler ensures the new Pod is assigned to nodes that satisfy these required affinity and anti-affinity rules based on existing Pods.
2. Soft Constraints (Scoring):
    - `podAffinity.preferredDuringSchedulingIgnoredDuringExecution` and `podAntiAffinity.preferredDuringSchedulingIgnoredDuringExecution`：
        - The scheduler scores nodes based on how well they meet these preferred affinity and anti-affinity rules to optimize Pod placement.
3. Ignored Fields:
    - Existing Pods' `podAffinity.preferredDuringSchedulingIgnoredDuringExecution`：
        - These preferred affinity rules are not considered during the scheduling decision for new Pods.
    - Existing Pods' `podAntiAffinity.preferredDuringSchedulingIgnoredDuringExecution`：

- Similarly, preferred anti-affinity rules of existing Pods are ignored during scheduling.

Scheduling a Group of Pods with Inter-pod Affinity to Themselves

If the current Pod being scheduled is the first in a series that have affinity to themselves, it is allowed to be scheduled if it passes all other affinity checks. This is determined by verifying that no other Pod in the cluster matches the namespace and selector of this Pod, that the Pod matches its own terms, and the chosen node matches all requested topologies. This ensures that there will not be a deadlock even if all the Pods have inter-pod affinity specified.

Pod Affinity Example

Consider the following Pod spec:

pods/pod-with-pod-affinity.yaml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: with-pod-affinity
spec:
  affinity:
    podAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
      - labelSelector:
          matchExpressions:
          - key: security
            operator: In
            values:
            - S1
        topologyKey: topology.kubernetes.io/zone
    podAntiAffinity:
      preferredDuringSchedulingIgnoredDuringExecution:
      - weight: 100
        podAffinityTerm:
          labelSelector:
            matchExpressions:
            - key: security
              operator: In
              values:
              - S2
          topologyKey: topology.kubernetes.io/zone
```

```
  containers:
  - name: with-pod-affinity
    image: registry.k8s.io/pause:3.8
```

This example defines one Pod affinity rule and one Pod anti-affinity rule. The Pod affinity rule uses the "hard" `requiredDuringSchedulingIgnoredDuringExecution`, while the anti-affinity rule uses the "soft" `preferredDuringSchedulingIgnoredDuringExecution`.

The affinity rule specifies that the scheduler is allowed to place the example Pod on a node only if that node belongs to a specific zone where other Pods have been labeled with `security=S1`. For instance, if we have a cluster with a designated zone, let's call it "Zone V," consisting of nodes labeled with `topology.kubernetes.io/zone=V`, the scheduler can assign the Pod to any node within Zone V, as long as there is at least one Pod within Zone V already labeled with `security=S1`. Conversely, if there are no Pods with `security=S1` labels in Zone V, the scheduler will not assign the example Pod to any node in that zone.

The anti-affinity rule specifies that the scheduler should try to avoid scheduling the Pod on a node if that node belongs to a specific zone where other Pods have been labeled with `security=S2`. For instance, if we have a cluster with a designated zone, let's call it "Zone R," consisting of nodes labeled with `topology.kubernetes.io/zone=R`, the scheduler should avoid assigning the Pod to any node within Zone R, as long as there is at least one Pod within Zone R already labeled with `security=S2`. Conversely, the anti-affinity rule does not impact scheduling into Zone R if there are no Pods with `security=S2` labels.

To get yourself more familiar with the examples of Pod affinity and anti-affinity, refer to the design proposal.

You can use the `In`, `NotIn`, `Exists` and `DoesNotExist` values in the `operator` field for Pod affinity and anti-affinity.

Read Operators to learn more about how these work.

In principle, the `topologyKey` can be any allowed label key with the following exceptions for performance and security reasons:

- For Pod affinity and anti-affinity, an empty `topologyKey` field is not allowed in both `requiredDuringSchedulingIgnoredDuringExecution` and `preferredDuringSchedulingIgnoredDuringExecution`.
- For `requiredDuringSchedulingIgnoredDuringExecution` Pod anti-affinity rules, the admission controller `LimitPodHardAntiAffinityTopology` limits `topologyKey` to `kubernetes.io/hostname`. You can modify or disable the admission controller if you want to allow custom topologies.

In addition to `labelSelector` and `topologyKey`, you can optionally specify a list of namespaces which the `labelSelector` should match against using the `namespaces` field at the same level as `labelSelector` and `topologyKey`. If omitted or empty, `namespaces` defaults to the namespace of the Pod where the affinity/anti-affinity definition appears.

Namespace Selector

FEATURE STATE: Kubernetes v1.24 [stable]

You can also select matching namespaces using `namespaceSelector`, which is a label query over the set of namespaces. The affinity term is applied to namespaces selected by both `namespaceSelector` and the `namespaces` field. Note that an empty `namespaceSelector` ({}) matches all namespaces, while a null or empty `namespaces` list and null `namespaceSelector` matches the namespace of the Pod where the rule is defined.

matchLabelKeys

FEATURE STATE: Kubernetes v1.33 [stable](enabled by default)

> **Note:**
> The `matchLabelKeys` field is a beta-level field and is enabled by default in Kubernetes 1.36. When you want to disable it, you have to disable it explicitly via the `MatchLabelKeysInPodAffinity` feature gate.

Kubernetes includes an optional `matchLabelKeys` field for Pod affinity or anti-affinity. The field specifies keys for the labels that should match with the incoming Pod's labels, when satisfying the Pod (anti)affinity.

The keys are used to look up values from the Pod labels; those key-value labels are combined (using `AND`) with the match restrictions defined using the `labelSelector` field. The combined filtering selects the set of existing Pods that will be taken into Pod (anti)affinity calculation.

> **Caution:**

> It's not recommended to use `matchLabelKeys` with labels that might be updated directly on pods. Even if you edit the pod's label that is specified at `matchLabelKeys` **directly**, (that is, not via a deployment), kube-apiserver doesn't reflect the label update onto the merged `labelSelector`.

A common use case is to use `matchLabelKeys` with `pod-template-hash` (set on Pods managed as part of a Deployment, where the value is unique for each revision). Using `pod-template-hash` in `matchLabelKeys` allows you to target the Pods that belong to the same revision as the incoming Pod, so that a rolling upgrade won't break affinity.

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: application-server
...
spec:
  template:
    spec:
      affinity:
        podAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
              - key: app
                operator: In
                values:
                - database
            topologyKey: topology.kubernetes.io/zone
            # Only Pods from a given rollout are taken into consideration when cal
            # If you update the Deployment, the replacement Pods follow their own
            # (if there are any defined in the new Pod template)
            matchLabelKeys:
            - pod-template-hash
```

mismatchLabelKeys

ⓘ **FEATURE STATE:** Kubernetes v1.33 [stable](enabled by default)

> **Note:**
> The `mismatchLabelKeys` field is a beta-level field and is enabled by default in Kubernetes 1.36. When you want to disable it, you have to disable it explicitly via the `MatchLabelKeysInPodAffinity` feature gate.

Kubernetes includes an optional `mismatchLabelKeys` field for Pod affinity or anti-affinity. The field specifies keys for the labels that should not match with the incoming Pod's labels, when satisfying the Pod (anti)affinity.

> **Caution:**
> It's not recommended to use `mismatchLabelKeys` with labels that might be updated directly on pods. Even if you edit the pod's label that is specified at `mismatchLabelKeys` **directly**, (that is, not via a deployment), kube-apiserver doesn't reflect the label update onto the merged `labelSelector`.

One example use case is to ensure Pods go to the topology domain (node, zone, etc) where only Pods from the same tenant or team are scheduled in. In other words, you want to avoid running Pods from two different tenants on the same topology domain at the same time.

```yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    # Assume that all relevant Pods have a "tenant" label set
    tenant: tenant-a
...
spec:
  affinity:
    podAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
      # ensure that Pods associated with this tenant land on the correct node pool
      - matchLabelKeys:
          - tenant
        labelSelector: {}
        topologyKey: node-pool
    podAntiAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
      # ensure that Pods associated with this tenant can't schedule to nodes used f
```

```
    - mismatchLabelKeys:
      - tenant # whatever the value of the "tenant" label for this Pod, prevent
               # scheduling to nodes in any pool where any Pod from a different
               # tenant is running.
      labelSelector:
        # We have to have the labelSelector which selects only Pods with the tena
        # otherwise this Pod would have anti-affinity against Pods from daemonset
        # which aren't supposed to have the tenant label.
        matchExpressions:
        - key: tenant
          operator: Exists
      topologyKey: node-pool
```

More practical use-cases

Inter-pod affinity and anti-affinity can be even more useful when they are used with higher level collections such as ReplicaSets, StatefulSets, Deployments, etc. These rules allow you to configure that a set of workloads should be co-located in the same defined topology; for example, preferring to place two related Pods onto the same node.

For example: imagine a three-node cluster. You use the cluster to run a web application and also an in-memory cache (such as Redis). For this example, also assume that latency between the web application and the memory cache should be as low as is practical. You could use inter-pod affinity and anti-affinity to co-locate the web servers with the cache as much as possible.

In the following example Deployment for the Redis cache, the replicas get the label `app=store`. The `podAntiAffinity` rule tells the scheduler to avoid placing multiple replicas with the `app=store` label on a single node. This creates each cache in a separate node.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: redis-cache
spec:
  selector:
    matchLabels:
      app: store
  replicas: 3
  template:
    metadata:
      labels:
```

```
        app: store
    spec:
      affinity:
        podAntiAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
              - key: app
                operator: In
                values:
                - store
            topologyKey: "kubernetes.io/hostname"
      containers:
      - name: redis-server
        image: redis:3.2-alpine
```

The following example Deployment for the web servers creates replicas with the label `app=web-store`. The Pod affinity rule tells the scheduler to place each replica on a node that has a Pod with the label `app=store`. The Pod anti-affinity rule tells the scheduler never to place multiple `app=web-store` servers on a single node.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-server
spec:
  selector:
    matchLabels:
      app: web-store
  replicas: 3
  template:
    metadata:
      labels:
        app: web-store
    spec:
      affinity:
        podAntiAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
              - key: app
                operator: In
                values:
                - web-store
```

```
            topologyKey: "kubernetes.io/hostname"
        podAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
              - key: app
                operator: In
                values:
                - store
            topologyKey: "kubernetes.io/hostname"
      containers:
      - name: web-app
        image: nginx:1.16-alpine
```

Creating the two preceding Deployments results in the following cluster layout, where each web server is co-located with a cache, on three separate nodes.

node-1	node-2	node-3
webserver-1	*webserver-2*	*webserver-3*
cache-1	*cache-2*	*cache-3*

The overall effect is that each cache instance is likely to be accessed by a single client that is running on the same node. This approach aims to minimize both skew (imbalanced load) and latency.

You might have other reasons to use Pod anti-affinity. See the ZooKeeper tutorial for an example of a StatefulSet configured with anti-affinity for high availability, using the same technique as this example.

nodeName

`nodeName` is a more direct form of node selection than affinity or `nodeSelector`. `nodeName` is a field in the Pod spec. If the `nodeName` field is not empty, the scheduler ignores the Pod and the kubelet on the named node tries to place the Pod on that node. Using `nodeName` overrules using `nodeSelector` or affinity and anti-affinity rules.

Some of the limitations of using `nodeName` to select nodes are:

- If the named node does not exist, the Pod will not run, and in some cases may be automatically deleted.

- If the named node does not have the resources to accommodate the Pod, the Pod will fail and its reason will indicate why, for example OutOfmemory or OutOfcpu.
- Node names in cloud environments are not always predictable or stable.

> **Warning:**
>
> `nodeName` is intended for use by custom schedulers or advanced use cases where you need to bypass any configured schedulers. Bypassing the schedulers might lead to failed Pods if the assigned Nodes get oversubscribed. You can use node affinity or the `nodeSelector` field to assign a Pod to a specific Node without bypassing the schedulers.

Here is an example of a Pod spec using the `nodeName` field:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
  - name: nginx
    image: nginx
  nodeName: kube-01
```

The above Pod will only run on the node `kube-01`.

nominatedNodeName

> ⓘ **FEATURE STATE:** Kubernetes v1.35 [beta](enabled by default)

`nominatedNodeName` can be used for external components to nominate node for a pending pod. This nomination is best effort: it might be ignored if the scheduler determines the pod cannot go to a nominated node.

Also, this field can be (over)written by the scheduler:

- If the scheduler finds a node to nominate via the preemption.

- If the scheduler decides where the pod is going, and move it to the binding cycle.
  - Note that, in this case, `nominatedNodeName` is put only when the pod has to go through `WaitOnPermit` or `PreBind` extension points.

Here is an example of a Pod status using the `nominatedNodeName` field:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
...
status:
  nominatedNodeName: kube-01
```

Pod topology spread constraints

You can use *topology spread constraints* to control how Pods are spread across your cluster among failure-domains such as regions, zones, nodes, or among any other topology domains that you define. You might do this to improve performance, expected availability, or overall utilization.

Read Pod topology spread constraints to learn more about how these work.

Pod topology labels

> ⓘ **FEATURE STATE:** Kubernetes v1.35 [beta](enabled by default)

Pods inherit the topology labels (`topology.kubernetes.io/zone` and `topology.kubernetes.io/region`) from their assigned Node if those labels are present. These labels can then be utilized via the Downward API to provide the workload with node topology awareness.

Here is an example of a Pod using downward API for it's zone and region:

```
apiVersion: v1
kind: Pod
metadata:
```

```
  name: pod-with-topology-labels
spec:
  containers:
    - name: app
      image: alpine
      command: ["sh", "-c", "env"]
      env:
        - name: MY_ZONE
          valueFrom:
            fieldRef:
              fieldPath: metadata.labels['topology.kubernetes.io/zone']
        - name: MY_REGION
          valueFrom:
            fieldRef:
              fieldPath: metadata.labels['topology.kubernetes.io/region']
```

Operators

The following are all the logical operators that you can use in the `operator` field for `nodeAffinity` and `podAffinity` mentioned above.

Operator	Behavior
`In`	The label value is present in the supplied set of strings
`NotIn`	The label value is not contained in the supplied set of strings
`Exists`	A label with this key exists on the object
`DoesNotExist`	No label with this key exists on the object

The following operators can only be used with `nodeAffinity` .

Operator	Behavior
`Gt`	The field value will be parsed as an integer, and the integer that results from parsing the value of a label named by this selector is greater than this integer
`Lt`	The field value will be parsed as an integer, and the integer that results from parsing the value of a label named by this selector is less than this integer

> **Note:**
> `Gt` and `Lt` operators will not work with non-integer values. If the given value doesn't parse as an integer, the Pod will fail to get scheduled. Also, `Gt` and `Lt` are not available for `podAffinity`.

What's next

- Read more about taints and tolerations.
- Read the design docs for node affinity and for inter-pod affinity/anti-affinity.
- Learn about how the topology manager takes part in node-level resource allocation decisions.
- Learn how to use nodeSelector.
- Learn how to use affinity and anti-affinity.

4 - Pod Overhead

ⓘ **FEATURE STATE:** Kubernetes v1.24 [stable]

When you run a Pod on a Node, the Pod itself takes an amount of system resources. These resources are additional to the resources needed to run the container(s) inside the Pod. In Kubernetes, *Pod Overhead* is a way to account for the resources consumed by the Pod infrastructure on top of the container requests & limits.

In Kubernetes, the Pod's overhead is set at admission time according to the overhead associated with the Pod's RuntimeClass.

A pod's overhead is considered in addition to the sum of container resource requests when scheduling a Pod. Similarly, the kubelet will include the Pod overhead when sizing the Pod cgroup, and when carrying out Pod eviction ranking.

Configuring Pod overhead

You need to make sure a `RuntimeClass` is utilized which defines the `overhead` field.

Usage example

To work with Pod overhead, you need a RuntimeClass that defines the `overhead` field. As an example, you could use the following RuntimeClass definition with a virtualization container runtime (in this example, Kata Containers combined with the Firecracker virtual machine monitor) that uses around 120MiB per Pod for the virtual machine and the guest OS:

```yaml
# You need to change this example to match the actual runtime name, and per-Pod
# resource overhead, that the container runtime is adding in your cluster.
apiVersion: node.k8s.io/v1
kind: RuntimeClass
metadata:
  name: kata-fc
handler: kata-fc
overhead:
  podFixed:
    memory: "120Mi"
    cpu: "250m"
```

Workloads which are created which specify the `kata-fc` RuntimeClass handler will take the memory and cpu overheads into account for resource quota calculations, node scheduling, as well as Pod cgroup sizing.

Consider running the given example workload, test-pod:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  runtimeClassName: kata-fc
  containers:
  - name: busybox-ctr
    image: busybox:1.28
    stdin: true
    tty: true
    resources:
      limits:
        cpu: 500m
        memory: 100Mi
  - name: nginx-ctr
    image: nginx
    resources:
      limits:
        cpu: 1500m
        memory: 100Mi
```

> **Note:**
> If only `limits` are specified in the pod definition, kubelet will deduce `requests` from those limits and set them to be the same as the defined `limits`.

At admission time the RuntimeClass admission controller updates the workload's PodSpec to include the `overhead` as described in the RuntimeClass. If the PodSpec already has this field defined, the Pod will be rejected. In the given example, since only the RuntimeClass name is specified, the admission controller mutates the Pod to include an `overhead`.

After the RuntimeClass admission controller has made modifications, you can check the updated Pod overhead value:

```
kubectl get pod test-pod -o jsonpath='{.spec.overhead}'
```

The output is:

```
map[cpu:250m memory:120Mi]
```

If a ResourceQuota is defined, the sum of container requests as well as the `overhead` field are counted.

When the kube-scheduler is deciding which node should run a new Pod, the scheduler considers that Pod's `overhead` as well as the sum of container requests for that Pod. For this example, the scheduler adds the requests and the overhead, then looks for a node that has 2.25 CPU and 320 MiB of memory available.

Once a Pod is scheduled to a node, the kubelet on that node creates a new cgroup for the Pod. It is within this pod that the underlying container runtime will create containers.

If the resource has a limit defined for each container (Guaranteed QoS or Burstable QoS with limits defined), the kubelet will set an upper limit for the pod cgroup associated with that resource (cpu.cfs_quota_us for CPU and memory.limit_in_bytes memory). This upper limit is based on the sum of the container limits plus the `overhead` defined in the PodSpec.

For CPU, if the Pod is Guaranteed or Burstable QoS, the kubelet will set `cpu.shares` based on the sum of container requests plus the `overhead` defined in the PodSpec.

Looking at our example, verify the container requests for the workload:

```
kubectl get pod test-pod -o jsonpath='{.spec.containers[*].resources.limits}'
```

The total container requests are 2000m CPU and 200MiB of memory:

```
map[cpu: 500m memory:100Mi] map[cpu:1500m memory:100Mi]
```

Check this against what is observed by the node:

```
kubectl describe node | grep test-pod -B2
```

The output shows requests for 2250m CPU, and for 320MiB of memory. The requests include Pod overhead:

```
Namespace    Name       CPU Requests  CPU Limits   Memory Requests  Memory Limits
---------    ----       ------------  ----------   ---------------  -------------
default      test-pod   2250m (56%)   2250m (56%)  320Mi (1%)       320Mi (1%)
```

Verify Pod cgroup limits

Check the Pod's memory cgroups on the node where the workload is running. In the following example, `crictl` is used on the node, which provides a CLI for CRI-compatible container runtimes. This is an advanced example to show Pod overhead behavior, and it is not expected that users should need to check cgroups directly on the node.

First, on the particular node, determine the Pod identifier:

```
# Run this on the node where the Pod is scheduled
POD_ID="$(sudo crictl pods --name test-pod -q)"
```

From this, you can determine the cgroup path for the Pod:

```
# Run this on the node where the Pod is scheduled
sudo crictl inspectp -o=json $POD_ID | grep cgroupsPath
```

The resulting cgroup path includes the Pod's `pause` container. The Pod level cgroup is one directory above.

```
"cgroupsPath": "/kubepods/podd7f4b509-cf94-4951-9417-d1087c92a5b2/7ccf55aee35dd16
```

In this specific case, the pod cgroup path is `kubepods/podd7f4b509-cf94-4951-9417-d1087c92a5b2`. Verify the Pod level cgroup setting for memory:

```
# Run this on the node where the Pod is scheduled.
# Also, change the name of the cgroup to match the cgroup allocated for your pod.
 cat /sys/fs/cgroup/memory/kubepods/podd7f4b509-cf94-4951-9417-d1087c92a5b2/memory
```

This is 320 MiB, as expected:

```
335544320
```

Observability

Some `kube_pod_overhead_*` metrics are available in kube-state-metrics to help identify when Pod overhead is being utilized and to help observe stability of workloads running with a defined overhead.

What's next

- Learn more about RuntimeClass
- Read the PodOverhead Design enhancement proposal for extra context

5 - Pod Scheduling Readiness

ⓘ **FEATURE STATE:** Kubernetes v1.30 [stable]

Pods were considered ready for scheduling once created. Kubernetes scheduler does its due diligence to find nodes to place all pending Pods. However, in a real-world case, some Pods may stay in a "miss-essential-resources" state for a long period. These Pods actually churn the scheduler (and downstream integrators like Cluster AutoScaler) in an unnecessary manner.

By specifying/removing a Pod's `.spec.schedulingGates`, you can control when a Pod is ready to be considered for scheduling.

Configuring Pod schedulingGates

The `schedulingGates` field contains a list of strings, and each string literal is perceived as a criteria that Pod should be satisfied before considered schedulable. This field can be initialized only when a Pod is created (either by the client, or mutated during admission). After creation, each schedulingGate can be removed in arbitrary order, but addition of a new scheduling gate is disallowed.

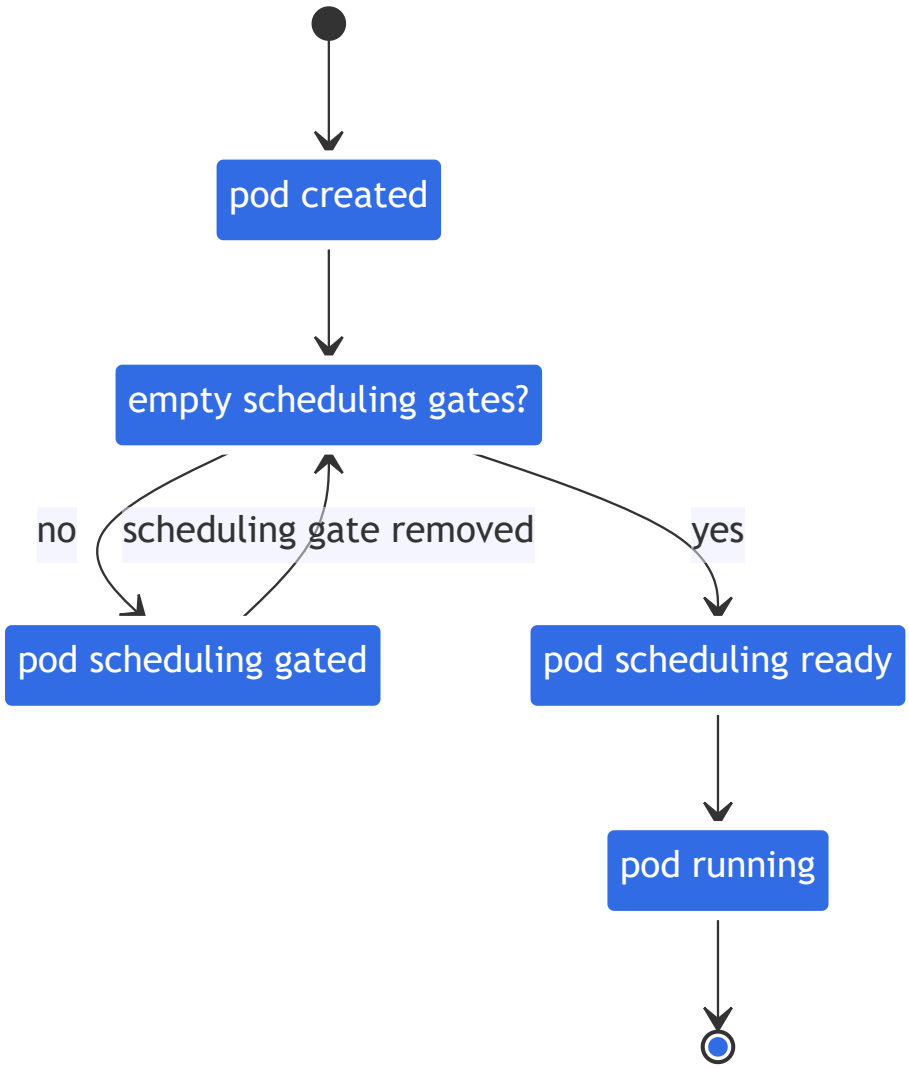


Figure. Pod SchedulingGates

Usage example

To mark a Pod not-ready for scheduling, you can create it with one or more scheduling gates like this:

pods/pod-with-scheduling-gates.yaml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  schedulingGates:
  - name: example.com/foo
  - name: example.com/bar
  containers:
  - name: pause
```

```
    image: registry.k8s.io/pause:3.6
```

After the Pod's creation, you can check its state using:

```
kubectl get pod test-pod
```

The output reveals it's in `SchedulingGated` state:

```
NAME       READY   STATUS            RESTARTS   AGE
test-pod   0/1     SchedulingGated   0          7s
```

You can also check its `schedulingGates` field by running:

```
kubectl get pod test-pod -o jsonpath='{.spec.schedulingGates}'
```

The output is:

```
[{"name":"example.com/foo"},{"name":"example.com/bar"}]
```

To inform scheduler this Pod is ready for scheduling, you can remove its `schedulingGates` entirely by reapplying a modified manifest:

pods/pod-without-scheduling-gates.yaml 

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
  - name: pause
    image: registry.k8s.io/pause:3.6
```

You can check if the `schedulingGates` is cleared by running:

```
kubectl get pod test-pod -o jsonpath='{.spec.schedulingGates}'
```

The output is expected to be empty. And you can check its latest status by running:

```
kubectl get pod test-pod -o wide
```

Given the test-pod doesn't request any CPU/memory resources, it's expected that this Pod's state get transited from previous `SchedulingGated` to `Running`:

```
NAME       READY   STATUS    RESTARTS   AGE   IP         NODE
test-pod   1/1     Running   0          15s   10.0.0.4   node-2
```

Observability

The metric `scheduler_pending_pods` comes with a new label `"gated"` to distinguish whether a Pod has been tried scheduling but claimed as unschedulable, or explicitly marked as not ready for scheduling. You can use `scheduler_pending_pods{queue="gated"}` to check the metric result.

Mutable Pod scheduling directives

You can mutate scheduling directives of Pods while they have scheduling gates, with certain constraints. At a high level, you can only tighten the scheduling directives of a Pod. In other words, the updated directives would cause the Pods to only be able to be scheduled on a subset of the nodes that it would previously match. More concretely, the rules for updating a Pod's scheduling directives are as follows:

1. For `.spec.nodeSelector`, only additions are allowed. If absent, it will be allowed to be set.

2. For `spec.affinity.nodeAffinity`, if nil, then setting anything is allowed.

3. If `NodeSelectorTerms` was empty, it will be allowed to be set. If not empty, then only additions of `NodeSelectorRequirements` to `matchExpressions` or `fieldExpressions` are allowed, and no changes to existing `matchExpressions` and `fieldExpressions` will be allowed. This is because the terms in `.requiredDuringSchedulingIgnoredDuringExecution.NodeSelectorTerms`, are ORed while

the expressions in `nodeSelectorTerms[].matchExpressions` and `nodeSelectorTerms[].fieldExpressions` are ANDed.

4. For `.preferredDuringSchedulingIgnoredDuringExecution`, all updates are allowed. This is because preferred terms are not authoritative, and so policy controllers don't validate those terms.

What's next

- Read the PodSchedulingReadiness KEP for more details

6 - Pod Topology Spread Constraints

You can use *topology spread constraints* to control how Pods are spread across your cluster among failure-domains such as regions, zones, nodes, and other user-defined topology domains. This can help to achieve high availability as well as efficient resource utilization.

You can set cluster-level constraints as a default, or configure topology spread constraints for individual workloads.

Motivation

Imagine that you have a cluster of up to twenty nodes, and you want to run a workload that automatically scales how many replicas it uses. There could be as few as two Pods or as many as fifteen. When there are only two Pods, you'd prefer not to have both of those Pods run on the same node: you would run the risk that a single node failure takes your workload offline.

In addition to this basic usage, there are some advanced usage examples that enable your workloads to benefit on high availability and cluster utilization.

As you scale up and run more Pods, a different concern becomes important. Imagine that you have three nodes running five Pods each. The nodes have enough capacity to run that many replicas; however, the clients that interact with this workload are split across three different datacenters (or infrastructure zones). Now you have less concern about a single node failure, but you notice that latency is higher than you'd like, and you are paying for network costs associated with sending network traffic between the different zones.

You decide that under normal operation you'd prefer to have a similar number of replicas scheduled into each infrastructure zone, and you'd like the cluster to self-heal in the case that there is a problem.

Pod topology spread constraints offer you a declarative way to configure that.

`topologySpreadConstraints` field

The Pod API includes a field, `spec.topologySpreadConstraints`. The usage of this field looks like the following:

```
---
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
spec:
  # Configure a topology spread constraint
  topologySpreadConstraints:
    - maxSkew: <integer>
      minDomains: <integer> # optional
      topologyKey: <string>
      whenUnsatisfiable: <string>
      labelSelector: <object>
      matchLabelKeys: <list> # optional; beta since v1.27
      nodeAffinityPolicy: [Honor|Ignore] # optional; beta since v1.26
      nodeTaintsPolicy: [Honor|Ignore] # optional; beta since v1.26
  ### other Pod fields go here
```

> **Note:**
> There can only be one `topologySpreadConstraint` for a given `topologyKey` and `whenUnsatisfiable` value. For example, if you have defined a `topologySpreadConstraint` that uses the `topologyKey` "kubernetes.io/hostname" and `whenUnsatisfiable` value "DoNotSchedule", you can only add another `topologySpreadConstraint` for the `topologyKey` "kubernetes.io/hostname" if you use a different `whenUnsatisfiable` value.

You can read more about this field by running `kubectl explain Pod.spec.topologySpreadConstraints` or refer to the scheduling section of the API reference for Pod.

Spread constraint definition

You can define one or multiple `topologySpreadConstraints` entries to instruct the kube-scheduler how to place each incoming Pod in relation to the existing Pods across your cluster. Those fields are:

- **maxSkew** describes the degree to which Pods may be unevenly distributed. You must specify this field and the number must be greater than zero. Its semantics differ according to the value of `whenUnsatisfiable`：
  - if you select `whenUnsatisfiable: DoNotSchedule`, then `maxSkew` defines the maximum permitted difference between the number of matching pods in the target topology and the *global minimum* (the minimum number of matching

pods in an eligible domain or zero if the number of eligible domains is less than MinDomains). For example, if you have 3 zones with 2, 2 and 1 matching pods respectively, `MaxSkew` is set to 1 then the global minimum is 1.

- if you select `whenUnsatisfiable: ScheduleAnyway`, the scheduler gives higher precedence to topologies that would help reduce the skew.

- **minDomains** indicates a minimum number of eligible domains. This field is optional. A domain is a particular instance of a topology. An eligible domain is a domain whose nodes match the node selector.

> **Note:**
> Before Kubernetes v1.30, the `minDomains` field was only available if the `MinDomainsInPodTopologySpread` feature gate was enabled (default since v1.28). In older Kubernetes clusters it might be explicitly disabled or the field might not be available.

  - The value of `minDomains` must be greater than 0, when specified. You can only specify `minDomains` in conjunction with `whenUnsatisfiable: DoNotSchedule`.
  - When the number of eligible domains with match topology keys is less than `minDomains`, Pod topology spread treats global minimum as 0, and then the calculation of `skew` is performed. The global minimum is the minimum number of matching Pods in an eligible domain, or zero if the number of eligible domains is less than `minDomains`.
  - When the number of eligible domains with matching topology keys equals or is greater than `minDomains`, this value has no effect on scheduling.
  - If you do not specify `minDomains`, the constraint behaves as if `minDomains` is 1.
- **topologyKey** is the key of node labels. Nodes that have a label with this key and identical values are considered to be in the same topology. We call each instance of a topology (in other words, a <key, value> pair) a domain. The scheduler will try to put a balanced number of pods into each domain. Also, we define an eligible domain as a domain whose nodes meet the requirements of nodeAffinityPolicy and nodeTaintsPolicy.

- **whenUnsatisfiable** indicates how to deal with a Pod if it doesn't satisfy the spread constraint:

  - `DoNotSchedule` (default) tells the scheduler not to schedule it.
  - `ScheduleAnyway` tells the scheduler to still schedule it while prioritizing nodes that minimize the skew.
- **labelSelector** is used to find matching Pods. Pods that match this label selector are counted to determine the number of Pods in their corresponding topology domain. See Label Selectors for more details.

- **matchLabelKeys** is a list of pod label keys to select the group of pods over which the spreading skew will be calculated. At a pod creation, the kube-apiserver uses those keys to lookup values from the incoming pod labels, and those key-value labels will be merged with any existing `labelSelector`. The same key is forbidden to exist in both `matchLabelKeys` and `labelSelector`. `matchLabelKeys` cannot be set when `labelSelector` isn't set. Keys that don't exist in the pod labels will be ignored. A null or empty list means only match against the `labelSelector`.

  > **Caution:**
  > It's not recommended to use `matchLabelKeys` with labels that might be updated directly on pods. Even if you edit the pod's label that is specified at `matchLabelKeys` **directly**, (that is, you edit the Pod and not a Deployment), kube-apiserver doesn't reflect the label update onto the merged `labelSelector`.

  With `matchLabelKeys`, you don't need to update the `pod.spec` between different revisions. The controller/operator just needs to set different values to the same label key for different revisions. For example, if you are configuring a Deployment, you can use the label keyed with pod-template-hash, which is added automatically by the Deployment controller, to distinguish between different revisions in a single Deployment.

  ```
  topologySpreadConstraints:
      - maxSkew: 1
        topologyKey: kubernetes.io/hostname
        whenUnsatisfiable: DoNotSchedule
        labelSelector:
          matchLabels:
            app: foo
        matchLabelKeys:
          - pod-template-hash
  ```

  > **Note:**
  > The `matchLabelKeys` field is a beta-level field and enabled by default in 1.27. You can disable it by disabling the `MatchLabelKeysInPodTopologySpread` feature gate.
  >
  > Before v1.34, `matchLabelKeys` was handled implicitly. Since v1.34, key-value labels corresponding to `matchLabelKeys` are explicitly merged into

> `labelSelector` . You can disable it and revert to the previous behavior by disabling the `MatchLabelKeysInPodTopologySpreadSelectorMerge` feature gate of kube-apiserver.

- **nodeAffinityPolicy** indicates how we will treat Pod's nodeAffinity/nodeSelector when calculating pod topology spread skew. Options are:
  - Honor: only nodes matching nodeAffinity/nodeSelector are included in the calculations.
  - Ignore: nodeAffinity/nodeSelector are ignored. All nodes are included in the calculations.

  If this value is null, the behavior is equivalent to the Honor policy.

> **Note:**
> The `nodeAffinityPolicy` became beta in 1.26 and graduated to GA in 1.33. It's enabled by default in beta, you can disable it by disabling the `NodeInclusionPolicyInPodTopologySpread` feature gate.

- **nodeTaintsPolicy** indicates how we will treat node taints when calculating pod topology spread skew. Options are:
  - Honor: nodes without taints, along with tainted nodes for which the incoming pod has a toleration, are included.
  - Ignore: node taints are ignored. All nodes are included.

  If this value is null, the behavior is equivalent to the Ignore policy.

> **Note:**
> The `nodeTaintsPolicy` became beta in 1.26 and graduated to GA in 1.33. It's enabled by default in beta, you can disable it by disabling the `NodeInclusionPolicyInPodTopologySpread` feature gate.

When a Pod defines more than one `topologySpreadConstraint` , those constraints are combined using a logical AND operation: the kube-scheduler looks for a node for the incoming Pod that satisfies all the configured constraints.

Node labels

Topology spread constraints rely on node labels to identify the topology domain(s) that each node is in. For example, a node might have labels:

```
  region: us-east-1
  zone: us-east-1a
```

> **Note:**
> For brevity, this example doesn't use the well-known label keys `topology.kubernetes.io/zone` and `topology.kubernetes.io/region`. However, those registered label keys are nonetheless recommended rather than the private (unqualified) label keys `region` and `zone` that are used here.
>
> You can't make a reliable assumption about the meaning of a private label key between different contexts.

Suppose you have a 4-node cluster with the following labels:

```
NAME    STATUS   ROLES    AGE     VERSION   LABELS
node1   Ready    <none>   4m26s   v1.16.0   node=node1,zone=zoneA
node2   Ready    <none>   3m58s   v1.16.0   node=node2,zone=zoneA
node3   Ready    <none>   3m17s   v1.16.0   node=node3,zone=zoneB
node4   Ready    <none>   2m43s   v1.16.0   node=node4,zone=zoneB
```

Then the cluster is logically viewed as below:

Consistency

You should set the same Pod topology spread constraints on all pods in a group.

Usually, if you are using a workload controller such as a Deployment, the pod template takes care of this for you. If you mix different spread constraints then Kubernetes follows the API definition of the field; however, the behavior is more likely to become confusing and troubleshooting is less straightforward.

You need a mechanism to ensure that all the nodes in a topology domain (such as a cloud provider region) are labeled consistently. To avoid you needing to manually label nodes, most clusters automatically populate well-known labels such as `kubernetes.io/hostname`. Check whether your cluster supports this.

Topology spread constraint examples

Example: one topology spread constraint

Suppose you have a 4-node cluster where 3 Pods labeled `foo: bar` are located in node1, node2 and node3 respectively:

If you want an incoming Pod to be evenly spread with existing Pods across zones, you can use a manifest similar to:

pods/topology-spread-constraints/one-constraint.yaml

```yaml
kind: Pod
apiVersion: v1
metadata:
  name: mypod
  labels:
    foo: bar
spec:
  topologySpreadConstraints:
  - maxSkew: 1
    topologyKey: zone
    whenUnsatisfiable: DoNotSchedule
    labelSelector:
      matchLabels:
        foo: bar
  containers:
  - name: pause
    image: registry.k8s.io/pause:3.1
```

From that manifest, `topologyKey: zone` implies the even distribution will only be applied to nodes that are labeled `zone: <any value>` (nodes that don't have a `zone` label are skipped). The field `whenUnsatisfiable: DoNotSchedule` tells the scheduler to let the incoming Pod stay pending if the scheduler can't find a way to satisfy the constraint.

If the scheduler placed this incoming Pod into zone `A`, the distribution of Pods would become `[3, 1]`. That means the actual skew is then 2 (calculated as `3 - 1`), which violates `maxSkew: 1`. To satisfy the constraints and context for this example, the incoming Pod can only be placed onto a node in zone `B`:

OR

You can tweak the Pod spec to meet various kinds of requirements:

- Change `maxSkew` to a bigger value - such as `2` - so that the incoming Pod can be placed into zone `A` as well.
- Change `topologyKey` to `node` so as to distribute the Pods evenly across nodes instead of zones. In the above example, if `maxSkew` remains `1`, the incoming Pod can only be placed onto the node `node4`.
- Change `whenUnsatisfiable: DoNotSchedule` to `whenUnsatisfiable: ScheduleAnyway` to ensure the incoming Pod to be always schedulable (suppose other scheduling APIs are satisfied). However, it's preferred to be placed into the topology domain which has fewer matching Pods. (Be aware that this preference is jointly normalized with other internal scheduling priorities such as resource usage ratio).

Example: multiple topology spread constraints

This builds upon the previous example. Suppose you have a 4-node cluster where 3 existing Pods labeled `foo: bar` are located on node1, node2 and node3 respectively:

You can combine two topology spread constraints to control the spread of Pods both by node and by zone:

pods/topology-spread-constraints/two-constraints.yaml

```yaml
kind: Pod
apiVersion: v1
metadata:
  name: mypod
  labels:
    foo: bar
spec:
  topologySpreadConstraints:
  - maxSkew: 1
    topologyKey: zone
    whenUnsatisfiable: DoNotSchedule
    labelSelector:
      matchLabels:
        foo: bar
  - maxSkew: 1
    topologyKey: node
    whenUnsatisfiable: DoNotSchedule
    labelSelector:
      matchLabels:
        foo: bar
  containers:
```

```
  - name: pause
    image: registry.k8s.io/pause:3.1
```

In this case, to match the first constraint, the incoming Pod can only be placed onto nodes in zone `B`; while in terms of the second constraint, the incoming Pod can only be scheduled to the node `node4`. The scheduler only considers options that satisfy all defined constraints, so the only valid placement is onto node `node4`.

Example: conflicting topology spread constraints

Multiple constraints can lead to conflicts. Suppose you have a 3-node cluster across 2 zones:

If you were to apply `two-constraints.yaml` (the manifest from the previous example) to **this** cluster, you would see that the Pod `mypod` stays in the `Pending` state. This happens because: to satisfy the first constraint, the Pod `mypod` can only be placed into zone `B`; while in terms of the second constraint, the Pod `mypod` can only schedule to node `node2`. The intersection of the two constraints returns an empty set, and the scheduler cannot place the Pod.

To overcome this situation, you can either increase the value of `maxSkew` or modify one of the constraints to use `whenUnsatisfiable: ScheduleAnyway`. Depending on circumstances, you might also decide to delete an existing Pod manually - for example, if you are troubleshooting why a bug-fix rollout is not making progress.

Interaction with node affinity and node selectors

The scheduler will skip the non-matching nodes from the skew calculations if the incoming Pod has `spec.nodeSelector` or `spec.affinity.nodeAffinity` defined.

Example: topology spread constraints with node affinity

Suppose you have a 5-node cluster ranging across zones A to C:

---

and you know that zone `C` must be excluded. In this case, you can compose a manifest as below, so that Pod `mypod` will be placed into zone `B` instead of zone `C`. Similarly, Kubernetes also respects `spec.nodeSelector`.

pods/topology-spread-constraints/one-constraint-with-nodeaffinity.yaml

```
kind: Pod
```

```yaml
apiVersion: v1
metadata:
  name: mypod
  labels:
    foo: bar
spec:
  topologySpreadConstraints:
  - maxSkew: 1
    topologyKey: zone
    whenUnsatisfiable: DoNotSchedule
    labelSelector:
      matchLabels:
        foo: bar
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
        - matchExpressions:
          - key: zone
            operator: NotIn
            values:
            - zoneC
  containers:
  - name: pause
    image: registry.k8s.io/pause:3.1
```

Implicit conventions

There are some implicit conventions worth noting here:

- Only the Pods holding the same namespace as the incoming Pod can be matching candidates.
- The scheduler only considers nodes that have all `topologySpreadConstraints[*].topologyKey` present at the same time. Nodes missing any of these `topologyKeys` are bypassed. This implies that:
  1. any Pods located on those bypassed nodes do not impact `maxSkew` calculation - in the above example, suppose the node `node1` does not have a label "zone", then the 2 Pods will be disregarded, hence the incoming Pod will be scheduled into zone `A`.
  2. the incoming Pod has no chances to be scheduled onto this kind of nodes - in the above example, suppose a node `node5` has the **mistyped** label `zone-typo`:

`zoneC` (and no `zone` label set). After node `node5` joins the cluster, it will be bypassed and Pods for this workload aren't scheduled there.

- Be aware of what will happen if the incoming Pod's `topologySpreadConstraints[*].labelSelector` doesn't match its own labels. In the above example, if you remove the incoming Pod's labels, it can still be placed onto nodes in zone `B`, since the constraints are still satisfied. However, after that placement, the degree of imbalance of the cluster remains unchanged - it's still zone `A` having 2 Pods labeled as `foo: bar`, and zone `B` having 1 Pod labeled as `foo: bar`. If this is not what you expect, update the workload's `topologySpreadConstraints[*].labelSelector` to match the labels in the pod template.

Cluster-level default constraints

It is possible to set default topology spread constraints for a cluster. Default topology spread constraints are applied to a Pod if, and only if:

- It doesn't define any constraints in its `.spec.topologySpreadConstraints`.
- It belongs to a Service, ReplicaSet, StatefulSet or ReplicationController.

Default constraints can be set as part of the `PodTopologySpread` plugin arguments in a scheduling profile. The constraints are specified with the same API above, except that `labelSelector` must be empty. The selectors are calculated from the Services, ReplicaSets, StatefulSets or ReplicationControllers that the Pod belongs to.

An example configuration might look like follows:

```yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration

profiles:
  - schedulerName: default-scheduler
    pluginConfig:
      - name: PodTopologySpread
        args:
          defaultConstraints:
            - maxSkew: 1
              topologyKey: topology.kubernetes.io/zone
              whenUnsatisfiable: ScheduleAnyway
          defaultingType: List
```

Built-in default constraints

FEATURE STATE: Kubernetes v1.24 [stable]

If you don't configure any cluster-level default constraints for pod topology spreading, then kube-scheduler acts as if you specified the following default topology constraints:

```
defaultConstraints:
  - maxSkew: 3
    topologyKey: "kubernetes.io/hostname"
    whenUnsatisfiable: ScheduleAnyway
  - maxSkew: 5
    topologyKey: "topology.kubernetes.io/zone"
    whenUnsatisfiable: ScheduleAnyway
```

Also, the legacy `SelectorSpread` plugin, which provides an equivalent behavior, is disabled by default.

> **Note:**
> The `PodTopologySpread` plugin does not score the nodes that don't have the topology keys specified in the spreading constraints. This might result in a different default behavior compared to the legacy `SelectorSpread` plugin when using the default topology constraints.
>
> If your nodes are not expected to have **both** `kubernetes.io/hostname` and `topology.kubernetes.io/zone` labels set, define your own constraints instead of using the Kubernetes defaults.

If you don't want to use the default Pod spreading constraints for your cluster, you can disable those defaults by setting `defaultingType` to `List` and leaving empty `defaultConstraints` in the `PodTopologySpread` plugin configuration:

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration

profiles:
  - schedulerName: default-scheduler
    pluginConfig:
```

```
      - name: PodTopologySpread
        args:
          defaultConstraints: []
          defaultingType: List
```

Comparison with podAffinity and podAntiAffinity

In Kubernetes, inter-Pod affinity and anti-affinity control how Pods are scheduled in relation to one another - either more packed or more scattered.

podAffinity

attracts Pods; you can try to pack any number of Pods into qualifying topology domain(s).

podAntiAffinity

repels Pods. If you set this to `requiredDuringSchedulingIgnoredDuringExecution` mode then only a single Pod can be scheduled into a single topology domain; if you choose `preferredDuringSchedulingIgnoredDuringExecution` then you lose the ability to enforce the constraint.

For finer control, you can specify topology spread constraints to distribute Pods across different topology domains - to achieve either high availability or cost-saving. This can also help on rolling update workloads and scaling out replicas smoothly.

For more context, see the Motivation section of the enhancement proposal about Pod topology spread constraints.

Known limitations

- There's no guarantee that the constraints remain satisfied when Pods are removed. For example, scaling down a Deployment may result in imbalanced Pods distribution.

  You can use a tool such as the Descheduler to rebalance the Pods distribution.

- Pods matched on tainted nodes are respected. See Issue 80921.

- The scheduler doesn't have prior knowledge of all the zones or other topology domains that a cluster has. They are determined from the existing nodes in the cluster. This could lead to a problem in autoscaled clusters, when a node pool (or node group) is scaled to zero nodes, and you're expecting the cluster to scale up,

because, in this case, those topology domains won't be considered until there is at least one node in them.

You can work around this by using a Node autoscaler that is aware of Pod topology spread constraints and is also aware of the overall set of topology domains.

- Pods that don't match their own labelSelector create "ghost pods". If a pod's labels don't match the `labelSelector` in its topology spread constraint, the pod won't count itself in spread calculations. This means:
  - Multiple such pods can just accumulate on the same topology (until matching pods are newly created/deleted) because those pod's schedule don't change a spreading calculation result.
  - The spreading constraint works in an unintended way, most likely not matching your expectations

  Ensure your pod's labels match the `labelSelector` in your spread constraints. Typically, a pod should match its own topology spread constraint selector.

What's next

- The blog article Introducing PodTopologySpread explains `maxSkew` in some detail, as well as covering some advanced usage examples.
- Read the scheduling section of the API reference for Pod.

7 - Taints and Tolerations

Node affinity is a property of Pods that *attracts* them to a set of nodes (either as a preference or a hard requirement). *Taints* are the opposite -- they allow a node to repel a set of pods.

Tolerations are applied to pods. Tolerations allow the scheduler to schedule pods with matching taints. Tolerations allow scheduling but don't guarantee scheduling: the scheduler also evaluates other parameters as part of its function.

Taints and tolerations work together to ensure that pods are not scheduled onto inappropriate nodes. One or more taints are applied to a node; this marks that the node should not accept any pods that do not tolerate the taints.

Concepts

You add a taint to a node using kubectl taint. For example,

```
kubectl taint nodes node1 key1=value1:NoSchedule
```

places a taint on node `node1`. The taint has key `key1`, value `value1`, and taint effect `NoSchedule`. This means that no pod will be able to schedule onto `node1` unless it has a matching toleration.

To remove the taint added by the command above, you can run:

```
kubectl taint nodes node1 key1=value1:NoSchedule-
```

You specify a toleration for a pod in the PodSpec. Both of the following tolerations "match" the taint created by the `kubectl taint` line above, and thus a pod with either toleration would be able to schedule onto `node1`:

```
tolerations:
- key: "key1"
  operator: "Equal"
  value: "value1"
  effect: "NoSchedule"
```

```yaml
tolerations:
- key: "key1"
  operator: "Exists"
  effect: "NoSchedule"
```

The default Kubernetes scheduler takes taints and tolerations into account when selecting a node to run a particular Pod. However, if you manually specify the `.spec.nodeName` for a Pod, that action bypasses the scheduler; the Pod is then bound onto the node where you assigned it, even if there are `NoSchedule` taints on that node that you selected. If this happens and the node also has a `NoExecute` taint set, the kubelet will eject the Pod unless there is an appropriate tolerance set.

Here's an example of a pod that has some tolerations defined:

pods/pod-with-toleration.yaml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  labels:
    env: test
spec:
  containers:
  - name: nginx
    image: nginx
    imagePullPolicy: IfNotPresent
  tolerations:
  - key: "example-key"
    operator: "Exists"
    effect: "NoSchedule"
```

The default value for `operator` is `Equal`.

A toleration "matches" a taint if the keys are the same and the effects are the same, and:

- the `operator` is `Exists` (in which case no `value` should be specified), or
- the `operator` is `Equal` and the values should be equal.

> **Note:**

> There are two special cases:
>
> If the `key` is empty, then the `operator` must be `Exists`, which matches all keys and values. Note that the `effect` still needs to be matched at the same time.
>
> An empty `effect` matches all effects with key `key1`.

The above example used the `effect` of `NoSchedule`. Alternatively, you can use the `effect` of `PreferNoSchedule`.

The allowed values for the `effect` field are:

`NoExecute`

This affects pods that are already running on the node as follows:

- Pods that do not tolerate the taint are evicted immediately
- Pods that tolerate the taint without specifying `tolerationSeconds` in their toleration specification remain bound forever
- Pods that tolerate the taint with a specified `tolerationSeconds` remain bound for the specified amount of time. After that time elapses, the node lifecycle controller evicts the Pods from the node.

`NoSchedule`

No new Pods will be scheduled on the tainted node unless they have a matching toleration. Pods currently running on the node are **not** evicted.

`PreferNoSchedule`

`PreferNoSchedule` is a "preference" or "soft" version of `NoSchedule`. The control plane will *try* to avoid placing a Pod that does not tolerate the taint on the node, but it is not guaranteed.

You can put multiple taints on the same node and multiple tolerations on the same pod. The way Kubernetes processes multiple taints and tolerations is like a filter: start with all of a node's taints, then ignore the ones for which the pod has a matching toleration; the remaining un-ignored taints have the indicated effects on the pod. In particular,

- if there is at least one un-ignored taint with effect `NoSchedule` then Kubernetes will not schedule the pod onto that node
- if there is no un-ignored taint with effect `NoSchedule` but there is at least one un-ignored taint with effect `PreferNoSchedule` then Kubernetes will *try* to not schedule the pod onto the node

- if there is at least one un-ignored taint with effect `NoExecute` then the pod will be evicted from the node (if it is already running on the node), and will not be scheduled onto the node (if it is not yet running on the node).

For example, imagine you taint a node like this

```
kubectl taint nodes node1 key1=value1:NoSchedule
kubectl taint nodes node1 key1=value1:NoExecute
kubectl taint nodes node1 key2=value2:NoSchedule
```

And a pod has two tolerations:

```yaml
tolerations:
- key: "key1"
  operator: "Equal"
  value: "value1"
  effect: "NoSchedule"
- key: "key1"
  operator: "Equal"
  value: "value1"
  effect: "NoExecute"
```

In this case, the pod will not be able to schedule onto the node, because there is no toleration matching the third taint. But it will be able to continue running if it is already running on the node when the taint is added, because the third taint is the only one of the three that is not tolerated by the pod.

Normally, if a taint with effect `NoExecute` is added to a node, then any pods that do not tolerate the taint will be evicted immediately, and pods that do tolerate the taint will never be evicted. However, a toleration with `NoExecute` effect can specify an optional `tolerationSeconds` field that dictates how long the pod will stay bound to the node after the taint is added. For example,

```yaml
tolerations:
- key: "key1"
  operator: "Equal"
  value: "value1"
  effect: "NoExecute"
  tolerationSeconds: 3600
```

means that if this pod is running and a matching taint is added to the node, then the pod will stay bound to the node for 3600 seconds, and then be evicted. If the taint is removed before that time, the pod will not be evicted.

Numeric comparison operators

ⓘ **FEATURE STATE:** Kubernetes v1.35 [alpha](disabled by default)

In addition to `Equal` and `Exists`, you can use numeric comparison operators (`Gt` and `Lt`) to match taints with integer values. This is useful for threshold-based scheduling, such as matching nodes by reliability level or SLA tier.

- `Gt` matches when the taint value is greater than the toleration value.
- `Lt` matches when the taint value is less than the toleration value.

For numeric operators, both the toleration and taint values must be valid integers. If either value cannot be parsed as an integer, the toleration does not match.

> **Note:**
> When you create a Pod that uses `Gt` or `Lt` tolerations operators, the API server validates that the toleration values are valid integers. Taint values on nodes are not validated at node registration time. If a node has a non-numeric taint value (for example, `servicelevel.organization.example/agreed-service-level=high:NoSchedule`), pods with numeric comparison operators will not match that taint and cannot schedule on that node.

For example, if nodes are tainted with a value representing a service level agreement (SLA):

```
kubectl taint nodes node1 servicelevel.organization.example/agreed-service-level=9
```

A pod can tolerate nodes with SLA greater than 900:

pods/pod-with-numeric-toleration.yaml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-numeric-toleration
  labels:
    env: test
spec:
  containers:
    - name: nginx
      image: nginx
      imagePullPolicy: IfNotPresent
  tolerations:
    - key: "servicelevel.organization.example/agreed-service-level"
      operator: "Gt"
      value: "900"
      effect: "NoSchedule"
```

This toleration matches the taint on `node1` because `950 > 900` (the taint value is greater than the toleration value for the `Gt` operator).
Similarly, you can use the `Lt` operator to match taints where the taint value is less than the toleration value:

```yaml
tolerations:
- key: "servicelevel.organization.example/agreed-service-level"
  operator: "Lt"
  value: "1000"
  effect: "NoSchedule"
```

> **Note:**
> When using numeric comparison operators:
>
> - Both the toleration and taint values must be valid signed 64-bit integers (zero leading numbers (e.g., "0550") are not allowed).
> - If a value cannot be parsed as an integer, the toleration does not match.
> - Numeric operators work with all taint effects: `NoSchedule`, `PreferNoSchedule`, and `NoExecute`.
> - For `PreferNoSchedule` with numeric operators: if a pod's toleration doesn't satisfy the numeric comparison (e.g., taint value < toleration value when using

`Gt`), the scheduler gives the node a lower priority but may still schedule there if no better options exist.

Warning:

Before disabling the `TaintTolerationComparisonOperators` feature gate:

- You should identify all workloads using the `Gt` or `Lt` operators to avoid controller hot-loops.
- Update all workload controller templates to use `Equal` or `Exists` operators instead
- Delete any pending pods that use `Gt` or `Lt` operators
- Monitor the `apiserver_request_total` metric for spikes in validation errors

Example Use Cases

Taints and tolerations are a flexible way to steer pods *away* from nodes or evict pods that shouldn't be running. A few of the use cases are

- **Dedicated Nodes**: If you want to dedicate a set of nodes for exclusive use by a particular set of users, you can add a taint to those nodes (say, `kubectl taint nodes nodename dedicated=groupName:NoSchedule`) and then add a corresponding toleration to their pods (this would be done most easily by writing a custom admission controller). The pods with the tolerations will then be allowed to use the tainted (dedicated) nodes as well as any other nodes in the cluster. If you want to dedicate the nodes to them *and* ensure they *only* use the dedicated nodes, then you should additionally add a label similar to the taint to the same set of nodes (e.g. `dedicated=groupName`), and the admission controller should additionally add a node affinity to require that the pods can only schedule onto nodes labeled with `dedicated=groupName`.

- **Nodes with Special Hardware**: In a cluster where a small subset of nodes have specialized hardware (for example GPUs), it is desirable to keep pods that don't need the specialized hardware off of those nodes, thus leaving room for later-arriving pods that do need the specialized hardware. This can be done by tainting the nodes that have the specialized hardware (e.g. `kubectl taint nodes nodename special=true:NoSchedule` or `kubectl taint nodes nodename special=true:PreferNoSchedule`) and adding a corresponding toleration to pods that use the special hardware. As in the dedicated nodes use case, it is probably easiest to apply the tolerations using a custom admission controller. For example, it is

recommended to use Extended Resources to represent the special hardware, taint your special hardware nodes with the extended resource name and run the ExtendedResourceToleration admission controller. Now, because the nodes are tainted, no pods without the toleration will schedule on them. But when you submit a pod that requests the extended resource, the `ExtendedResourceToleration` admission controller will automatically add the correct toleration to the pod and that pod will schedule on the special hardware nodes. This will make sure that these special hardware nodes are dedicated for pods requesting such hardware and you don't have to manually add tolerations to your pods.

- **Taint based Evictions**: A per-pod-configurable eviction behavior when there are node problems, which is described in the next section.

Taint based Evictions

FEATURE STATE: Kubernetes v1.18 [stable]

The node controller automatically taints a Node when certain conditions are true. The following taints are built in:

- `node.kubernetes.io/not-ready` : Node is not ready. This corresponds to the NodeCondition `Ready` being " `False` ".
- `node.kubernetes.io/unreachable` : Node is unreachable from the node controller. This corresponds to the NodeCondition `Ready` being " `Unknown` ".
- `node.kubernetes.io/memory-pressure` : Node has memory pressure.
- `node.kubernetes.io/disk-pressure` : Node has disk pressure.
- `node.kubernetes.io/pid-pressure` : Node has PID pressure.
- `node.kubernetes.io/network-unavailable` : Node's network is unavailable.
- `node.kubernetes.io/unschedulable` : Node is unschedulable.
- `node.cloudprovider.kubernetes.io/uninitialized` : When the kubelet is started with an "external" cloud provider, this taint is set on a node to mark it as unusable. After a controller from the cloud-controller-manager initializes this node, the kubelet removes this taint.

In case a node is to be drained, the node controller or the kubelet adds relevant taints with `NoExecute` effect. This effect is added by default for the `node.kubernetes.io/not-ready` and `node.kubernetes.io/unreachable` taints. If the fault condition returns to normal, the kubelet or node controller can remove the relevant taint(s).

In some cases when the node is unreachable, the API server is unable to communicate with the kubelet on the node. The decision to delete the pods cannot be communicated

to the kubelet until communication with the API server is re-established. In the meantime, the pods that are scheduled for deletion may continue to run on the partitioned node.

> **Note:**
> The control plane limits the rate of adding new taints to nodes. This rate limiting manages the number of evictions that are triggered when many nodes become unreachable at once (for example: if there is a network disruption).

You can specify `tolerationSeconds` for a Pod to define how long that Pod stays bound to a failing or unresponsive Node.

For example, you might want to keep an application with a lot of local state bound to node for a long time in the event of network partition, hoping that the partition will recover and thus the pod eviction can be avoided. The toleration you set for that Pod might look like:

```
tolerations:
- key: "node.kubernetes.io/unreachable"
  operator: "Exists"
  effect: "NoExecute"
  tolerationSeconds: 6000
```

> **Note:**
> Kubernetes automatically adds a toleration for `node.kubernetes.io/not-ready` and `node.kubernetes.io/unreachable` with `tolerationSeconds=300`, unless you, or a controller, set those tolerations explicitly.
>
> These automatically-added tolerations mean that Pods remain bound to Nodes for 5 minutes after one of these problems is detected.

DaemonSet pods are created with `NoExecute` tolerations for the following taints with no `tolerationSeconds`:

- `node.kubernetes.io/unreachable`
- `node.kubernetes.io/not-ready`

This ensures that DaemonSet pods are never evicted due to these problems.

> **Note:**
> The node controller was responsible for adding taints to nodes and evicting pods. But after 1.29, the taint-based eviction implementation has been moved out of node controller into a separate, and independent component called taint-eviction-controller. Users can optionally disable taint-based eviction by setting `--controllers=-taint-eviction-controller` in kube-controller-manager.

Taint Nodes by Condition

The control plane, using the node controller, automatically creates taints with a `NoSchedule` effect for node conditions.

The scheduler checks taints, not node conditions, when it makes scheduling decisions. This ensures that node conditions don't directly affect scheduling. For example, if the `DiskPressure` node condition is active, the control plane adds the `node.kubernetes.io/disk-pressure` taint and does not schedule new pods onto the affected node. If the `MemoryPressure` node condition is active, the control plane adds the `node.kubernetes.io/memory-pressure` taint.

You can ignore node conditions for newly created pods by adding the corresponding Pod tolerations. The control plane also adds the `node.kubernetes.io/memory-pressure` toleration on pods that have a QoS class other than `BestEffort`. This is because Kubernetes treats pods in the `Guaranteed` or `Burstable` QoS classes (even pods with no memory request set) as if they are able to cope with memory pressure, while new `BestEffort` pods are not scheduled onto the affected node.

The DaemonSet controller automatically adds the following `NoSchedule` tolerations to all daemons, to prevent DaemonSets from breaking.

- `node.kubernetes.io/memory-pressure`
- `node.kubernetes.io/disk-pressure`
- `node.kubernetes.io/pid-pressure` (1.14 or later)
- `node.kubernetes.io/unschedulable` (1.10 or later)
- `node.kubernetes.io/network-unavailable` (*host network only*)

Adding these tolerations ensures backward compatibility. You can also add arbitrary tolerations to DaemonSets.

Device taints and tolerations

Instead of tainting entire nodes, administrators can also taint individual devices when the cluster uses dynamic resource allocation to manage special hardware. The advantage is that tainting can be targeted towards exactly the hardware that is faulty or needs maintenance. Tolerations are also supported and can be specified when requesting devices. Like taints they apply to all pods which share the same allocated device.

What's next

- Read about Node-pressure Eviction and how you can configure it
- Read about Pod Priority
- Read about device taints and tolerations

8 - Scheduling Framework

ⓘ **FEATURE STATE:** Kubernetes v1.19 [stable]

The *scheduling framework* is a pluggable architecture for the Kubernetes scheduler. It consists of a set of "plugin" APIs that are compiled directly into the scheduler. These APIs allow most scheduling features to be implemented as plugins, while keeping the scheduling "core" lightweight and maintainable. Refer to the design proposal of the scheduling framework for more technical information on the design of the framework.

Framework workflow

The Scheduling Framework defines a few extension points. Scheduler plugins register to be invoked at one or more extension points. Some of these plugins can change the scheduling decisions and some are informational only.

Each attempt to schedule one Pod is split into two phases, the **scheduling cycle** and the **binding cycle**.

Scheduling cycle & binding cycle

The scheduling cycle selects a node for the Pod, and the binding cycle applies that decision to the cluster. Together, a scheduling cycle and binding cycle are referred to as a "scheduling context".

Scheduling cycles are run serially, while binding cycles may run concurrently.

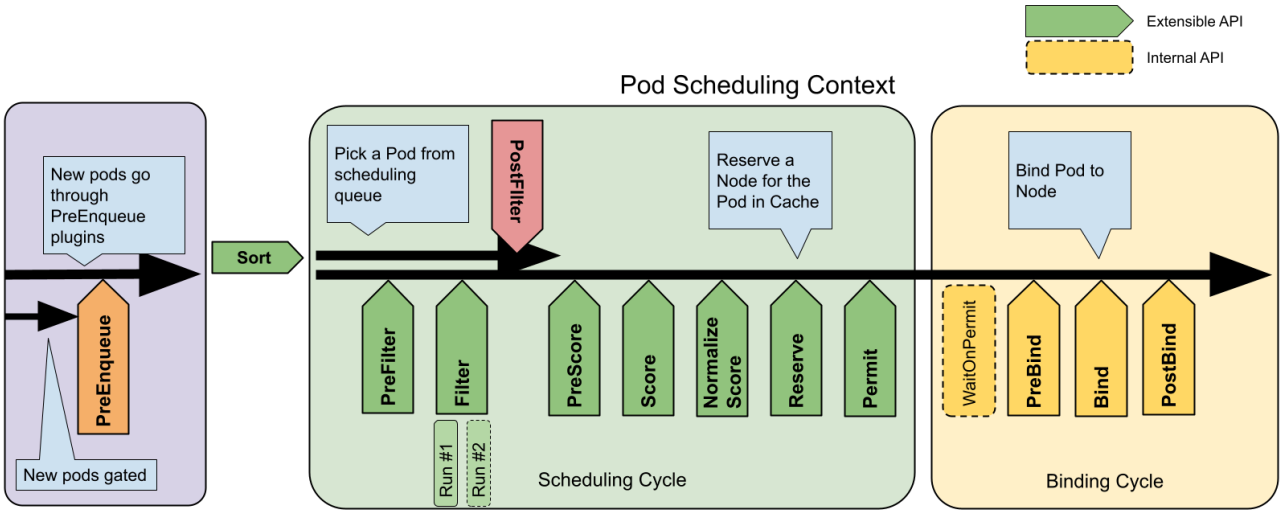
A scheduling or binding cycle can be aborted if the Pod is determined to be unschedulable or if there is an internal error. The Pod will be returned to the queue and retried.

Interfaces

The following picture shows the scheduling context of a Pod and the interfaces that the scheduling framework exposes.

One plugin may implement multiple interfaces to perform more complex or stateful tasks.

Some interfaces match the scheduler extension points which can be configured through Scheduler Configuration.



Scheduling framework extension points

PreEnqueue

These plugins are called prior to adding Pods to the internal active queue, where Pods are marked as ready for scheduling.

Only when all PreEnqueue plugins return `Success`, the Pod is allowed to enter the active queue. Otherwise, it's placed in the internal unschedulable Pods list, and doesn't get an `Unschedulable` condition.

For more details about how internal scheduler queues work, read Scheduling queue in kube-scheduler.

EnqueueExtension

EnqueueExtension is the interface where the plugin can control whether to retry scheduling of Pods rejected by this plugin, based on changes in the cluster. Plugins that implement PreEnqueue, PreFilter, Filter, Reserve or Permit should implement this interface.

QueueingHint

> ⓘ **FEATURE STATE:** Kubernetes v1.34 [stable](enabled by default)

QueueingHint is a callback function for deciding whether a Pod can be requeued to the active queue or backoff queue. It's executed every time a certain kind of event or change happens in the cluster. When the QueueingHint finds that the event might

make the Pod schedulable, the Pod is put into the active queue or the backoff queue so that the scheduler will retry the scheduling of the Pod.

QueueSort

These plugins are used to sort Pods in the scheduling queue. A queue sort plugin essentially provides a `Less(Pod1, Pod2)` function. Only one queue sort plugin may be enabled at a time.

PreFilter

These plugins are used to pre-process info about the Pod, or to check certain conditions that the cluster or the Pod must meet. If a PreFilter plugin returns an error, the scheduling cycle is aborted.

Filter

These plugins are used to filter out nodes that cannot run the Pod. For each node, the scheduler will call filter plugins in their configured order. If any filter plugin marks the node as infeasible, the remaining plugins will not be called for that node. Nodes may be evaluated concurrently.

PostFilter

These plugins are called after the Filter phase, but only when no feasible nodes were found for the pod. Plugins are called in their configured order. If any postFilter plugin marks the node as `Schedulable`, the remaining plugins will not be called. A typical PostFilter implementation is preemption, which tries to make the pod schedulable by preempting other Pods.

PreScore

These plugins are used to perform "pre-scoring" work, which generates a sharable state for Score plugins to use. If a PreScore plugin returns an error, the scheduling cycle is aborted.

Score

These plugins are used to rank nodes that have passed the filtering phase. The scheduler will call each scoring plugin for each node. There will be a well defined range of integers representing the minimum and maximum scores. After the NormalizeScore

phase, the scheduler will combine node scores from all plugins according to the configured plugin weights.

Capacity scoring

> ⓘ **FEATURE STATE:** Kubernetes v1.33 [alpha](disabled by default)

The feature gate `VolumeCapacityPriority` was used in v1.32 to support storage that are statically provisioned. Starting from v1.33, the new feature gate `StorageCapacityScoring` replaces the old `VolumeCapacityPriority` gate with added support to dynamically provisioned storage. When `StorageCapacityScoring` is enabled, the VolumeBinding plugin in the kube-scheduler is extended to score Nodes based on the storage capacity on each of them. This feature is applicable to CSI volumes that supported Storage Capacity, including local storage backed by a CSI driver.

NormalizeScore

These plugins are used to modify scores before the scheduler computes a final ranking of Nodes. A plugin that registers for this extension point will be called with the Score results from the same plugin. This is called once per plugin per scheduling cycle.

For example, suppose a plugin `BlinkingLightScorer` ranks Nodes based on how many blinking lights they have.

```
func ScoreNode(_ *v1.pod, n *v1.Node) (int, error) {
    return getBlinkingLightCount(n)
}
```

However, the maximum count of blinking lights may be small compared to `NodeScoreMax`. To fix this, `BlinkingLightScorer` should also register for this extension point.

```
func NormalizeScores(scores map[string]int) {
    highest := 0
    for _, score := range scores {
        highest = max(highest, score)
    }
    for node, score := range scores {
        scores[node] = score*NodeScoreMax/highest
```

```
    }
}
```

If any NormalizeScore plugin returns an error, the scheduling cycle is aborted.

> **Note:**
> Plugins wishing to perform "pre-reserve" work should use the NormalizeScore extension point.

Reserve

A plugin that implements the Reserve interface has two methods, namely `Reserve` and `Unreserve`, that back two informational scheduling phases called Reserve and Unreserve, respectively. Plugins which maintain runtime state (aka "stateful plugins") should use these phases to be notified by the scheduler when resources on a node are being reserved and unreserved for a given Pod.

The Reserve phase happens before the scheduler actually binds a Pod to its designated node. It exists to prevent race conditions while the scheduler waits for the bind to succeed. The `Reserve` method of each Reserve plugin may succeed or fail; if one `Reserve` method call fails, subsequent plugins are not executed and the Reserve phase is considered to have failed. If the `Reserve` method of all plugins succeed, the Reserve phase is considered to be successful and the rest of the scheduling cycle and the binding cycle are executed.

The Unreserve phase is triggered if the Reserve phase or a later phase fails. When this happens, the `Unreserve` method of **all** Reserve plugins will be executed in the reverse order of `Reserve` method calls. This phase exists to clean up the state associated with the reserved Pod.

> **Caution:**
> The implementation of the `Unreserve` method in Reserve plugins must be idempotent and may not fail.

Permit

Permit plugins are invoked at the end of the scheduling cycle for each Pod, to prevent or delay the binding to the candidate node. A permit plugin can do one of the three things:

1. **approve**
   Once all Permit plugins approve a Pod, it is sent for binding.

2. **deny**
   If any Permit plugin denies a Pod, it is returned to the scheduling queue. This will trigger the Unreserve phase in Reserve plugins.

3. **wait** (with a timeout)
   If a Permit plugin returns "wait", then the Pod is kept in an internal "waiting" Pods list, and the binding cycle of this Pod starts but directly blocks until it gets approved. If a timeout occurs, **wait** becomes **deny** and the Pod is returned to the scheduling queue, triggering the Unreserve phase in Reserve plugins.

> **Note:**
> While any plugin can access the list of "waiting" Pods and approve them (see `FrameworkHandle`), we expect only the permit plugins to approve binding of reserved Pods that are in "waiting" state. Once a Pod is approved, it is sent to the PreBind phase.

PreBind

These plugins are used to perform any work required before a Pod is bound. For example, a pre-bind plugin may provision a network volume and mount it on the target node before allowing the Pod to run there.

If any PreBind plugin returns an error, the Pod is rejected and returned to the scheduling queue.

Bind

These plugins are used to bind a Pod to a Node. Bind plugins will not be called until all PreBind plugins have completed. Each bind plugin is called in the configured order. A bind plugin may choose whether or not to handle the given Pod. If a bind plugin chooses to handle a Pod, **the remaining bind plugins are skipped**.

PostBind

This is an informational interface. Post-bind plugins are called after a Pod is successfully bound. This is the end of a binding cycle, and can be used to clean up associated resources.

Plugin API

There are two steps to the plugin API. First, plugins must register and get configured, then they use the extension point interfaces. Extension point interfaces have the following form.

```
type Plugin interface {
    Name() string
}

type QueueSortPlugin interface {
    Plugin
    Less(*v1.pod, *v1.pod) bool
}

type PreFilterPlugin interface {
    Plugin
    PreFilter(context.Context, *framework.CycleState, *v1.pod) error
}

// ...
```

Plugin configuration

You can enable or disable plugins in the scheduler configuration. If you are using Kubernetes v1.18 or later, most scheduling plugins are in use and enabled by default.

In addition to default plugins, you can also implement your own scheduling plugins and get them configured along with default plugins. You can visit scheduler-plugins for more details.

If you are using Kubernetes v1.18 or later, you can configure a set of plugins as a scheduler profile and then define multiple profiles to fit various kinds of workload. Learn more at multiple profiles.

9 - Dynamic Resource Allocation

ⓘ **FEATURE STATE:** Kubernetes v1.35 [stable](enabled by default)

This page describes *dynamic resource allocation (DRA)* in Kubernetes.

About DRA

DRA is a Kubernetes feature that lets you request and share resources among Pods. These resources are often attached devices like hardware accelerators.

With DRA, device drivers and cluster admins define device *classes* that are available to *claim* in workloads. Kubernetes allocates matching devices to specific claims and places the corresponding Pods on nodes that can access the allocated devices.

Allocating resources with DRA is a similar experience to dynamic volume provisioning, in which you use PersistentVolumeClaims to claim storage capacity from storage classes and request the claimed capacity in your Pods.

Benefits of DRA

DRA provides a flexible way to categorize, request, and use devices in your cluster. Using DRA provides benefits like the following:

- **Flexible device filtering**: use common expression language (CEL) to perform fine-grained filtering for specific device attributes.
- **Device sharing**: share the same resource with multiple containers or Pods by referencing the corresponding resource claim.
- **Centralized device categorization**: device drivers and cluster admins can use device classes to provide app operators with hardware categories that are optimized for various use cases. For example, you can create a cost-optimized device class for general-purpose workloads, and a high-performance device class for critical jobs.
- **Simplified Pod requests**: with DRA, app operators don't need to specify device quantities in Pod resource requests. Instead, the Pod references a resource claim, and the device configuration in that claim applies to the Pod.

These benefits provide significant improvements in the device allocation workflow when compared to device plugins, which require per-container device requests, don't support device sharing, and don't support expression-based device filtering.

Types of DRA users

The workflow of using DRA to allocate devices involves the following types of users:

- **Device owner**: responsible for devices. Device owners might be commercial vendors, the cluster operator, or another entity. To use DRA, devices must have DRA-compatible drivers that do the following:
  - Create ResourceSlices that provide Kubernetes with information about nodes and resources.
  - Update ResourceSlices when resource capacity in the cluster changes.
  - Optionally, create DeviceClasses that workload operators can use to claim devices.
- **Cluster admin**: responsible for configuring clusters and nodes, attaching devices, installing drivers, and similar tasks. To use DRA, cluster admins do the following:
  - Attach devices to nodes.
  - Install device drivers that support DRA.
  - Optionally, create DeviceClasses that workload operators can use to claim devices.
- **Workload operator**: responsible for deploying and managing workloads in the cluster. To use DRA to allocate devices to Pods, workload operators do the following:
  - Create ResourceClaims or ResourceClaimTemplates to request specific configurations within DeviceClasses.
  - Deploy workloads that use specific ResourceClaims or ResourceClaimTemplates.

DRA terminology

DRA uses the following Kubernetes API kinds to provide the core allocation functionality. All of these API kinds are included in the `resource.k8s.io/v1` API group.

DeviceClass

Defines a category of devices that can be claimed and how to select specific device attributes in claims. The DeviceClass parameters can match zero or more devices in ResourceSlices. To claim devices from a DeviceClass, ResourceClaims select specific device attributes.

ResourceClaim

Describes a request for access to attached resources, such as devices, in the cluster. ResourceClaims provide Pods with access to a specific resource. ResourceClaims can be created by workload operators or generated by Kubernetes based on a ResourceClaimTemplate.

ResourceClaimTemplate

Defines a template that Kubernetes uses to create per-Pod ResourceClaims for a workload. ResourceClaimTemplates provide Pods with access to separate, similar resources. Each ResourceClaim that Kubernetes generates from the template is bound to a specific Pod. When the Pod terminates, Kubernetes deletes the corresponding ResourceClaim.

ResourceSlice

Represents one or more resources that are attached to nodes, such as devices. Drivers create and manage ResourceSlices in the cluster. When a ResourceClaim is created and used in a Pod, Kubernetes uses ResourceSlices to find nodes that have access to the claimed resources. Kubernetes allocates resources to the ResourceClaim and schedules the Pod onto a node that can access the resources.

DeviceClass

A DeviceClass lets cluster admins or device drivers define categories of devices in the cluster. DeviceClasses tell operators what devices they can request and how they can request those devices. You can use common expression language (CEL) to select devices based on specific attributes. A ResourceClaim that references the DeviceClass can then request specific configurations within the DeviceClass.

To create a DeviceClass, see Set Up DRA in a Cluster.

ResourceClaims and ResourceClaimTemplates

A ResourceClaim defines the resources that a workload needs. Every ResourceClaim has *requests* that reference a DeviceClass and select devices from that DeviceClass. ResourceClaims can also use *selectors* to filter for devices that meet specific requirements, and can use *constraints* to limit the devices that can satisfy a request. ResourceClaims can be created by workload operators or can be generated by Kubernetes based on a ResourceClaimTemplate. A ResourceClaimTemplate defines a template that Kubernetes can use to auto-generate ResourceClaims for Pods.

Use cases for ResourceClaims and ResourceClaimTemplates

The method that you use depends on your requirements, as follows:

- **ResourceClaim**: you want multiple Pods to share access to specific devices. You manually manage the lifecycle of ResourceClaims that you create.
- **ResourceClaimTemplate**: you want Pods to have independent access to separate, similarly-configured devices. Kubernetes generates ResourceClaims from the specification in the ResourceClaimTemplate. The lifetime of each generated ResourceClaim is bound to the lifetime of the corresponding Pod.
- **PodGroup ResourceClaimTemplate**: you want PodGroups to have independent access to separate, similarly-configured devices that can be shared by their Pods. Kubernetes generates one ResourceClaim for the PodGroup from the specification in the ResourceClaimTemplate. The lifetime of each generated ResourceClaim is bound to the lifetime of the corresponding PodGroup. This requires the `DRAWorkloadResourceClaims` feature to be enabled.

When you define a workload, you can use Common Expression Language (CEL) to filter for specific device attributes or capacity. The available parameters for filtering depend on the device and the drivers.

If you directly reference a specific ResourceClaim in a Pod, that ResourceClaim must already exist in the same namespace as the Pod. If the ResourceClaim doesn't exist in the namespace, the Pod won't schedule. This behavior is similar to how a PersistentVolumeClaim must exist in the same namespace as a Pod that references it.

You can reference an auto-generated ResourceClaim in a Pod, but this isn't recommended because auto-generated ResourceClaims are bound to the lifetime of the Pod or PodGroup that triggered the generation.

To learn how to claim resources using one of these methods, see Allocate Devices to Workloads with DRA.

Prioritized list

> ⓘ **FEATURE STATE:** Kubernetes v1.36 [stable](enabled by default)

You can provide a prioritized list of subrequests for requests in a ResourceClaim or ResourceClaimTemplate. The scheduler will then select the first subrequest that can be allocated. This allows users to specify alternative devices that can be used by the workload if the primary choice is not available.

In the example below, the ResourceClaimTemplate requested a device with the color black and the size large. If a device with those attributes is not available, the pod cannot be scheduled. With the prioritized list feature, a second alternative can be specified, which requests two devices with the color white and size small. The large

black device will be allocated if it is available. If it is not, but two small white devices are available, the pod will still be able to run.

```
apiVersion: resource.k8s.io/v1
kind: ResourceClaimTemplate
metadata:
  name: prioritized-list-claim-template
spec:
  spec:
    devices:
      requests:
      - name: req-0
        firstAvailable:
        - name: large-black
          deviceClassName: resource.example.com
          selectors:
          - cel:
              expression: |-
                device.attributes["resource-driver.example.com"].color == "black" &
                device.attributes["resource-driver.example.com"].size == "large"
        - name: small-white
          deviceClassName: resource.example.com
          selectors:
          - cel:
              expression: |-
                device.attributes["resource-driver.example.com"].color == "white" &
                device.attributes["resource-driver.example.com"].size == "small"
          count: 2
```

If the pod is eligible for multiple nodes in the cluster, the scheduler will use the index of chosen subrequests from any prioritized lists as one of the inputs when it scores each node. So nodes that can allocate devices requested in a higher ranked subrequest are more likely to be chosen than nodes that can only allocate devices for lower ranked subrequests.

The decision is made on a per-Pod basis, so if the Pod is a member of a ReplicaSet or similar grouping, you cannot rely on all the members of the group having the same subrequest chosen. Your workload must be able to accommodate this.

Workload ResourceClaims

ⓘ **FEATURE STATE:** Kubernetes v1.36 [alpha](disabled by default)

When you organize Pods with the Workload API, you can reserve ResourceClaims for entire PodGroups instead of individual Pods and generate ResourceClaimTemplates for a PodGroup instead of a single Pod, allowing the Pods within a PodGroup to share access to devices allocated to the generated ResourceClaim.

This feature targets two problems:

- The ResourceClaim API's `status.reservedFor` list can only contain 256 items. Since kube-scheduler only records individual Pods in that list, only 256 Pods can share a ResourceClaim. By allowing PodGroups to be recorded in `status.reservedFor`, many more than 256 Pods can share a ResourceClaim.
- Pods can only share a ResourceClaim when its exact name is known. For complex workloads that replicate *groups* of Pods, ResourceClaims shared by the Pods in each group need to be created and deleted explicitly when the set of groups scales up and down. By generating ResourceClaims for each PodGroup, a single ResourceClaimTemplate can form the basis for ResourceClaims that are both replicated automatically and shareable among the Pods in a PodGroup.

The PodGroup API defines a `spec.resourceClaims` field with the same structure and similar meaning as the `spec.resourceClaims` field in the Pod API:

```yaml
apiVersion: scheduling.k8s.io/v1alpha2
kind: PodGroup
metadata:
  name: training-group
  namespace: some-ns
spec:
  ...
  resourceClaims:
  - name: pg-claim
    resourceClaimName: my-pg-claim
  - name: pg-claim-template
    resourceClaimTemplateName: my-pg-template
```

Like claims made by Pods, claims for PodGroups defining a `resourceClaimName` refer to a ResourceClaim by name. Claims defining a `resourceClaimTemplateName` refer to a ResourceClaimTemplate which replicates into one ResourceClaim for the entire PodGroup that can be shared amongst its Pods.

When a Pod defines a claim with a `name`, `resourceClaimName`, and `resourceClaimTemplateName` that all match one of its PodGroup's `spec.resourceClaims`, then kube-scheduler reserves the ResourceClaim for the PodGroup instead of the Pod. If the Pod's claim does not match one made by its PodGroup, then kube-scheduler reserves the ResourceClaim for the Pod. In either case, reservation is recorded in the ResourceClaim's `status.reservedFor`. PodGroup reservations and the corresponding resource allocation persist in the ResourceClaim until the PodGroup is deleted, even if the group no longer has any Pods.

When a Pod claim matching a PodGroup claim defines a `resourceClaimTemplateName`, then one ResourceClaim is generated for the PodGroup. Other Pods in the group defining the same claim will share that generated ResourceClaim instead of prompting a new ResourceClaim to be generated for each Pod. Whether or not a `resourceClaimTemplateName` claim matches a PodGroup claim, the name of the generated ResourceClaim is recorded in the Pod's `status.resourceClaimStatuses`.

ResourceClaims generated from a ResourceClaimTemplate for a PodGroup follow the lifecycle of the PodGroup. The ResourceClaim is first created when both the PodGroup and its ResourceClaimTemplate exist. The ResourceClaim is deleted after the PodGroup has been deleted and the ResourceClaim is no longer reserved.

Consider the following example:

```yaml
apiVersion: scheduling.k8s.io/v1alpha2
kind: PodGroup
metadata:
  name: training-group
  namespace: some-ns
spec:
  ...
  resourceClaims:
  - name: pg-claim
    resourceClaimName: my-pg-claim
  - name: pg-claim-template
    resourceClaimTemplateName: my-pg-template
---
apiVersion: v1
kind: Pod
metadata:
  name: training-group-pod-1
  namespace: some-ns
spec:
  ...
  schedulingGroup:
    podGroupName: training-group
```

```
  resourceClaims:
  - name: pod-claim
    resourceClaimName: my-pod-claim
  - name: pod-claim-template
    resourceClaimTemplateName: my-pod-template
  - name: pg-claim
    resourceClaimName: my-pg-claim
  - name: pg-claim-template
    resourceClaimTemplateName: my-pg-template
```

In this example, the `training-group` PodGroup has one Pod named `training-group-pod-1`. The Pod's `pod-claim` and `pod-claim-template` claims do not match any claim made by the PodGroup, so those claims are not affected by the PodGroup: ResourceClaim `my-pod-claim` becomes reserved for the Pod and a ResourceClaim is generated from ResourceClaimTemplate `my-pod-template` and also becomes reserved for the Pod. The `pg-claim` and `pg-claim-template` do match claims made by the PodGroup. ResourceClaim `my-pg-claim` becomes reserved for the PodGroup and a ResourceClaim is generated from ResourceClaimTemplate `my-pg-template` and also becomes reserved for the PodGroup.

Associating ResourceClaims with Workload API resources is an *alpha feature* and only enabled when the `DRAWorkloadResourceClaims` feature gate is enabled in the kube-apiserver, kube-controller-manager, kube-scheduler, and kubelet.

ResourceSlice

Each ResourceSlice represents one or more devices in a pool. The pool is managed by a device driver, which creates and manages ResourceSlices. The resources in a pool might be represented by a single ResourceSlice or span multiple ResourceSlices.

ResourceSlices provide useful information to device users and to the scheduler, and are crucial for dynamic resource allocation. Every ResourceSlice must include the following information:

- **Resource pool**: a group of one or more resources that the driver manages. The pool can span more than one ResourceSlice. Changes to the resources in a pool must be propagated across all of the ResourceSlices in that pool. The device driver that manages the pool is responsible for ensuring that this propagation happens.
- **Devices**: devices in the managed pool. A ResourceSlice can list every device in a pool or a subset of the devices in a pool. The ResourceSlice defines device information like attributes, versions, and capacity. Device users can select devices for allocation by filtering for device information in ResourceClaims or in DeviceClasses.

- **Nodes**: the nodes that can access the resources. Drivers can choose which nodes can access the resources, whether that's all of the nodes in the cluster, a single named node, or nodes that have specific node labels.

Drivers use a controller to reconcile ResourceSlices in the cluster with the information that the driver has to publish. This controller overwrites any manual changes, such as cluster users creating or modifying ResourceSlices.

Consider the following example ResourceSlice:

```yaml
apiVersion: resource.k8s.io/v1
kind: ResourceSlice
metadata:
  name: cat-slice
spec:
  driver: "resource-driver.example.com"
  pool:
    generation: 1
    name: "black-cat-pool"
    resourceSliceCount: 1
  # The allNodes field defines whether any node in the cluster can access the devic
  allNodes: true
  devices:
  - name: "large-black-cat"
    attributes:
      color:
        string: "black"
      size:
        string: "large"
      cat:
        bool: true
```

This ResourceSlice is managed by the `resource-driver.example.com` driver in the `black-cat-pool` pool. The `allNodes: true` field indicates that any node in the cluster can access the devices. There's one device in the ResourceSlice, named `large-black-cat`, with the following attributes:

- `color`：`black`
- `size`：`large`
- `cat`：`true`

A DeviceClass could select this ResourceSlice by using these attributes, and a ResourceClaim could filter for specific devices in that DeviceClass.

Naming and prioritization

The order in which the Kubernetes scheduler evaluates devices for allocation is determined by the lexicographical sorting of ResourceSlice and resource pool names. The scheduler uses a first-fit strategy, meaning it selects the first available device that satisfies the claim's requirements.

This allows the priority of resource allocation to be influenced by the names assigned to pools and ResourceSlices. Note that pools without binding conditions are always evaluated before those with binding conditions, regardless of their names.

For drivers built using the `k8s.io/dynamic-resources/kubeletplugin` Go package or the ResourceSlice controller from that module, these components automatically handle ResourceSlice naming to ensure they are evaluated in the order specified by the driver.

How resource allocation with DRA works

The following sections describe the workflow for the various types of DRA users and for the Kubernetes system during dynamic resource allocation.

Workflow for users

1. **Driver creation**: device owners or third-party entities create drivers that can create and manage ResourceSlices in the cluster. These drivers optionally also create DeviceClasses that define a category of devices and how to request them.
2. **Cluster configuration**: cluster admins create clusters, attach devices to nodes, and install the DRA device drivers. Cluster admins optionally create DeviceClasses that define categories of devices and how to request them.
3. **Resource claims**: workload operators create ResourceClaimTemplates or ResourceClaims that request specific device configurations within a DeviceClass. In the same step, workload operators modify their Kubernetes manifests to request those ResourceClaimTemplates or ResourceClaims.

Workflow for Kubernetes

1. **ResourceSlice creation**: drivers in the cluster create ResourceSlices that represent one or more devices in a managed pool of similar devices.

2. **Workload creation**: the cluster control plane checks new workloads for references to ResourceClaimTemplates or to specific ResourceClaims.

    - If the workload uses a ResourceClaimTemplate, a controller named the `resourceclaim-controller` generates ResourceClaims for the workload.

    - If the workload uses a specific ResourceClaim, Kubernetes checks whether that ResourceClaim exists in the cluster. If the ResourceClaim doesn't exist, the Pods won't deploy.
3. **ResourceSlice filtering**: for every Pod, Kubernetes checks the ResourceSlices in the cluster to find a device that satisfies all of the following criteria:
    - The nodes that can access the resources are eligible to run the Pod.
    - The ResourceSlice has unallocated resources that match the requirements of the Pod's ResourceClaim.
4. **Resource allocation**: after finding an eligible ResourceSlice for a Pod's ResourceClaim, the Kubernetes scheduler updates the ResourceClaim with the allocation details. The scheduler uses a first-fit strategy and evaluates pools and ResourceSlices in lexicographical order by their names. Drivers can prioritize specific slices or pools by naming them appropriately. For details, see Naming and prioritization.
5. **Pod scheduling**: when resource allocation is complete, the scheduler places the Pod on a node that can access the allocated resource. The device driver and the kubelet on that node configure the device and the Pod's access to the device.

Observability of dynamic resources

You can check the status of dynamically allocated resources by using any of the following methods:

- kubelet device metrics
- ResourceClaim status
- Device health monitoring

kubelet device metrics

The `PodResourcesLister` kubelet gRPC service lets you monitor in-use devices. The `DynamicResource` message provides information that's specific to dynamic resource allocation, such as the device name and the claim name. For details, see Monitoring device plugin resources.

ResourceClaim device status

ⓘ **FEATURE STATE:** Kubernetes v1.33 [beta](enabled by default)

DRA drivers can report driver-specific device status data for each allocated device in the `status.devices` field of a ResourceClaim. For example, the driver might list the IP addresses that are assigned to a network interface device. Updating this field requires specific synthetic RBAC permissions, see Hardening Guide - Dynamic Resource Allocation and Harden Dynamic Resource Allocation in Your Cluster.

The accuracy of the information that a driver adds to a ResourceClaim `status.devices` field depends on the driver. Evaluate drivers to decide whether you can rely on this field as the only source of device information.

If you disable the `DRAResourceClaimDeviceStatus` feature gate, the `status.devices` field automatically gets cleared when storing the ResourceClaim. A ResourceClaim device status is supported when it is possible, from a DRA driver, to update an existing ResourceClaim where the `status.devices` field is set.

For details about the `status.devices` field, see the ResourceClaim API reference.

Device Health Monitoring

> ⓘ **FEATURE STATE:** Kubernetes v1.36 [beta](enabled by default)

Kubernetes provides a mechanism for monitoring and reporting the health of dynamically allocated infrastructure resources. For stateful applications running on specialized hardware, it is critical to know when a device has failed or become unhealthy. It is also helpful to find out if the device recovers.

To use this functionality, the `ResourceHealthStatus` feature gate must be enabled (beta and enabled by default since v1.36), and the DRA driver must implement the `DRAResourceHealth` gRPC service.

When a DRA driver detects that an allocated device has become unhealthy, it reports this status back to the kubelet. This health information is then exposed directly in the Pod's status. The kubelet populates the `allocatedResourcesStatus` field in the status of each container, detailing the health of each device assigned to that container. Each resource health entry can include an optional `message` field with additional human-readable context about the health status, such as error details or failure reasons.

If the kubelet does not receive a health update from a DRA driver within a timeout period, the device's health status is marked as "Unknown". DRA drivers can configure this timeout on a per-device basis by setting the `health_check_timeout_seconds` field in the `DeviceHealth` gRPC message. If not specified, the kubelet uses a default timeout of 30 seconds. This allows different hardware types (for example, GPUs, FPGAs, or storage

devices) to use appropriate timeout values based on their health-reporting characteristics.

This provides crucial visibility for users and controllers to react to hardware failures. For a Pod that is failing, you can inspect this status to determine if the failure was related to an unhealthy device.

> **Note:**
> Device health status is not updated in the Pod status after a Pod has terminated (for example, in Failed state).

Pre-scheduled Pods

When you - or another API client - create a Pod with `spec.nodeName` already set, the scheduler gets bypassed. If some ResourceClaim needed by that Pod does not exist yet, is not allocated or not reserved for the Pod, then the kubelet will fail to run the Pod and re-check periodically because those requirements might still get fulfilled later.

Such a situation can also arise when support for dynamic resource allocation was not enabled in the scheduler at the time when the Pod got scheduled (version skew, configuration, feature gate, etc.). kube-controller-manager detects this and tries to make the Pod runnable by reserving the required ResourceClaims. However, this only works if those were allocated by the scheduler for some other pod.

It is better to avoid bypassing the scheduler because a Pod that is assigned to a node blocks normal resources (RAM, CPU) that then cannot be used for other Pods while the Pod is stuck. To make a Pod run on a specific node while still going through the normal scheduling flow, create the Pod with a node selector that exactly matches the desired node:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-cats
spec:
  nodeSelector:
    kubernetes.io/hostname: name-of-the-intended-node
  ...
```

You may also be able to mutate the incoming Pod, at admission time, to unset the `.spec.nodeName` field and to use a node selector instead.

Limitations

- The Kubernetes scheduler doesn't support preemption for DRA resources. This means that an existing Pod that's running on a node and is using DRA resources can't be preempted by a higher-priority Pod that also needs DRA resources. The high-priority Pod will remain in a pending state until the device becomes available, which happens when the conflicting Pod terminates or is manually deleted.

DRA beta features

The following sections describe DRA features that support advanced use cases. Usage of them is optional and may only be relevant with DRA drivers that support them.

Some of them are available in the Alpha or Beta feature stage. Those depend on feature gates and may depend on additional API groups. For more information, see Set up DRA in the cluster.

Admin access

ⓘ **FEATURE STATE:** Kubernetes v1.36 [stable](enabled by default)

You can mark a request in a ResourceClaim or ResourceClaimTemplate as having privileged features for maintenance and troubleshooting tasks. A request with admin access grants access to in-use devices and may enable additional permissions when making the device available in a container:

```yaml
apiVersion: resource.k8s.io/v1
kind: ResourceClaimTemplate
metadata:
  name: large-black-cat-claim-template
spec:
  spec:
    devices:
      requests:
      - name: req-0
```

```
      exactly:
        deviceClassName: resource.example.com
        allocationMode: All
        adminAccess: true
```

Admin access is a privileged mode and should not be granted to regular users in multi-tenant clusters. Only users authorized to create ResourceClaim or ResourceClaimTemplate objects in namespaces labeled with `resource.kubernetes.io/admin-access: "true"` (case-sensitive) can use the `adminAccess` field. This ensures that non-admin users cannot misuse the feature.

Admin access is a *beta feature* and is enabled by default with the `DRAAdminAccess` feature gate in the kube-apiserver, kube-scheduler, and kubelet.

Granular status authorization

FEATURE STATE: Kubernetes v1.36 [beta](enabled by default)

Starting in Kubernetes v1.36, DRA enforces fine-grained authorization checks for updates to `ResourceClaim` status by using synthetic subresources and node-aware verbs.

For security hardening guidance, including RBAC examples for scheduler and DRA drivers, see Hardening Guide - Dynamic Resource Allocation.

For a step-by-step cluster administrator procedure, see Harden Dynamic Resource Allocation in Your Cluster.

DRA alpha features

The following sections describe DRA features that are available in the Alpha feature stage. They depend on enabling feature gates and may depend on additional API groups. For more information, see Set up DRA in the cluster.

Extended resource allocation by DRA

FEATURE STATE: Kubernetes v1.36 [beta](enabled by default)

You can provide an extended resource name for a DeviceClass. The scheduler will then select the devices matching the class for the extended resource requests. This allows users to continue using extended resource requests in a pod to request either extended resources provided by device plugin, or DRA devices. The same extended resource can be provided either by device plugin, or DRA on one single cluster node. The same extended resource can be provided by device plugin on some nodes, and DRA on other nodes in the same cluster.

In the example below, the DeviceClass is given an extendedResourceName `example.com/gpu`. If a pod requested for the extended resource `example.com/gpu: 2`, it can be scheduled to a node with two or more devices matching the DeviceClass.

```
apiVersion: resource.k8s.io/v1
kind: DeviceClass
metadata:
  name: gpu.example.com
spec:
  selectors:
  - cel:
      expression: device.driver == 'gpu.example.com' && device.attributes['gpu.exar
        == 'gpu'
  extendedResourceName: example.com/gpu
```

In addition, users can use a special extended resource to allocate devices without having to explicitly create a ResourceClaim. Using the extended resource name prefix `deviceclass.resource.kubernetes.io/` and the DeviceClass name. This works for any DeviceClass, even if it does not specify an extended resource name. The resulting ResourceClaim will contain a request for an `ExactCount` of the specified number of devices of that DeviceClass.

Extended resource allocation by DRA is a *beta feature* and is enabled by default with the `DRAExtendedResource` feature gate in the kube-apiserver, kube-scheduler, kube-controller-manager, and kubelet.

Partitionable devices

ⓘ **FEATURE STATE:** Kubernetes v1.36 [beta](enabled by default)

Devices represented in DRA don't necessarily have to be a single unit connected to a single machine, but can also be a logical device comprised of multiple devices connected to multiple machines. These devices might consume overlapping resources of the underlying phyical devices, meaning that when one logical device is allocated other devices will no longer be available.

In the ResourceSlice API, this is represented as a list of named CounterSets, each of which contains a set of named counters. The counters represent the resources available on the physical device that are used by the logical devices advertised through DRA.

Logical devices can specify the ConsumesCounters list. Each entry contains a reference to a CounterSet and a set of named counters with the amounts they will consume. So for a device to be allocatable, the referenced counter sets must have sufficient quantity for the counters referenced by the device.

CounterSets must be specified in separate ResourceSlices from devices. Devices can consume counters from any CounterSet defined in the same resource pool as the device.

Here is an example of two devices, each consuming 6Gi of memory from a shared counter with 8Gi of memory. Thus, only one of the devices can be allocated at any point in time. The scheduler handles this and it is transparent to the consumer as the ResourceClaim API is not affected.

```
apiVersion: resource.k8s.io/v1
kind: ResourceSlice
metadata:
  name: resourceslice-with-countersets
spec:
  nodeName: worker-1
  pool:
    name: pool
    generation: 1
    resourceSliceCount: 2
  driver: dra.example.com
  sharedCounters:
  - name: gpu-1-counters
    counters:
      memory:
        value: 8Gi
---
apiVersion: resource.k8s.io/v1
kind: ResourceSlice
metadata:
```

```
  name: resourceslice-with-devices
spec:
  nodeName: worker-1
  pool:
    name: pool
    generation: 1
    resourceSliceCount: 2
  driver: dra.example.com
  devices:
  - name: device-1
    consumesCounters:
    - counterSet: gpu-1-counters
      counters:
        memory:
          value: 6Gi
  - name: device-2
    consumesCounters:
    - counterSet: gpu-1-counters
      counters:
        memory:
          value: 6Gi
```

Partitionable devices is a *beta feature* and enabled when the `DRAPartitionableDevices` feature gate is kept enabled in the kube-apiserver and kube-scheduler.

Consumable capacity

> ⓘ **FEATURE STATE:** Kubernetes v1.36 [beta](enabled by default)

The consumable capacity feature allows the same devices to be consumed by multiple independent ResourceClaims, with the Kubernetes scheduler managing how much of the device's capacity is used up by each claim. This is analogous to how Pods can share the resources on a Node; ResourceClaims can share the resources on a Device.

The device driver can set `allowMultipleAllocations` field added in `.spec.devices` of `ResourceSlice` to allow allocating that device to multiple independent ResourceClaims or to multiple requests within a ResourceClaim.

Users can set `capacity` field added in `spec.devices.requests` of `ResourceClaim` to specify the device resource requirements for each allocation.

For the device that allows multiple allocations, the requested capacity is drawn from — or consumed from — its total capacity, a concept known as **consumable capacity**. Then, the scheduler ensures that the aggregate consumed capacity across all claims does not exceed the device's overall capacity. Furthermore, driver authors can use the `requestPolicy` constraints on individual device capacities to control how those capacities are consumed. For example, the driver author can specify that a given capacity is only consumed in increments of 1Gi.

Here is an example of a network device which allows multiple allocations and contains a consumable bandwidth capacity.

```yaml
kind: ResourceSlice
apiVersion: resource.k8s.io/v1
metadata:
  name: resourceslice
spec:
  nodeName: worker-1
  pool:
    name: pool
    generation: 1
    resourceSliceCount: 1
  driver: dra.example.com
  devices:
  - name: eth1
    allowMultipleAllocations: true
    attributes:
      name:
        string: "eth1"
    capacity:
      bandwidth:
        requestPolicy:
          default: "1M"
          validRange:
            min: "1M"
            step: "8"
        value: "10G"
```

The consumable capacity can be requested as shown in the below example.

```yaml
apiVersion: resource.k8s.io/v1
kind: ResourceClaimTemplate
metadata:
```

```
  name: bandwidth-claim-template
spec:
  spec:
    devices:
      requests:
      - name: req-0
        exactly:
          deviceClassName: resource.example.com
          capacity:
            requests:
              bandwidth: 1G
```

The allocation result will include the consumed capacity and the identifier of the share.

```
apiVersion: resource.k8s.io/v1
kind: ResourceClaim
...
status:
  allocation:
    devices:
      results:
      - consumedCapacity:
          bandwidth: 1G
        device: eth1
        shareID: "a671734a-e8e5-11e4-8fde-42010af09327"
```

In this example, a multiply-allocatable device was chosen. However, any `resource.example.com` device with at least the requested 1G bandwidth could have met the requirement. If a non-multiply-allocatable device were chosen, the allocation would have resulted in the entire device. To force the use of a only multiply-allocatable devices, you can use the CEL criteria `device.allowMultipleAllocations == true`.

DistinctAttribute constraint

When requesting multiple devices in a ResourceClaim, you can use the DistinctAttribute constraint to ensure that each allocated device has a different value for a specified attribute. This constraint was introduced with the consumable capacity feature.

The DistinctAttribute constraint is particularly useful when working with multiply-allocatable devices. It prevents the scheduler from allocating the same device multiple times within a single ResourceClaim, even when that device allows multiple allocations.

Beyond preventing duplicate allocations, this constraint helps optimize performance by ensuring devices are distributed based on their attributes. For example, you can use it to distribute devices across different NUMA nodes to optimize memory bandwidth and reduce contention.

Device taints and tolerations

> ⓘ **FEATURE STATE:** Kubernetes v1.36 [beta](enabled by default)

Device taints are similar to node taints: a taint has a string key, a string value, and an effect. The effect is applied to the ResourceClaim which is using a tainted device and to all Pods referencing that ResourceClaim. The "NoSchedule" effect prevents scheduling those Pods. Tainted devices are ignored when trying to allocate a ResourceClaim because using them would prevent scheduling of Pods.

The "NoExecute" effect implies "NoSchedule" and in addition causes eviction of all Pods which have been scheduled already. This eviction is implemented in the device taint eviction controller in kube-controller-manager by deleting affected Pods.

The "None" effect is ignored by the scheduler and eviction controller. DRA drivers can use it to communicate exceptions to admins or other controllers, like for example degraded health of a device. Admins can also use it to do dry-runs of pod eviction in DeviceTaintRules (more on that below).

ResourceClaims can tolerate taints. If a taint is tolerated, its effect does not apply. An empty toleration matches all taints. A toleration can be limited to certain effects and/or match certain key/value pairs. A toleration can check that a certain key exists, regardless which value it has, or it can check for specific values of a key. For more information on this matching see the node taint concepts.

Eviction can be delayed by tolerating a taint for a certain duration. That delay starts at the time when a taint gets added to a device, which is recorded in a field of the taint.

Taints apply as described above also to ResourceClaims allocating "all" devices on a node. All devices must be untainted or all of their taints must be tolerated. Allocating a device with admin access (described above) is not exempt either. An admin using that mode must explicitly tolerate all taints to access tainted devices.

Device taints and tolerations is a *beta feature* and enabled when the `DRADeviceTaints` feature gate is kept enabled in the kube-apiserver, kube-controller-manager and kube-scheduler. To use DeviceTaintRules, the `resource.k8s.io/v1beta2` API version must be enabled together with the `DRADeviceTaintRules` feature gate. In contrast to

`DRADeviceTaints` , `DRADeviceTaintRules` is off by default because of this dependency on the beta API group, which has to be off by default.

You can add taints to devices in the following ways, by using the DeviceTaintRule API kind.

Taints set by the driver

A DRA driver can add taints to the device information that it publishes in ResourceSlices. Consult the documentation of a DRA driver to learn whether the driver uses taints and what their keys and values are.

Taints set by an admin

> ⓘ **FEATURE STATE:** Kubernetes v1.36 [beta](disabled by default)

An admin or a control plane component can taint devices without having to tell the DRA driver to include taints in its device information in ResourceSlices. They do that by creating DeviceTaintRules. Each DeviceTaintRule adds one taint to devices which match the device selector. Without such a selector, no devices are tainted. This makes it harder to accidentally evict all pods using ResourceClaims when leaving out the selector by mistake.

Devices can be selected by giving the name of a DeviceClass, driver, pool, and/or device. The DeviceClass selects all devices that are selected by the selectors in that DeviceClass. With just the driver name, an admin can taint all devices managed by that driver, for example while doing some kind of maintenance of that driver across the entire cluster. Adding a pool name can limit the taint to a single node, if the driver manages node-local devices.

Finally, adding the device name can select one specific device. The device name and pool name can also be used alone, if desired. For example, drivers for node-local devices are encouraged to use the node name as their pool name. Then tainting with that pool name automatically taints all devices on a node.

Drivers might use stable names like "gpu-0" that hide which specific device is currently assigned to that name. To support tainting a specific hardware instance, CEL selectors can be used in a DeviceTaintRule to match a vendor-specific unique ID attribute, if the driver supports one for its hardware.

The taint applies as long as the DeviceTaintRule exists. It can be modified and and removed at any time. Here is one example of a DeviceTaintRule for a fictional DRA

driver:

```
apiVersion: resource.k8s.io/v1beta2
kind: DeviceTaintRule
metadata:
  name: example
spec:
  # The entire hardware installation for this
  # particular driver is broken.
  # Evict all pods and don't schedule new ones.
  deviceSelector:
    driver: dra.example.com
  taint:
    key: dra.example.com/unhealthy
    value: Broken
    effect: NoExecute
```

The kube-apiserver automatically tracks when this taint was created by setting the `timeAdded` field in the `spec`. The toleration period starts at that time stamp. During updates which change the effect (see simulated eviction flow below), the kube-apiserver automatically updates the time stamp. Users can control the time stamp explicitly by setting the field when creating a DeviceTaintRule and by changing it to some different value when updating.

The status contains a condition added by the eviction controller:

```
kubectl describe devicetaintrules
```

```
Name:         example
...
Spec:
  Device Selector:
    Driver:  dra.example.com
  Taint:
    Effect:      NoExecute
    Key:         dra.example.com/unhealthy
    Time Added:  2025-11-05T18:15:37Z
    Value:       Broken
Status:
  Conditions:
    Last Transition Time:  2025-11-05T18:15:37Z
    Message:               1 pod evicted since starting the controller.
    Observed Generation:   1
```

```
    Reason:                Completed
    Status:                False
    Type:                  EvictionInProgress
Events:                    <none>
```

Pods get evicted by deleting them. Usually this happens very quickly, except when a toleration for the taint delays it for a certain period or when there are very many pods which need to be evicted. When it takes longer, the message provides information about the current status:

```
2 pods need to be evicted in 2 different namespaces. 1 pod evicted since starting t
```

The condition can be used to check whether an eviction is currently active:

```
kubectl wait --for=condition=EvictionInProgress=false DeviceTaintRule/example
```

Beware of the potential race between scheduler and controller observing the new taint at different times, which can lead to pods still being scheduled at a time when the controller thinks that there are none which need to be evicted and thus sets this condition to `False`. In practice, this race is made very unlikely by updating the status only after an intentional delay of a few seconds.

For `effect: None`, the message provides information about the number of affected devices, how many of those are allocated, and how many pods would be evicted if the effect was `NoExecute`. This can be used to do a dry-run before actually triggering eviction:

- Create a DeviceTaintRule with the desired selectors and `effect: None`.
- Review the message:

  ```
  3 published devices selected. 1 allocated device selected.
  1 pod would be evicted in 1 namespace if the effect was NoExecute.
  This information will not be updated again. Recreate the DeviceTaintRule to tr
  ```

  Published devices are those listed in ResourceSlices. Tainting them prevents allocation for new pods. Only allocated devices cause eviction of the pods using them.

- Edit the DeviceTaintRule and change the effect into `NoExecute`.

Resource pool status

ⓘ **FEATURE STATE:** Kubernetes v1.36 [alpha](disabled by default)

You can query the availability of devices in resource pools using the ResourcePoolStatusRequest API. This provides visibility into how many devices are available, allocated, or unavailable across your cluster's DRA resource pools.

To check resource pool status:

1. Create a ResourcePoolStatusRequest specifying the driver name (required) and optionally a limit on the number of pools returned. You can also limit it to a single pool by specifying a pool name:

```yaml
apiVersion: resource.k8s.io/v1beta2
kind: ResourcePoolStatusRequest
metadata:
  name: check-gpus
spec:
  driver: example.com/gpu
  # Optional: filter to a specific pool
  # poolName: my-pool
  # Optional: limit number of pools returned (default: 100, max: 1000)
  # limit: 10
```

2. Wait for the controller to process the request:

```
kubectl wait --for=condition=Complete resourcepoolstatusrequest/check-gpus --t
```

3. Read the status to see pool availability:

```
kubectl get resourcepoolstatusrequest/check-gpus -o yaml
```

The status includes:

- `poolCount` : total number of pools matching the filter (may exceed the number of pools listed if truncated by the limit).
- `pools` : a list of pool details, each containing:
  - `driver` and `poolName` : identify the pool.
  - `generation` : the latest pool generation observed across ResourceSlices.
  - `resourceSliceCount` : the number of ResourceSlices making up the pool.
  - `totalDevices` : total devices in the pool.
  - `allocatedDevices` : devices currently allocated to claims.
  - `availableDevices` : devices available for allocation (totalDevices - allocatedDevices - unavailableDevices).
  - `unavailableDevices` : devices not available due to taints or other conditions.
  - `nodeName` : the node associated with the pool, if any.
  - `validationError` : set when the pool's data could not be fully validated (for example, during a generation rollout). When set, device count fields may be unset.
- `conditions` : includes `Complete` (success) or `Failed` (error) condition types.

4. Delete the request when done:

```
kubectl delete resourcepoolstatusrequest/check-gpus
```

ResourcePoolStatusRequest objects are processed once by a controller in kube-controller-manager. The spec is immutable once created, and the entire object becomes immutable once the status is populated. To get updated availability data, delete and recreate the request. Completed requests are automatically cleaned up after 1 hour.

This feature requires explicit RBAC permissions on the ResourcePoolStatusRequest resource. No default ClusterRoles include this permission.

Resource pool status is an *alpha feature* and only enabled when the `DRAResourcePoolStatus` feature gate is enabled in the kube-apiserver and kube-controller-manager.

Device binding conditions

ⓘ **FEATURE STATE:** Kubernetes v1.36 [beta](enabled by default)

Device Binding Conditions allow the Kubernetes scheduler to delay Pod binding until external resources, such as fabric-attached GPUs or reprogrammable FPGAs, are confirmed to be ready.

This waiting behavior is implemented in the PreBind phase of the scheduling framework. During this phase, the scheduler checks whether all required device conditions are satisfied before proceeding with binding.

This improves scheduling reliability by avoiding premature binding and enables coordination with external device controllers.

To use this feature, device drivers (typically managed by driver owners) must publish the following fields in the `Device` section of a `ResourceSlice`. Cluster administrators must enable the `DRADeviceBindingConditions` and `DRAResourceClaimDeviceStatus` feature gates for the scheduler to honor these fields.

`bindingConditions`

A list of *condition types* that must be set to True (in the `.status.conditions` field of the associated ResourceClaim) before the Pod can be bound. These conditions typically represent readiness signals, such as DeviceAttached or DeviceInitialized.

`bindingFailureConditions`

A list of condition types that, if set to True in status.conditions field of the associated ResourceClaim, indicate a failure state. If any of these conditions are True, the scheduler will abort binding and reschedule the Pod.

`bindsToNode`

if set to `true`, the scheduler records the selected node name in the `status.allocation.nodeSelector` field of the ResourceClaim. This does not affect the Pod's `spec.nodeSelector`. Instead, it sets a node selector inside the ResourceClaim, which external controllers can use to perform node-specific operations such as device attachment or preparation.

All condition types listed in bindingConditions and bindingFailureConditions are evaluated from the `status.conditions` field of the ResourceClaim. External controllers are responsible for updating these conditions using standard Kubernetes condition semantics (`type` , `status` , `reason` , `message` , `lastTransitionTime`).

The scheduler waits up to **600 seconds** (default) for all `bindingConditions` to become `True` . If the timeout is reached or any `bindingFailureConditions` are `True` , the scheduler clears the allocation and reschedules the Pod. A cluster administration can configure this timeout duration by editing the kube-scheduler configuration file.

An example of configuring this timeout in `KubeSchedulerConfiguration` is given below:

```yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: default-scheduler
  pluginConfig:
  - name: DynamicResources
    args:
      apiVersion: kubescheduler.config.k8s.io/v1
      kind: DynamicResourcesArgs
      bindingTimeout: 60s
```

Example

Here is an example of a ResourceSlice that you might see in a cluster where there's a DRA driver in use, and that driver supports binding conditions:

```yaml
apiVersion: resource.k8s.io/v1
kind: ResourceSlice
metadata:
  name: gpu-slice-1
spec:
  driver: dra.example.com
  nodeSelector:
    nodeSelectorTerms:
    - matchExpressions:
      - key: accelerator-type
        operator: In
        values:
        - "high-performance"
  pool:
    name: gpu-pool
    generation: 1
    resourceSliceCount: 1
  devices:
    - name: gpu-1
      attributes:
        vendor:
          string: "example"
        model:
          string: "example-gpu"
      bindsToNode: true
      bindingConditions:
        - dra.example.com/is-prepared
```

```
    bindingFailureConditions:
      - dra.example.com/preparing-failed
```

This example ResourceSlice has the following properties:

- The ResourceSlice targets nodes labeled with `accelerator-type=high-performance`, so that the scheduler uses only a specific set of eligible nodes.
- The scheduler selects one node from the selected group (for example, `node-3`) and sets the `status.allocation.nodeSelector` field in the ResourceClaim to that node name.
- The `dra.example.com/is-prepared` binding condition indicates that the device `gpu-1` must be prepared (the `is-prepared` condition has a status of `True`) before binding.
- If the `gpu-1` device preparation fails (the `preparing-failed` condition has a status of `True`), the scheduler aborts binding.
- The scheduler waits up to 600 seconds (default) for the device to become ready.
- External controllers can use the node selector in the ResourceClaim to perform node-specific setup on the selected node.

Device binding conditions is a *beta feature* and is enabled by default, controlled by the `DRADeviceBindingConditions` feature gate in the kube-apiserver and kube-scheduler.

Node allocatable resources

> ⓘ **FEATURE STATE:** Kubernetes v1.36 [alpha](disabled by default)

Devices managed by DRA can have an underlying footprint composed of node-allocatable resources, such as `cpu`, `memory`, `hugepages`, or `ephemeral-storage`. This feature integrates these DRA-based requests into the scheduler's standard accounting alongside regular Pod `spec` requests for these resources.

Users (PodSpec authors) can use a mixture of Pod-level resources, container-level resources, and resource claims with associated node-allocatable resources. These devices represent resources like CPUs or memory directly, or they could be accelerators, network interface cards, or other devices that require some host resources when allocated. The DRA driver will populate information in the ResourceSlice that tells the scheduler how to calculate the node allocatable resources when the device is allocated to a ResourceClaim. PodSpec authors do not need to make that calculation themselves.

When authoring a PodSpec using claims for these types of devices, there are a few things to be aware of:

- When Pod-level resources are used, the sum of all container and claim resources must not exceed the Pod-level resources; otherwise, the Pod will fail to schedule.
- A container's total resource requirement is the sum of its container-level resources and any node-allocatable resources from its associated resource claims.
- Claims that consume node allocatable resources cannot be shared between Pods.

Details for DRA Driver Authors

DRA drivers declare this node allocatable resource footprint using the `nodeAllocatableResourceMappings` field on devices within a ResourceSlice. This mapping translates the requested DRA device or capacity into standard resources that are tracked in the node's `status.allocatable` (note that extended resources are not supported for this mapping). This is useful both for drivers that directly expose native resources (like a CPU or Memory DRA driver) and for devices that require auxiliary node dependencies (like an accelerator that needs host memory).

This mapping defines the translation of the requested DRA device or capacity units to the corresponding quantity of the node-allocatable resource. The scheduler calculates the exact quantity using:

- **Device-based scaling:** If `capacityKey` is not set, the `allocationMultiplier` multiplies the device count allocated to the claim. The `allocationMultiplier` defaults to 1 if not specified.
- **Capacity-based scaling:** If `capacityKey` is set, it references a capacity name defined in the device's `capacity` map. The scheduler looks up the amount of that capacity consumed by the claim and multiplies it by the `allocationMultiplier` .

Example: CPU DRA Driver (Capacity-based scaling)

Here is an example where a CPU DRA driver exposes a CPU socket as a pool of 128 CPUs using DRA consumable capacity. The `capacityKey` links the consumed `cpu.example.com/cpu` capacity directly to the node's standard `cpu` allocatable resource:

```yaml
apiVersion: resource.k8s.io/v1
kind: ResourceSlice
metadata:
  name: my-node-cpus
spec:
  driver: cpu.example.com
  nodeName: my-node
```

```
  pool:
    name: socket-cpus
    generation: 1
    resourceSliceCount: 1
  devices:
  - name: socket0cpus
    allowMultipleAllocations: true
    capacity:
      "cpu.example.com/cpu": "128"
    nodeAllocatableResourceMappings:
      cpu:
        capacityKey: "cpu.example.com/cpu"
        # allocationMultiplier defaults to 1 if omitted
  - name: socket1cpus
    allowMultipleAllocations: true
    capacity:
      "cpu.example.com/cpu": "128"
    nodeAllocatableResourceMappings:
      cpu:
        capacityKey: "cpu.example.com/cpu"
        # allocationMultiplier defaults to 1 if omitted
```

Example: Accelerator with Auxiliary Resources (Device-based scaling)

Here is an example of a resource slice where an accelerator requires an additional 8Gi of memory per device instance to function:

```
apiVersion: resource.k8s.io/v1
kind: ResourceSlice
metadata:
  name: my-node-xpus
spec:
  driver: xpu.example.com
  nodeName: my-node
  pool:
    name: xpu-pool
    generation: 1
    resourceSliceCount: 1
  devices:
  - name: xpu-model-x-001
    attributes:
      example.com/model:
        string: "model-x"
    nodeAllocatableResourceMappings:
```

```
      memory:
        allocationMultiplier: "8Gi"
```

After a Pod is successfully bound to the node, the exact quantities of node-allocatable resources allocated via DRA are included in the Pod's `status.nodeAllocatableResourceClaimStatuses` field.

Node-allocatable resources is an alpha feature and is enabled when the `DRANodeAllocatableResources` feature gate is enabled in the kube-apiserver, kube-scheduler, and kubelet. In the alpha phase, the kubelet does not account for these resources when determining QoS classes, configuring cgroups, or making eviction decisions.

DRA device metadata in containers

FEATURE STATE: Kubernetes v1.36 [alpha]

DRA drivers can expose device metadata such as device attributes (PCI bus addresses or mdevUUID for mediated devices) or network configuration directly to containers as JSON files. This lets applications inside the container discover information about allocated devices without querying the Kubernetes API or building custom controllers.

KEP-5304 defines a device metadata protocol that drivers must follow so applications inside the container see a consistent layout across drivers and clusters. The DRA kubelet plugin library implements this protocol for you; the rest of this section describes how to use it.

Device metadata follows the same rules as device access: it is available inside a container only when that container requests the device in its container specification, and not otherwise. For how to request DRA devices in Pods and containers, see Request devices in workloads using DRA.

Device metadata protocol

The protocol consists of four rules:

1. **File paths.** Metadata files live inside containers under `/var/run/kubernetes.io/dra-device-attributes`. For a directly referenced ResourceClaim the path is `resourceclaims/<claimName>/<requestName>/<driverName>-metadata.json`; for a claim created from a ResourceClaimTemplate the path is `resourceclaimtemplates/<podClaimName>/<requestName>/<driverName>-metadata.json` (where `podClaimName` is `pod.spec.resourceClaims[].name`).

In cases where the ResourceClaim request uses the prioritized list feature, only the top-level request name is used for the `<requestName>` segment in the file path (that is, the `/<subrequest>` portion is dropped). Inside the JSON file, the `requests[].name` field carries the full `<request>/<subrequest>` reference (for example, `gpu/high-memory`) so that consumers can identify which alternative was allocated.

The path constants are defined in `k8s.io/dynamic-resource-allocation/api/metadata`.

2. **JSON API.** Each file is a stream of one or more `DeviceMetadata` objects serialized as versioned JSON with `apiVersion` and `kind`, following Kubernetes API conventions. The same metadata is encoded once per supported API version (newest first). All objects in the stream are semantically equivalent; consumers should use the first object they can decode.

3. **Generation.** When a driver updates a metadata file the embedded `metadata.generation` field must increase so consumers can detect changes.

4. **Container exposure.** Files are typically exposed via CDI bind-mounts, but other mechanisms are permitted as long as the file appears at the correct path and is read-only inside the container.

How device metadata works

Device metadata is a driver-side feature that does not require any Kubernetes API changes or feature gates. Using the DRA kubelet plugin library is a common way to implement a driver, but drivers can be built in other ways as well. Drivers that use the kubelet plugin enable this feature by passing the `EnableDeviceMetadata` and `MetadataVersions` options when starting the plugin. `MetadataVersions` specifies which API versions are serialized into the metadata file and must be set explicitly by the driver. Check the documentation of your DRA driver to learn whether device metadata is supported and how to enable it.

When device metadata is enabled, the driver generates metadata files and CDI bind-mount specifications while preparing the allocated devices for the pod, before the consuming containers start. The metadata appears inside containers at the well-known paths as defined above.

When a single request allocates devices from multiple DRA drivers, each driver writes its own metadata file. Containers enumerate `*-metadata.json` files in the request directory to discover all devices.

The Go package `k8s.io/dynamic-resource-allocation/devicemetadata` provides utilities for reading and decoding these metadata files by applications inside the container.

Metadata schema

Each metadata file conforms to the `DeviceMetadata` API (`metadata.resource.k8s.io/v1alpha1`). The following example shows a metadata file for a GPU device allocated through a ResourceClaimTemplate:


```
{
  "kind": "DeviceMetadata",
  "apiVersion": "metadata.resource.k8s.io/v1alpha1",
  "metadata": {
    "name": "pod0-gpu-2kqrd",
    "namespace": "gpu-test1",
    "uid": "c7e7b22e-239b-4498-b27c-7f1344481e14",
    "generation": 1
  },
  "podClaimName": "gpu",
  "requests": [
    {
      "name": "gpu",
      "devices": [
        {
          "driver": "gpu.example.com",
          "pool": "worker-0",
          "name": "gpu-0",
          "attributes": {
            "driverVersion": {
              "version": "1.0.0"
            },
            "index": {
              "int": 0
            },
            "model": {
              "string": "LATEST-GPU-MODEL"
            },
            "uuid": {
              "string": "gpu-18db0e85-99e9-c746-8531-ffeb86328b39"
            }
          }
        }
      ]
    }
  ]
}
```


Immediate and deferred metadata

Drivers provide metadata in one of two ways:

Immediate

The driver populates metadata while preparing the claim on the node and writes the metadata file before the container starts. This is typical for GPU drivers where device information is known at preparation time.

Deferred

In some cases, for example a network driver, the device information is not available during device allocation time but becomes available after the pod sandbox is created. In those cases the driver creates the CDI mount with an empty metadata file and writes the actual metadata later via an NRI hook that runs before the container starts. This ensures applications never see a missing or partially written file. Each update must increment `metadata.generation` so consumers can detect changes. The `MetadataUpdater` API in the DRA kubelet plugin library handles generation bookkeeping automatically for driver authors.

In both cases, metadata remains available to each consuming container for the lifetime of that container. Metadata files are cleaned up after all containers in the Pod have terminated.

To learn how to use device metadata in your workloads, see Access DRA device metadata.

Custom drivers

Custom, hand-crafted drivers that do not use the DRA kubelet plugin library must implement the device metadata protocol themselves. That means writing `DeviceMetadata` JSON at the correct file paths, incrementing `metadata.generation` on every update, and exposing the files read-only inside the container through CDI or an equivalent mechanism.

List type attributes

ⓘ **FEATURE STATE:** Kubernetes v1.36 [alpha](disabled by default)

This feature improves the ResourceSlice API, allowing DRA drivers to specify list values for device attributes instead of only scalars. This is useful for modeling more complex internal node topologies, for example when a CPU has adjacency to multiple PCIe roots.

For ResourceClaim authors (end users), this means that the `matchAttribute` and `distinctAttribute` work better for these cases.

- `matchAttribute` — the two attributes must have a *non-empty list intersection*, rather than be identical (scalar values are treated as single-item lists). This just means that if one driver publishes a single value for, say, the PCIe root, and another driver publishes a list, the constraint is met as long as the single value appears somewhere in the list.
- `distinctAttribute` — the attribute values must be *pairwise-disjoint* (no value shared between any two devices)

To help ResourceClaim authors use attributes that may be lists inside CEL expressions, this feature also introduces an `includes()` CEL function.

```
# Scalar attribute (backward compatible)
# assume: device.attributes["dra.example.com"].model = "model-a"
device.attributes["dra.example.com"].model.includes("model-a")  # true
device.attributes["dra.example.com"].model.includes("model-b")  # false

# List-type attribute (requires DRAListTypeAttributes)
# assume: device.attributes["dra.example.com"].supported-models= ["model-a", "model
device.attributes["dra.example.com"].supported-models.includes("model-a")  # true
device.attributes["dra.example.com"].supported-models.includes("model-c")  # false
```

Details for DRA Driver Authors

By default, each `DeviceAttribute` holds exactly one scalar value: a boolean, an integer, a string, or a semantic version string. The `DRAListTypeAttributes` feature gate extends `DeviceAttribute` with four list-type fields, allowing a device to advertise multiple values for a single attribute:

- **`bools`** — a list of boolean values
- **`ints`** — a list of 64-bit integer values
- **`strings`** — a list of strings (each at most 64 characters)
- **`versions`** — a list of semantic version strings per semver.org spec 2.0.0 (each at most 64 characters)

The total number of individual attribute values per device (scalar fields plus all list elements combined) is limited to **48**. When any device in a ResourceSlice uses this feature or other advanced features such as taints, the ResourceSlice will be limited to at most **64** devices. use list-type attributes or other advanced features such as taints.

Here is an example of a device advertising multiple supported models using a list-type string attribute:

```yaml
kind: ResourceSlice
apiVersion: resource.k8s.io/v1
metadata:
  name: example-resourceslice
spec:
  nodeName: worker-1
  pool:
    name: pool
    generation: 1
    resourceSliceCount: 1
  driver: dra.example.com
  devices:
  - name: gpu-0
    attributes:
      dra.example.com/supported-models:
        strings:
        - model-a
        - model-b
```

List type attributes is an *alpha feature* and only enabled when the `DRAListTypeAttributes` feature gate is enabled in the kube-apiserver and kube-scheduler.

What's next

- Set Up DRA in a Cluster
- Allocate devices to workloads using DRA
- Access DRA device metadata
- For more information on the design, see the Dynamic Resource Allocation with Structured Parameters KEP.

10 - Gang Scheduling

ⓘ **FEATURE STATE:** Kubernetes v1.35 [alpha](disabled by default)

Gang scheduling ensures that a group of Pods are scheduled on an "all-or-nothing" basis. If the cluster cannot accommodate the entire group (or a defined minimum number of Pods), none of the Pods are bound to a node.

This feature depends on the PodGroup API. Ensure the `GenericWorkload` feature gate and the `scheduling.k8s.io/v1alpha2` API group are enabled in the cluster.

How it works

When the `GangScheduling` plugin is enabled, the scheduler alters the lifecycle for Pods belonging to a PodGroup that has a `gang` scheduling policy. The process follows these steps for each PodGroup:

1. The scheduler holds Pods in the `PreEnqueue` phase until:
   - The referenced PodGroup object exists.
   - The number of `Pods` created for the `PodGroup` is at least equal to `minCount`.

   `Pods` do not enter the active scheduling queue until both conditions are met.
2. Once the quorum is met, the scheduler attempts to find placements for all Pods in the group. It utilizes the PodGroup scheduling cycle to make a single, atomic scheduling decision. `GangScheduling` plugin implements a `Permit` extension point that is evaluated for each schedulable Pod during the cycle. This is used to determine whether the `minCount` constraint is satisfied, by comparing the number of successfully placed pods against the `minCount` value.
3. If the scheduler finds valid placements for at least the `minCount` number of Pods, it allows those successfully placed Pods to be bound to their assigned nodes. If it cannot find enough placements to satisfy the `minCount` requirement, none of the Pods are scheduled. Instead, they are moved to the unschedulable queue to wait for cluster resources to free up, allowing other workloads to be scheduled in the meantime.

What's next

- Learn about the PodGroup API and its lifecycle.

- Read about PodGroup scheduling policies.
- Read about PodGroup scheduling.

11 - Scheduler Performance Tuning

ⓘ **FEATURE STATE:** Kubernetes v1.14 [beta]

kube-scheduler is the Kubernetes default scheduler. It is responsible for placement of Pods on Nodes in a cluster.

Nodes in a cluster that meet the scheduling requirements of a Pod are called *feasible* Nodes for the Pod. The scheduler finds feasible Nodes for a Pod and then runs a set of functions to score the feasible Nodes, picking a Node with the highest score among the feasible ones to run the Pod. The scheduler then notifies the API server about this decision in a process called *Binding*.

This page explains performance tuning optimizations that are relevant for large Kubernetes clusters.

In large clusters, you can tune the scheduler's behaviour balancing scheduling outcomes between latency (new Pods are placed quickly) and accuracy (the scheduler rarely makes poor placement decisions).

You configure this tuning setting via kube-scheduler setting `percentageOfNodesToScore`. This KubeSchedulerConfiguration setting determines a threshold for scheduling nodes in your cluster.

Setting the threshold

The `percentageOfNodesToScore` option accepts whole numeric values between 0 and 100. The value 0 is a special number which indicates that the kube-scheduler should use its compiled-in default. If you set `percentageOfNodesToScore` above 100, kube-scheduler acts as if you had set a value of 100.

To change the value, edit the kube-scheduler configuration file and then restart the scheduler. In many cases, the configuration file can be found at `/etc/kubernetes/config/kube-scheduler.yaml`.

After you have made this change, you can run

```
kubectl get pods -n kube-system | grep kube-scheduler
```

to verify that the kube-scheduler component is healthy.

Node scoring threshold

To improve scheduling performance, the kube-scheduler can stop looking for feasible nodes once it has found enough of them. In large clusters, this saves time compared to a naive approach that would consider every node.

You specify a threshold for how many nodes are enough, as a whole number percentage of all the nodes in your cluster. The kube-scheduler converts this into an integer number of nodes. During scheduling, if the kube-scheduler has identified enough feasible nodes to exceed the configured percentage, the kube-scheduler stops searching for more feasible nodes and moves on to the scoring phase.

How the scheduler iterates over Nodes describes the process in detail.

Default threshold

If you don't specify a threshold, Kubernetes calculates a figure using a linear formula that yields 50% for a 100-node cluster and yields 10% for a 5000-node cluster. The lower bound for the automatic value is 5%.

This means that the kube-scheduler always scores at least 5% of your cluster no matter how large the cluster is, unless you have explicitly set `percentageOfNodesToScore` to be smaller than 5.

If you want the scheduler to score all nodes in your cluster, set `percentageOfNodesToScore` to 100.

Example

Below is an example configuration that sets `percentageOfNodesToScore` to 50%.

```
apiVersion: kubescheduler.config.k8s.io/v1alpha1
kind: KubeSchedulerConfiguration
algorithmSource:
  provider: DefaultProvider

...

percentageOfNodesToScore: 50
```

Tuning percentageOfNodesToScore

`percentageOfNodesToScore` must be a value between 1 and 100 with the default value being calculated based on the cluster size. There is also a hardcoded minimum value of 100 nodes.

> **Note:**
> In clusters with less than 100 feasible nodes, the scheduler still checks all the nodes because there are not enough feasible nodes to stop the scheduler's search early.
>
> In a small cluster, if you set a low value for `percentageOfNodesToScore`, your change will have no or little effect, for a similar reason.
>
> If your cluster has several hundred Nodes or fewer, leave this configuration option at its default value. Making changes is unlikely to improve the scheduler's performance significantly.

An important detail to consider when setting this value is that when a smaller number of nodes in a cluster are checked for feasibility, some nodes are not sent to be scored for a given Pod. As a result, a Node which could possibly score a higher value for running the given Pod might not even be passed to the scoring phase. This would result in a less than ideal placement of the Pod.

You should avoid setting `percentageOfNodesToScore` very low so that kube-scheduler does not make frequent, poor Pod placement decisions. Avoid setting the percentage to anything below 10%, unless the scheduler's throughput is critical for your application and the score of nodes is not important. In other words, you prefer to run the Pod on any Node as long as it is feasible.

How the scheduler iterates over Nodes

This section is intended for those who want to understand the internal details of this feature.

In order to give all the Nodes in a cluster a fair chance of being considered for running Pods, the scheduler iterates over the nodes in a round robin fashion. You can imagine that Nodes are in an array. The scheduler starts from the start of the array and checks feasibility of the nodes until it finds enough Nodes as specified by `percentageOfNodesToScore`. For the next Pod, the scheduler continues from the point in the Node array that it stopped at when checking feasibility of Nodes for the previous Pod.

If Nodes are in multiple zones, the scheduler iterates over Nodes in various zones to ensure that Nodes from different zones are considered in the feasibility checks. As an example, consider six nodes in two zones:

```
Zone 1: Node 1, Node 2, Node 3, Node 4
Zone 2: Node 5, Node 6
```

The Scheduler evaluates feasibility of the nodes in this order:

```
Node 1, Node 5, Node 2, Node 6, Node 3, Node 4
```

After going over all the Nodes, it goes back to Node 1.

Enabling Opportunistic Batching

ⓘ **FEATURE STATE:** Kubernetes v1.35 [beta](enabled by default)

When scheduling large workloads, pod definitions are typically identical and require the scheduler to perform the same operations over and over again. The Opportunistic Batching feature allows the scheduler to reuse the filtering and scoring results between scheduling cycles which greatly speeds up the scheduling process.

Basically, this feature works like:

1. The scheduler schedules pod-1 and caches the scheduling result.
2. The scheduler schedules pod-2, 3, ... with the cached results.
3. The cache expires after 0.5 second. The scheduler schedules the next pod which builds a new cache.

Pods with equivalent scheduling constraints have to come to the scheduling cycle back to back. When the scheduler schedules a pod with different constraints, the cache is not used, but replaced with a new one.

We apply this batching scheduling to specific pods that:

1. Don't have inter pod affinity/anti-affinity
2. Don't have topology spread constraints
3. Don't have DRA (i.e., don't have any Resource Claims)
4. Don't request extended resources that are backed by DRA

5. Scheduled exclusively on nodes (i.e., placing more than one pod on one node invalidates the cache)

Also, to enable this feature, the scheduler configuration needs to:

1. Disable default topology spread (set empty)
2. Set `IgnorePreferredTermsOfExistingPods` of InterPodAffinityArgs to `true` to make the batching more efficient

Note that whenever:

1. Existing pods use pod affinity constraints that match any of the scheduled pods' labels, the feature may bring no benefit
2. Custom plugins are used, they need to implement the Signature extension point

The restrictions and conditions are expected to evolve in future releases.

What's next

- Check the kube-scheduler configuration reference (v1)

12 - PodGroup Scheduling

ⓘ **FEATURE STATE:** Kubernetes v1.35 [alpha](disabled by default)

The standard Kubernetes scheduler evaluates Pods sequentially. When multiple workloads, such as machine learning training jobs, are submitted concurrently, this sequential evaluation can lead to resource deadlocks. For example, two competing workloads might each schedule a subset of their Pods, consuming cluster capacity but leaving neither workload with enough resources to fully start.

The PodGroup scheduling cycle evaluates a group of Pods as a single unit. The scheduler attempts to find placements for all Pods in the group simultaneously. If it cannot find sufficient resources to satisfy the entire group's requirements, none of the Pods are bound.

Additionally, treating the group as a unified entity establishes a foundational architecture that simplifies the implementation of other group-based scheduling features.

This feature depends on the Workload API. Ensure the `GenericWorkload` feature gate and the `scheduling.k8s.io/v1alpha1` API group are enabled in the cluster.

PodGroup scheduling cycle

To support scheduling a group of Pods together, the kube-scheduler uses the **PodGroup scheduling cycle**. Instead of processing Pods individually and holding them at a `WaitOnPermit` gate, the scheduler evaluates the entire group of pending Pods belonging to a specific PodGroup collectively. Rather than executing separate scheduling cycles for each Pod, it evaluates feasibility for the entire group and moves directly to the binding phase afterwards.

When the scheduler pops a Pod belonging to a PodGroup, it retrieves all other queued Pods in that group. It then sorts them deterministically based on priority and the time they were initially observed by the scheduler, and initiates the PodGroup scheduling cycle as follows:

1. **Snapshotting the cluster state:** When the scheduler begins evaluating a PodGroup, it takes a single snapshot of the cluster state that lasts for the entire duration of the cycle. This ensures the evaluation remains consistent for the whole group and prevents race conditions with other events.

2. **Finding feasible placements:** The scheduler runs the PodGroup scheduling algorithm to find valid Node placements for the Pods in the group.

3. **Atomic decision:** Depending on the algorithm's outcome, the scheduling decision is applied atomically for the entire PodGroup.

    - **Success:** If the scheduler finds sufficient resources and valid placements for the Pods (e.g., satisfying the `minCount` constraint for gang scheduling), those Pods proceed directly to the binding cycle with their selected nodes. Any remaining unschedulable Pods are returned to the scheduling queue to wait for available resources so they can join the already scheduled Pods.

      Furthermore, if new Pods are added to a PodGroup after others have already been scheduled, the cycle evaluates the new Pods while accounting for the existing ones.

    - **Failure:** If the scheduler cannot find enough resources to make the PodGroup feasible (e.g., failing to meet the `minCount` constraint), the entire PodGroup is considered unschedulable. No Pods are bound, but instead, all are returned to the scheduling queue. Standard scheduling backoff logic applies, allowing the PodGroup to be retried later.

By using this single-cycle approach, the scheduler avoids inefficient bottlenecks where partially scheduled groups reserve cluster capacity while waiting indefinitely for the rest of their group to fit.

PodGroup scheduling algorithm

The default PodGroup scheduling algorithm relies heavily on the baseline Pod-based scheduling algorithm. It iterates over the Pods and performs the following for each:

1. Finds a feasible node using the standard per-Pod filtering and scoring phases.
    - If the Pod fits, it is temporarily assumed and reserved on the selected node until the end of the scheduling algorithm.
    - If the Pod cannot fit, the scheduler attempts preemption by running the `PostFilter` extension point.
2. Checks whether the schedulable Pods meet the group's scheduling criteria (e.g., the `minCount` for gang scheduling) using the `Permit` extension point. If it returns a `Success` status for any Pod, the PodGroup is deemed feasible. If the algorithm processes all Pods without achieving a `Success` status, the PodGroup is considered unschedulable.

Placement scheduling algorithm

ⓘ **FEATURE STATE:** Kubernetes v1.36 [alpha](disabled by default)

Placement scheduling algorithm is an alternative PodGroup scheduling algorithm, which uses scheduling plugins to find the optimal placement for the considered PodGroup. Users can accomodate the algorithm to their specific needs by using and configuring plugins.

The algorithm proceeds in three main phases for a given PodGroup:

Phase 1: Candidate placement generation

Generates candidate *placements* (subsets of nodes, that are theoretically feasible for PodGroup assignment), for example based on the PodGroup's scheduling constraints (which can be defined in the PodGroup object).

This phase executes as extension point: `PlacementGeneratePlugin`.

Phase 2: Pod-level filtering and feasibility check

Validates each proposed placement, by running a default PodGroup scheduling algorithm, to see if the required number of Pods from the PodGroup can fit. If they can, the placement is marked as feasible.

Phase 3: Placement scoring and selection

Scores all feasible placements to select the optimal domain for the PodGroup.

This phase executes as extension point: `PlacementScorePlugin`.

Limitations

The PodGroup scheduling algorithm relies on specific Pod sorting and may fail to find a valid placement that could have been discovered by processing the group's Pods in a different order. In particular:

- For basic **homogeneous** Pod groups (i.e., those where all Pods have identical scheduling requirements and lack inter-Pod dependencies like affinity, anti-affinity, or topology spread constraints), the algorithm is expected to find a placement if one exists.

- For **heterogeneous** Pod groups, finding a valid placement is not guaranteed.
- For Pod groups with **inter-Pod dependencies**, finding a valid placement is not guaranteed.

In addition to the above, for cases involving **intra-group dependencies** (e.g., when the schedulability of one Pod depends on another group member via inter-Pod affinity), this algorithm may fail to find a placement regardless of cluster state due to its deterministic processing order.

For consistent behavior throughout the entire cycle, the algorithm requires that all Pods belonging to a single PodGroup share the same `.spec.schedulerName`. This requirement is validated before the cycle starts, and the PodGroup is rejected if the constraint is not met.

PodGroup conditions

After a PodGroup scheduling cycle completes, the scheduler updates conditions on the PodGroup's `status.conditions`:

- `PodGroupScheduled`: reports whether the PodGroup has been successfully scheduled.
- `DisruptionTarget`: indicates the PodGroup is about to be terminated due to a disruption such as preemption.

`PodGroupScheduled`

When the scheduling cycle succeeds, the condition is set to `True` with reason `Scheduled`. For `gang` policy PodGroups, this means at least `minCount` Pods were placed.

When scheduling fails, the condition is set to `False` with one of the following reasons:

- `Unschedulable` — the group could not be placed due to resource constraints, affinity or anti-affinity rules, or insufficient capacity for the gang.
- `SchedulerError` — scheduling failed because of an internal scheduler error (for example, while parsing scheduling constraints such as `nodeAffinity`).

`DisruptionTarget`

When the scheduler preempts a PodGroup to make room for higher-priority PodGroups or Pods, it sets this condition to `True` with reason `PreemptionByScheduler`.

You can check conditions with:

```
kubectl get podgroup <name> -o jsonpath='{.status.conditions}'
```

What's next

- Learn about the Workload API.
- See how to reference a Workload in a Pod.
- Read about gang scheduling.

13 - Resource Bin Packing

> **Note:**
> This article applies to resource bin packing in context of scheduling of a single pod. For bin packing when scheduling pod groups, please read the article about Topology-aware Scheduling.

In the scheduling-plugin `NodeResourcesFit` of kube-scheduler, there are two scoring strategies that support the bin packing of resources: `MostAllocated` and `RequestedToCapacityRatio`.

Enabling bin packing using MostAllocated strategy

The `MostAllocated` strategy scores the nodes based on the utilization of resources, favoring the ones with higher allocation. For each resource type, you can set a weight to modify its influence in the node score.

To set the `MostAllocated` strategy for the `NodeResourcesFit` plugin, use a scheduler configuration similar to the following:

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- pluginConfig:
  - args:
      scoringStrategy:
        resources:
        - name: cpu
          weight: 1
        - name: memory
          weight: 1
        - name: intel.com/foo
          weight: 3
        - name: intel.com/bar
          weight: 3
        type: MostAllocated
    name: NodeResourcesFit
```

With this configuration, nodes are scored using a weighted average of utilization across all four resources. Because `intel.com/foo` and `intel.com/bar` each carry a weight of `3` versus `1` for CPU and memory, the utilization of those extended resources has three times more influence on the final node score. The scheduler selects the highest-scoring node, aiming to schedule pods on highly utilized nodes. This helps prepare for scale-down of the least utilized nodes.

To learn more about other parameters and their default configuration, see the API documentation for `NodeResourcesFitArgs` .

Enabling bin packing using RequestedToCapacityRatio

The `RequestedToCapacityRatio` strategy allows the users to specify the resources along with weights for each resource to score nodes based on the request to capacity ratio. This allows users to bin pack extended resources by using appropriate parameters to improve the utilization of scarce resources in large clusters. It favors nodes according to a configured function of the allocated resources. The behavior of the `RequestedToCapacityRatio` in the `NodeResourcesFit` score function can be controlled by the scoringStrategy field. Within the `scoringStrategy` field, you can configure two parameters: `requestedToCapacityRatio` and `resources` . The `shape` in the `requestedToCapacityRatio` parameter allows the user to tune the function as least requested or most requested based on `utilization` and `score` values. The `resources` parameter comprises both the `name` of the resource to be considered during scoring and its corresponding `weight` , which specifies the weight of each resource.

Below is an example configuration that sets the bin packing behavior for extended resources `intel.com/foo` and `intel.com/bar` using the `requestedToCapacityRatio` field.

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- pluginConfig:
  - args:
      scoringStrategy:
        resources:
        - name: intel.com/foo
          weight: 3
        - name: intel.com/bar
          weight: 3
        requestedToCapacityRatio:
          shape:
```

```
          - utilization: 0
            score: 0
          - utilization: 100
            score: 10
        type: RequestedToCapacityRatio
    name: NodeResourcesFit
```

In this example, only the extended resources `intel.com/foo` and `intel.com/bar` are listed in `resources`. The `NodeResourcesFit` plugin therefore scores nodes based solely on the utilization of those two resources; CPU and memory do not contribute to the score from this plugin. Because the configured shape assigns a higher score as utilization increases (`score: 0` at `utilization: 0` rising to `score: 10` at `utilization: 100`), the scheduler prefers nodes where more of these extended resources are already in use, bin-packing requests for them onto as few nodes as possible.

To include CPU and memory in this scoring strategy, add them to the `resources` list. Note that all resources in the list share the same `shape` function, so doing so will apply the same bin-packing curve to those resources as well.

Referencing the `KubeSchedulerConfiguration` file with the kube-scheduler flag `--config=/path/to/config/file` will pass the configuration to the scheduler.

To learn more about other parameters and their default configuration, see the API documentation for `NodeResourcesFitArgs`.

Tuning the score function

`shape` is used to specify the behavior of the `RequestedToCapacityRatio` function.

```
shape:
  - utilization: 0
    score: 0
  - utilization: 100
    score: 10
```

The above arguments give the node a `score` of 0 if `utilization` is 0% and 10 for `utilization` 100%, thus enabling bin packing behavior. To enable least requested the score value must be reversed as follows.

```
shape:
  - utilization: 0
```

```
    score: 10
  - utilization: 100
    score: 0
```

`resources` is an optional parameter which defaults to:

```
resources:
  - name: cpu
    weight: 1
  - name: memory
    weight: 1
```

It can be used to add extended resources as follows:

```
resources:
  - name: intel.com/foo
    weight: 5
  - name: cpu
    weight: 3
  - name: memory
    weight: 1
```

The `weight` parameter is optional and is set to 1 if not specified. Also, the `weight` cannot be set to a negative value.

Node scoring for capacity allocation

This section is intended for those who want to understand the internal details of this feature. Below is an example of how the node score is calculated for a given set of values.

Requested resources:

```
intel.com/foo : 2
memory: 256MB
cpu: 2
```

Resource weights:

```
intel.com/foo : 5
memory: 1
cpu: 3
```

FunctionShapePoint {{0, 0}, {100, 10}}

Node 1 spec:

```
Available:
  intel.com/foo: 4
  memory: 1 GB
  cpu: 8

Used:
  intel.com/foo: 1
  memory: 256MB
  cpu: 1
```

Node score:

```
intel.com/foo  = resourceScoringFunction((2+1),4)
               = (100 - ((4-3)*100/4))
               = (100 - 25)
               = 75                      # requested + used = 75% * available
               = rawScoringFunction(75)
               = 7                       # floor(75/10)

memory         = resourceScoringFunction((256+256),1024)
               = (100 -((1024-512)*100/1024))
               = 50                      # requested + used = 50% * available
               = rawScoringFunction(50)
               = 5                       # floor(50/10)

cpu            = resourceScoringFunction((2+1),8)
               = (100 -((8-3)*100/8))
               = 37.5                    # requested + used = 37.5% * available
               = rawScoringFunction(37.5)
               = 3                       # floor(37.5/10)

NodeScore   =  ((7 * 5) + (5 * 1) + (3 * 3)) / (5 + 1 + 3)
            =  5
```

Node 2 spec:

```
Available:
  intel.com/foo: 8
  memory: 1GB
  cpu: 8
Used:
  intel.com/foo: 2
  memory: 512MB
  cpu: 6
```

Node score:

```
intel.com/foo  = resourceScoringFunction((2+2),8)
               =  (100 - ((8-4)*100/8)
               =  (100 - 50)
               =  50
               =  rawScoringFunction(50)
               = 5

memory         = resourceScoringFunction((256+512),1024)
               = (100 -((1024-768)*100/1024))
               = 75
               = rawScoringFunction(75)
               = 7

cpu            = resourceScoringFunction((2+6),8)
               = (100 -((8-8)*100/8))
               = 100
               = rawScoringFunction(100)
               = 10

NodeScore   =  ((5 * 5) + (7 * 1) + (10 * 3)) / (5 + 1 + 3)
            =  7
```

What's next

- Read more about the scheduling framework
- Read more about scheduler configuration

14 - Workload-Aware Preemption

ⓘ **FEATURE STATE:** Kubernetes v1.36 [alpha](disabled by default)

Workload-aware preemption introduces a preemption mechanism specifically designed for PodGroups. When a PodGroup cannot be scheduled, the scheduler utilizes a preemption logic that tries to make scheduling of this PodGroup possible. This approach is used exclusively during PodGroup scheduling and replaces the default preemption mechanism for pods from a given PodGroup.

When this feature is enabled, the scheduler treats the PodGroup as a single preemptor unit, rather than evaluating individual pods from a PodGroup in isolation. To make room for the pending pods in the group, it searches for victims across the entire cluster, and knows how to treat and preempt other PodGroups as victims according to their disruption modes.

This feature depends on the Gang Scheduling and the Workload API. Ensure the `GenericWorkload` and `GangScheduling` feature gates and the `scheduling.k8s.io/v1alpha2` API group are enabled in the cluster.

How it works

The workload-aware preemption process follows the same principles as default preemption with a few differences:

1. Cluster-wide domain: Instead of evaluating preemption node by node, the scheduler evaluates the entire cluster as a single domain. It selects a set of victims across multiple nodes that can be removed to make enough room for the preemptor PodGroup to be scheduled.

2. Victim importance hierarchy: The scheduler decides which preemption units (individual pods or PodGroups) are more critical and should be spared from preemption using a strict hierarchy:

    - Priority: Higher priority units are always more important.
    - Workload type: PodGroups are considered more important than individual Pods of the same priority.
    - Group size (PodGroups): If both units are PodGroups, the one with more members (larger size) is considered more important.
    - Start time: Units that started earlier are more important.

3. Pod group priority and disruption: The scheduler considers the specific priority and disruption mode of a PodGroup to evaluate if and how its pods can be preempted during preemption events.

> **Note:**
> When scheduling a single Pod, the default pod preemption applies. As of 1.36, when the scheduler performs a default preemption for a single Pod and it attempts to preempt a Pod belonging to a PodGroup, it does **not** respect the `priority` or `disruptionMode` fields of that PodGroup.

What's next

- Learn more about PodGroup Priority and Disruption.
- Learn about the Workload API.
- Read more about Gang scheduling.

15 - Pod Priority and Preemption

ⓘ **FEATURE STATE:** Kubernetes v1.14 [stable]

Pods can have *priority*. Priority indicates the importance of a Pod relative to other Pods. If a Pod cannot be scheduled, the scheduler tries to preempt (evict) lower priority Pods to make scheduling of the pending Pod possible.

> **Warning:**
> In a cluster where not all users are trusted, a malicious user could create Pods at the highest possible priorities, causing other Pods to be evicted/not get scheduled. An administrator can use ResourceQuota to prevent users from creating pods at high priorities.
>
> See limit Priority Class consumption by default for details.

How to use priority and preemption

To use priority and preemption:

1. Add one or more PriorityClasses.
2. Create Pods with `priorityClassName` set to one of the added PriorityClasses. Of course you do not need to create the Pods directly; normally you would add `priorityClassName` to the Pod template of a collection object like a Deployment.

Keep reading for more information about these steps.

> **Note:**
> Kubernetes already ships with two PriorityClasses: `system-cluster-critical` and `system-node-critical`. These are common classes and are used to ensure that critical components are always scheduled first.

PriorityClass

A PriorityClass is a non-namespaced object that defines a mapping from a priority class name to the integer value of the priority. The name is specified in the `name` field of the PriorityClass object's metadata. The value is specified in the required `value` field. The higher the value, the higher the priority. The name of a PriorityClass object must be a valid DNS subdomain name, and it cannot be prefixed with `system-` .

A PriorityClass object can have any 32-bit integer value smaller than or equal to 1 billion. This means that the range of values for a PriorityClass object is from -2147483648 to 1000000000 inclusive. Larger numbers are reserved for built-in PriorityClasses that represent critical system Pods. A cluster admin should create one PriorityClass object for each such mapping that they want.

PriorityClass also has two optional fields: `globalDefault` and `description` . The `globalDefault` field indicates that the value of this PriorityClass should be used for Pods without a `priorityClassName` . Only one PriorityClass with `globalDefault` set to true can exist in the system. If there is no PriorityClass with `globalDefault` set, the priority of Pods with no `priorityClassName` is zero.

The `description` field is an arbitrary string. It is meant to tell users of the cluster when they should use this PriorityClass.

Notes about PodPriority and existing clusters

- If you upgrade an existing cluster without this feature, the priority of your existing Pods is effectively zero.
- Addition of a PriorityClass with `globalDefault` set to `true` does not change the priorities of existing Pods. The value of such a PriorityClass is used only for Pods created after the PriorityClass is added.
- If you delete a PriorityClass, existing Pods that use the name of the deleted PriorityClass remain unchanged, but you cannot create more Pods that use the name of the deleted PriorityClass.

Example PriorityClass

```yaml
apiVersion: scheduling.k8s.io/v1
kind: PriorityClass
metadata:
  name: high-priority
value: 1000000
globalDefault: false
description: "This priority class should be used for XYZ service pods only."
```

Non-preempting PriorityClass

FEATURE STATE: Kubernetes v1.24 [stable]

Pods with `preemptionPolicy: Never` will be placed in the scheduling queue ahead of lower-priority pods, but they cannot preempt other pods. A non-preempting pod waiting to be scheduled will stay in the scheduling queue, until sufficient resources are free, and it can be scheduled. Non-preempting pods, like other pods, are subject to scheduler back-off. This means that if the scheduler tries these pods and they cannot be scheduled, they will be retried with lower frequency, allowing other pods with lower priority to be scheduled before them.

Non-preempting pods may still be preempted by other, high-priority pods.

`preemptionPolicy` defaults to `PreemptLowerPriority`, which will allow pods of that PriorityClass to preempt lower-priority pods (as is existing default behavior). If `preemptionPolicy` is set to `Never`, pods in that PriorityClass will be non-preempting.

An example use case is for data science workloads. A user may submit a job that they want to be prioritized above other workloads, but do not wish to discard existing work by preempting running pods. The high priority job with `preemptionPolicy: Never` will be scheduled ahead of other queued pods, as soon as sufficient cluster resources "naturally" become free.

Example Non-preempting PriorityClass

```yaml
apiVersion: scheduling.k8s.io/v1
kind: PriorityClass
metadata:
  name: high-priority-nonpreempting
value: 1000000
preemptionPolicy: Never
globalDefault: false
description: "This priority class will not cause other pods to be preempted."
```

Pod priority

After you have one or more PriorityClasses, you can create Pods that specify one of those PriorityClass names in their specifications. The priority admission controller uses

the `priorityClassName` field and populates the integer value of the priority. If the priority class is not found, the Pod is rejected.

The following YAML is an example of a Pod configuration that uses the PriorityClass created in the preceding example. The priority admission controller checks the specification and resolves the priority of the Pod to 1000000.

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  labels:
    env: test
spec:
  containers:
  - name: nginx
    image: nginx
    imagePullPolicy: IfNotPresent
  priorityClassName: high-priority
```

Effect of Pod priority on scheduling order

When Pod priority is enabled, the scheduler orders pending Pods by their priority and a pending Pod is placed ahead of other pending Pods with lower priority in the scheduling queue. As a result, the higher priority Pod may be scheduled sooner than Pods with lower priority if its scheduling requirements are met. If such Pod cannot be scheduled, the scheduler will continue and try to schedule other lower priority Pods.

Preemption

When Pods are created, they go to a queue and wait to be scheduled. The scheduler picks a Pod from the queue and tries to schedule it on a Node. If no Node is found that satisfies all the specified requirements of the Pod, preemption logic is triggered for the pending Pod. Let's call the pending Pod P. Preemption logic tries to find a Node where removal of one or more Pods with lower priority than P would enable P to be scheduled on that Node. If such a Node is found, one or more lower priority Pods get evicted from the Node. After the Pods are gone, P can be scheduled on the Node.

User exposed information

When Pod P preempts one or more Pods on Node N, `nominatedNodeName` field of Pod P's status is set to the name of Node N. This field helps the scheduler track resources reserved for Pod P and also gives users information about preemptions in their clusters.

Please note that Pod P is not necessarily scheduled to the "nominated Node". The scheduler always tries the "nominated Node" before iterating over any other nodes. After victim Pods are preempted, they get their graceful termination period. If another node becomes available while scheduler is waiting for the victim Pods to terminate, scheduler may use the other node to schedule Pod P. As a result `nominatedNodeName` and `nodeName` of Pod spec are not always the same. Also, if the scheduler preempts Pods on Node N, but then a higher priority Pod than Pod P arrives, the scheduler may give Node N to the new higher priority Pod. In such a case, scheduler clears `nominatedNodeName` of Pod P. By doing this, scheduler makes Pod P eligible to preempt Pods on another Node.

Limitations of preemption

Graceful termination of preemption victims

When Pods are preempted, the victims get their graceful termination period. They have that much time to finish their work and exit. If they don't, they are killed. This graceful termination period creates a time gap between the point that the scheduler preempts Pods and the time when the pending Pod (P) can be scheduled on the Node (N). In the meantime, the scheduler keeps scheduling other pending Pods. As victims exit or get terminated, the scheduler tries to schedule Pods in the pending queue. Therefore, there is usually a time gap between the point that scheduler preempts victims and the time that Pod P is scheduled. In order to minimize this gap, one can set graceful termination period of lower priority Pods to zero or a small number.

PodDisruptionBudget is supported, but not guaranteed

A PodDisruptionBudget (PDB) allows application owners to limit the number of Pods of a replicated application that are down simultaneously from voluntary disruptions. Kubernetes supports PDB when preempting Pods, but respecting PDB is best effort. The scheduler tries to find victims whose PDB are not violated by preemption, but if no such victims are found, preemption will still happen, and lower priority Pods will be removed despite their PDBs being violated.

Inter-Pod affinity on lower-priority Pods

A Node is considered for preemption only when the answer to this question is yes: "If all the Pods with lower priority than the pending Pod are removed from the Node, can

the pending Pod be scheduled on the Node?"

> **Note:**
> Preemption does not necessarily remove all lower-priority Pods. If the pending Pod can be scheduled by removing fewer than all lower-priority Pods, then only a portion of the lower-priority Pods are removed. Even so, the answer to the preceding question must be yes. If the answer is no, the Node is not considered for preemption.

If a pending Pod has inter-pod affinity to one or more of the lower-priority Pods on the Node, the inter-Pod affinity rule cannot be satisfied in the absence of those lower-priority Pods. In this case, the scheduler does not preempt any Pods on the Node. Instead, it looks for another Node. The scheduler might find a suitable Node or it might not. There is no guarantee that the pending Pod can be scheduled.

Our recommended solution for this problem is to create inter-Pod affinity only towards equal or higher priority Pods.

Cross node preemption

Suppose a Node N is being considered for preemption so that a pending Pod P can be scheduled on N. P might become feasible on N only if a Pod on another Node is preempted. Here's an example:

- Pod P is being considered for Node N.
- Pod Q is running on another Node in the same Zone as Node N.
- Pod P has Zone-wide anti-affinity with Pod Q (`topologyKey: topology.kubernetes.io/zone`).
- There are no other cases of anti-affinity between Pod P and other Pods in the Zone.
- In order to schedule Pod P on Node N, Pod Q can be preempted, but scheduler does not perform cross-node preemption. So, Pod P will be deemed unschedulable on Node N.

If Pod Q were removed from its Node, the Pod anti-affinity violation would be gone, and Pod P could possibly be scheduled on Node N.

We may consider adding cross Node preemption in future versions if there is enough demand and if we find an algorithm with reasonable performance.

Troubleshooting

Pod priority and preemption can have unwanted side effects. Here are some examples of potential problems and ways to deal with them.

Pods are preempted unnecessarily

Preemption removes existing Pods from a cluster under resource pressure to make room for higher priority pending Pods. If you give high priorities to certain Pods by mistake, these unintentionally high priority Pods may cause preemption in your cluster. Pod priority is specified by setting the `priorityClassName` field in the Pod's specification. The integer value for priority is then resolved and populated to the `priority` field of `podSpec`.

To address the problem, you can change the `priorityClassName` for those Pods to use lower priority classes, or leave that field empty. An empty `priorityClassName` is resolved to zero by default.

When a Pod is preempted, there will be events recorded for the preempted Pod. Preemption should happen only when a cluster does not have enough resources for a Pod. In such cases, preemption happens only when the priority of the pending Pod (preemptor) is higher than the victim Pods. Preemption must not happen when there is no pending Pod, or when the pending Pods have equal or lower priority than the victims. If preemption happens in such scenarios, please file an issue.

Pods are preempted, but the preemptor is not scheduled

When pods are preempted, they receive their requested graceful termination period, which is by default 30 seconds. If the victim Pods do not terminate within this period, they are forcibly terminated. Once all the victims go away, the preemptor Pod can be scheduled.

While the preemptor Pod is waiting for the victims to go away, a higher priority Pod may be created that fits on the same Node. In this case, the scheduler will schedule the higher priority Pod instead of the preemptor.

This is expected behavior: the Pod with the higher priority should take the place of a Pod with a lower priority.

Higher priority Pods are preempted before lower priority pods

The scheduler tries to find nodes that can run a pending Pod. If no node is found, the scheduler tries to remove Pods with lower priority from an arbitrary node in order to

make room for the pending pod. If a node with low priority Pods is not feasible to run the pending Pod, the scheduler may choose another node with higher priority Pods (compared to the Pods on the other node) for preemption. The victims must still have lower priority than the preemptor Pod.

When there are multiple nodes available for preemption, the scheduler tries to choose the node with a set of Pods with lowest priority. However, if such Pods have PodDisruptionBudget that would be violated if they are preempted then the scheduler may choose another node with higher priority Pods.

When multiple nodes exist for preemption and none of the above scenarios apply, the scheduler chooses a node with the lowest priority.

Interactions between Pod priority and quality of service

Pod priority and QoS class are two orthogonal features with few interactions and no default restrictions on setting the priority of a Pod based on its QoS classes. The scheduler's preemption logic does not consider QoS when choosing preemption targets. Preemption considers Pod priority and attempts to choose a set of targets with the lowest priority. Higher-priority Pods are considered for preemption only if the removal of the lowest priority Pods is not sufficient to allow the scheduler to schedule the preemptor Pod, or if the lowest priority Pods are protected by `PodDisruptionBudget`.

The kubelet uses Priority to determine pod order for node-pressure eviction. You can use the QoS class to estimate the order in which pods are most likely to get evicted. The kubelet ranks pods for eviction based on the following factors:

1. Whether the starved resource usage exceeds requests
2. Pod Priority
3. Amount of resource usage relative to requests

See Pod selection for kubelet eviction for more details.

kubelet node-pressure eviction does not evict Pods when their usage does not exceed their requests. If a Pod with lower priority is not exceeding its requests, it won't be evicted. Another Pod with higher priority that exceeds its requests may be evicted.

What's next

- Read about using ResourceQuotas in connection with PriorityClasses: limit Priority Class consumption by default

- Learn about Pod Disruption
- Learn about API-initiated Eviction
- Learn about Node-pressure Eviction

16 - Node-pressure Eviction

Node-pressure eviction is the process by which the kubelet proactively terminates pods to reclaim resource on nodes.

The kubelet monitors resources like memory, disk space, and filesystem inodes on your cluster's nodes. When one or more of these resources reach specific consumption levels, the kubelet can proactively fail one or more pods on the node to reclaim resources and prevent starvation.

During a node-pressure eviction, the kubelet sets the phase for the selected pods to `Failed`, and terminates the Pod.

Node-pressure eviction is not the same as API-initiated eviction.

The kubelet does not respect your configured PodDisruptionBudget or the pod's `terminationGracePeriodSeconds`. If you use soft eviction thresholds, the kubelet respects your configured `eviction-max-pod-grace-period`. If you use hard eviction thresholds, the kubelet uses a `0s` grace period (immediate shutdown) for termination.

Self healing behavior

The kubelet attempts to reclaim node-level resources before it terminates end-user pods. For example, it removes unused container images when disk resources are starved.

If the pods are managed by a workload management object (such as StatefulSet or Deployment) that replaces failed pods, the control plane (`kube-controller-manager`) creates new pods in place of the evicted pods.

Self healing for static pods

If you are running a static pod on a node that is under resource pressure, the kubelet may evict that static Pod. The kubelet then tries to create a replacement, because static Pods always represent an intent to run a Pod on that node.

The kubelet takes the *priority* of the static pod into account when creating a replacement. If the static pod manifest specifies a low priority, and there are higher-priority Pods defined within the cluster's control plane, and the node is under resource pressure, the kubelet may not be able to make room for that static pod. The kubelet continues to attempt to run all static pods even when there is resource pressure on a node.

Eviction signals and thresholds

The kubelet uses various parameters to make eviction decisions, like the following:

- Eviction signals
- Eviction thresholds
- Monitoring intervals

Eviction signals

Eviction signals are the current state of a particular resource at a specific point in time. The kubelet uses eviction signals to make eviction decisions by comparing the signals to eviction thresholds, which are the minimum amount of the resource that should be available on the node.

The kubelet uses the following eviction signals:

Eviction Signal	Description	Linu Only
`memory.available`	`memory.available` := `node.status.capacity[memory]` - `node.stats.memory.workingSet`	
`nodefs.available`	`nodefs.available` := `node.stats.fs.available`	
`nodefs.inodesFree`	`nodefs.inodesFree` := `node.stats.fs.inodesFree`	•
`imagefs.available`	`imagefs.available` := `node.stats.runtime.imagefs.available`	
`imagefs.inodesFree`	`imagefs.inodesFree` := `node.stats.runtime.imagefs.inodesFree`	•
`containerfs.available`	`containerfs.available` := `node.stats.runtime.containerfs.available`	
`containerfs.inodesFree`	`containerfs.inodesFree` := `node.stats.runtime.containerfs.inodesFree`	•
`pid.available`	`pid.available` := `node.stats.rlimit.maxpid` - `node.stats.rlimit.curproc`	•

In this table, the **Description** column shows how kubelet gets the value of the signal. Each signal supports either a percentage or a literal value. The kubelet calculates the percentage value relative to the total capacity associated with the signal.

Memory signals

On Linux nodes, the value for `memory.available` is derived from the cgroupfs instead of tools like `free -m`. This is important because `free -m` does not work in a container, and if users use the node allocatable feature, out of resource decisions are made local to the end user Pod part of the cgroup hierarchy as well as the root node. This script or cgroupv2 script reproduces the same set of steps that the kubelet performs to calculate `memory.available`. The kubelet excludes inactive_file (the number of bytes of file-backed memory on the inactive LRU list) from its calculation, as it assumes that memory is reclaimable under pressure.

On Windows nodes, the value for `memory.available` is derived from the node's global memory commit levels (queried through the `GetPerformanceInfo()` system call) by subtracting the node's global `CommitTotal` from the node's `CommitLimit`. Please note that `CommitLimit` can change if the node's page-file size changes!

Filesystem signals

The kubelet recognizes three specific filesystem identifiers that can be used with eviction signals (`<identifier>.inodesFree` or `<identifier>.available`):

1. `nodefs`: The node's main filesystem, used for local disk volumes, emptyDir volumes not backed by memory, log storage, ephemeral storage, and more. For example, `nodefs` contains `/var/lib/kubelet`.

2. `imagefs`: An optional filesystem that container runtimes can use to store container images (which are the read-only layers). If there is no separate `containerfs`, the image filesystem also stores container writable layers.

3. `containerfs`: An optional filesystem that container runtimes can use to store container writable layers. When `containerfs` is used, the `imagefs` filesystem can be split to only store images (read-only layers) and nothing else.

These identifiers describe the filesystems as the kubelet observes them. They do not always mean three different mount points: in common layouts, two or all three identifiers can refer to the same underlying filesystem.

> **Note:**

> **FEATURE STATE:** Kubernetes v1.31 [beta]
> (enabled by default)
>
> The *split image filesystem* feature adds new eviction signals, thresholds, and metrics for `containerfs`. To use `containerfs`, the Kubernetes release v1.36 requires the `KubeletSeparateDiskGC` feature gate to be enabled. For Kubernetes v1.36, only CRI-O (v1.29 or higher) offers `containerfs` filesystem support.

The kubelet supports three common layouts for container filesystems:

- Everything is on the single `nodefs`, also referred to as "rootfs" or simply "root". In this layout, `nodefs`, `imagefs`, and `containerfs` all refer to the same underlying filesystem.
- Container runtime storage is on a dedicated disk, separate from the root filesystem. In this layout, `imagefs` and `containerfs` refer to the same underlying filesystem, which stores both image layers and container writable layers. This is often referred to as "split disk" (or "separate disk") filesystem.
- Container writable layers are on `nodefs`, and the container images (read-only layers) are stored on a separate `imagefs`. In this layout, `containerfs` and `nodefs` refer to the same underlying filesystem. This is often referred to as a "split image" filesystem.

The kubelet will attempt to auto-discover these filesystems with their current configuration directly from the underlying container runtime and will ignore other local node filesystems.

The kubelet does not support other container filesystems or storage configurations, and it does not currently support multiple filesystems for images and containers.

Deprecated kubelet garbage collection features

Some kubelet garbage collection features are deprecated in favor of eviction:

Existing Flag	Rationale
`--maximum-dead-containers`	deprecated once old logs are stored outside of container's context
`--maximum-dead-containers-per-container`	deprecated once old logs are stored outside of container's context

Existing Flag	Rationale
`--minimum-container-ttl-duration`	deprecated once old logs are stored outside of container's context

Eviction thresholds

You can specify custom eviction thresholds for the kubelet to use when it makes eviction decisions. You can configure soft and hard eviction thresholds.

Eviction thresholds have the form `[eviction-signal][operator][quantity]`, where:

- `eviction-signal` is the eviction signal to use.
- `operator` is the relational operator you want, such as `<` (less than).
- `quantity` is the eviction threshold amount, such as `1Gi`. The value of `quantity` must match the quantity representation used by Kubernetes. You can use either literal values or percentages (`%`).

For example, if a node has 10GiB of total memory and you want trigger eviction if the available memory falls below 1GiB, you can define the eviction threshold as either `memory.available<10%` or `memory.available<1Gi` (you cannot use both).

Soft eviction thresholds

A soft eviction threshold pairs an eviction threshold with a required administrator-specified grace period. The kubelet does not evict pods until the grace period is exceeded. The kubelet returns an error on startup if you do not specify a grace period.

You can specify both a soft eviction threshold grace period and a maximum allowed pod termination grace period for kubelet to use during evictions. If you specify a maximum allowed grace period and the soft eviction threshold is met, the kubelet uses the lesser of the two grace periods. If you do not specify a maximum allowed grace period, the kubelet kills evicted pods immediately without graceful termination.

You can use the following flags to configure soft eviction thresholds:

- `eviction-soft`: A set of eviction thresholds like `memory.available<1.5Gi` that can trigger pod eviction if held over the specified grace period.
- `eviction-soft-grace-period`: A set of eviction grace periods like `memory.available=1m30s` that define how long a soft eviction threshold must hold before triggering a Pod eviction.
- `eviction-max-pod-grace-period`: The maximum allowed grace period (in seconds) to use when terminating pods in response to a soft eviction threshold being met.

Hard eviction thresholds

A hard eviction threshold has no grace period. When a hard eviction threshold is met, the kubelet kills pods immediately without graceful termination to reclaim the starved resource.

You can use the `eviction-hard` flag to configure a set of hard eviction thresholds like `memory.available<1Gi`.

The kubelet has the following default hard eviction thresholds:

- `memory.available<100Mi` (Linux nodes)
- `memory.available<500Mi` (Windows nodes)
- `nodefs.available<10%`
- `imagefs.available<15%`
- `nodefs.inodesFree<5%` (Linux nodes)
- `imagefs.inodesFree<5%` (Linux nodes)

These default values of hard eviction thresholds will only be set if none of the parameters is changed. If you change the value of any parameter, then the values of other parameters will not be inherited as the default values and will be set to zero. In order to provide custom values, you should provide all the thresholds respectively. You can also set the kubelet config MergeDefaultEvictionSettings to true in the kubelet configuration file. If set to true and any parameter is changed, then the other parameters will inherit their default values instead of 0.

The `containerfs.available` and `containerfs.inodesFree` (Linux nodes) default eviction thresholds will be set as follows:

- If `containerfs` and `nodefs` refer to the same underlying filesystem, then `containerfs` thresholds are set the same as `nodefs`.
- If `containerfs` and `imagefs` refer to the same underlying filesystem, then `containerfs` thresholds are set the same as `imagefs`.

Setting custom overrides for thresholds related to `containerfs` is not supported, and a warning will be issued if an attempt to do so is made; any provided custom values will be ignored.

Eviction monitoring interval

The kubelet evaluates eviction thresholds based on its configured `housekeeping-interval`, which defaults to `10s`.

Node conditions

The kubelet reports node conditions to reflect that the node is under pressure because hard or soft eviction threshold is met, independent of configured grace periods.

The kubelet maps eviction signals to node conditions as follows:

Node Condition	Eviction Signal	Description
`MemoryPressure`	`memory.available`	Available memory on the node has satisfied an eviction threshold
`DiskPressure`	`nodefs.available`, `nodefs.inodesFree`, `imagefs.available`, `imagefs.inodesFree`, `containerfs.available`, or `containerfs.inodesFree`	Available disk space and inodes on either the node's root filesystem, image filesystem, or container filesystem has satisfied an eviction threshold
`PIDPressure`	`pid.available`	Available processes identifiers on the (Linux) node has fallen below an eviction threshold

The control plane also maps these node conditions to taints.

The kubelet updates the node conditions based on the configured `--node-status-update-frequency`, which defaults to `10s`.

Node condition oscillation

In some cases, nodes oscillate above and below soft eviction thresholds without holding for the defined grace periods. This causes the reported node condition to constantly switch between `true` and `false`, leading to bad eviction decisions.

To protect against oscillation, you can use the `eviction-pressure-transition-period` flag, which controls how long the kubelet must wait before transitioning a node condition to a different state. The transition period has a default value of `5m`.

Reclaiming node level resources

The kubelet tries to reclaim node-level resources before it evicts end-user pods.

When a `DiskPressure` node condition is reported, the kubelet reclaims node-level resources based on the filesystems on the node.

Without `imagefs` or `containerfs`

If the node only has a `nodefs` filesystem that meets eviction thresholds, the kubelet frees up disk space in the following order:

1. Garbage collect dead pods and containers.
2. Delete unused images.

With `imagefs`

If the node has a dedicated `imagefs` filesystem for container runtimes to use, the kubelet does the following:

- If the `nodefs` filesystem meets the eviction thresholds, the kubelet garbage collects dead pods and containers.
- If the `imagefs` filesystem meets the eviction thresholds, the kubelet deletes all unused images.

With `imagefs` and `containerfs`

If the node has a dedicated `containerfs` alongside the `imagefs` filesystem configured for the container runtimes to use, then kubelet will attempt to reclaim resources as follows:

- If the `containerfs` filesystem meets the eviction thresholds, the kubelet garbage collects dead pods and containers.
- If the `imagefs` filesystem meets the eviction thresholds, the kubelet deletes all unused images.

Pod selection for kubelet eviction

If the kubelet's attempts to reclaim node-level resources don't bring the eviction signal below the threshold, the kubelet begins to evict end-user pods.

The kubelet uses the following parameters to determine the pod eviction order:

1. Whether the pod's resource usage exceeds requests
2. Pod Priority
3. The pod's resource usage relative to requests

As a result, kubelet ranks and evicts pods in the following order:

1. `BestEffort` or `Burstable` pods where the usage exceeds requests. These pods are evicted based on their Priority and then by how much their usage level exceeds the request.

2. `Guaranteed` pods and `Burstable` pods where the usage is less than requests are evicted last, based on their Priority.

> **Note:**
> The kubelet does not use the pod's QoS class to determine the eviction order. You can use the QoS class to estimate the most likely pod eviction order when reclaiming resources like memory. QoS classification does not apply to EphemeralStorage requests, so the above scenario will not apply if the node is, for example, under `DiskPressure`.

`Guaranteed` pods are guaranteed only when requests and limits are specified for all the containers and they are equal. These pods will never be evicted because of another pod's resource consumption. If a system daemon (such as `kubelet` and `journald`) is consuming more resources than were reserved via `system-reserved` or `kube-reserved` allocations, and the node only has `Guaranteed` or `Burstable` pods using less resources than requests left on it, then the kubelet must choose to evict one of these pods to preserve node stability and to limit the impact of resource starvation on other pods. In this case, it will choose to evict pods of lowest Priority first.

If you are running a static pod and want to avoid having it evicted under resource pressure, set the `priority` field for that Pod directly. Static pods do not support the `priorityClassName` field.

When the kubelet evicts pods in response to inode or process ID starvation, it uses the Pods' relative priority to determine the eviction order, because inodes and PIDs have no requests.

The kubelet sorts pods differently based on whether the node has a dedicated `imagefs` or `containerfs` filesystem:

Without `imagefs` or `containerfs` (`nodefs` and `imagefs` use the same filesystem)

- If `nodefs` triggers evictions, the kubelet sorts pods based on their total disk usage (`local volumes + logs and a writable layer of all containers`).

With `imagefs` (`nodefs` and `imagefs` filesystems are separate)

- If `nodefs` triggers evictions, the kubelet sorts pods based on `nodefs` usage (`local volumes + logs of all containers`).
- If `imagefs` triggers evictions, the kubelet sorts pods based on the writable layer usage of all containers.

With `imagefs` and `containerfs` (`imagefs` and `containerfs` have been split)

- If `containerfs` triggers evictions, the kubelet sorts pods based on `containerfs` usage (`local volumes + logs and a writable layer of all containers`).
- If `imagefs` triggers evictions, the kubelet sorts pods based on the `storage of images` rank, which represents the disk usage of a given image.

Minimum eviction reclaim

> **Note:**
> As of Kubernetes v1.36, you cannot set a custom value for the `containerfs.available` metric. The configuration for this specific metric will be set automatically to reflect values set for either the `nodefs` or `imagefs`, depending on the configuration.

In some cases, pod eviction only reclaims a small amount of the starved resource. This can lead to the kubelet repeatedly hitting the configured eviction thresholds and triggering multiple evictions.

You can use the `--eviction-minimum-reclaim` flag or a kubelet config file to configure a minimum reclaim amount for each resource. When the kubelet notices that a resource is starved, it continues to reclaim that resource until it reclaims the quantity you specify.

For example, the following configuration sets minimum reclaim amounts:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
evictionHard:
  memory.available: "500Mi"
  nodefs.available: "1Gi"
  imagefs.available: "100Gi"
```

```
evictionMinimumReclaim:
  memory.available: "0Mi"
  nodefs.available: "500Mi"
  imagefs.available: "2Gi"
```

In this example, if the `nodefs.available` signal meets the eviction threshold, the kubelet reclaims the resource until the signal reaches the threshold of 1GiB, and then continues to reclaim the minimum amount of 500MiB, until the available nodefs storage value reaches 1.5GiB.

Similarly, the kubelet tries to reclaim the `imagefs` resource until the `imagefs.available` value reaches `102Gi`, representing 102 GiB of available container image storage. If the amount of storage that the kubelet could reclaim is less than 2GiB, the kubelet doesn't reclaim anything.

The default `eviction-minimum-reclaim` is `0` for all resources.

Node out of memory behavior

If the node experiences an *out of memory* (OOM) event prior to the kubelet being able to reclaim memory, the node depends on the oom_killer to respond.

The kubelet sets an `oom_score_adj` value for each container based on the QoS for the pod.

Quality of Service	oom_score_adj
`Guaranteed`	-997
`BestEffort`	1000
`Burstable`	*min(max(2, 1000 - (1000 × memoryRequestBytes) / machineMemoryCapacityBytes), 999)*

> **Note:**
> The kubelet also sets an `oom_score_adj` value of `-997` for any containers in Pods that have `system-node-critical` Priority.

If the kubelet can't reclaim memory before a node experiences OOM, the `oom_killer` calculates an `oom_score` based on the percentage of memory it's using on the node,

and then adds the `oom_score_adj` to get an effective `oom_score` for each container. It then kills the container with the highest score.

This means that containers in low QoS pods that consume a large amount of memory relative to their scheduling requests are killed first.

Unlike pod eviction, if a container is OOM killed, the kubelet can restart it based on its `restartPolicy`.

Good practices

The following sections describe good practice for eviction configuration.

Schedulable resources and eviction policies

When you configure the kubelet with an eviction policy, you should make sure that the scheduler will not schedule pods if they will trigger eviction because they immediately induce memory pressure.

Consider the following scenario:

- Node memory capacity: 10GiB
- Operator wants to reserve 10% of memory capacity for system daemons (kernel, `kubelet`, etc.)
- Operator wants to evict Pods at 95% memory utilization to reduce incidence of system OOM.

For this to work, the kubelet is launched as follows:

```
--eviction-hard=memory.available<500Mi
--system-reserved=memory=1.5Gi
```

In this configuration, the `--system-reserved` flag reserves 1.5GiB of memory for the system, which is `10% of the total memory + the eviction threshold amount`.

The node can reach the eviction threshold if a pod is using more than its request, or if the system is using more than 1GiB of memory, which makes the `memory.available` signal fall below 500MiB and triggers the threshold.

DaemonSets and node-pressure eviction

Pod priority is a major factor in making eviction decisions. If you do not want the kubelet to evict pods that belong to a DaemonSet, give those pods a high enough

priority by specifying a suitable `priorityClassName` in the pod spec. You can also use a lower priority, or the default, to only allow pods from that DaemonSet to run when there are enough resources.

Known issues

The following sections describe known issues related to out of resource handling.

kubelet may not observe memory pressure right away

By default, the kubelet polls cAdvisor to collect memory usage stats at a regular interval. If memory usage increases within that window rapidly, the kubelet may not observe `MemoryPressure` fast enough, and the OOM killer will still be invoked.

You can use the `--kernel-memcg-notification` flag to enable the `memcg` notification API on the kubelet to get notified immediately when a threshold is crossed.

If you are not trying to achieve extreme utilization, but a sensible measure of overcommit, a viable workaround for this issue is to use the `--kube-reserved` and `--system-reserved` flags to allocate memory for the system.

active_file memory is not considered as available memory

On Linux, the kernel tracks the number of bytes of file-backed memory on active least recently used (LRU) list as the `active_file` statistic. The kubelet treats `active_file` memory areas as not reclaimable. For workloads that make intensive use of block-backed local storage, including ephemeral local storage, kernel-level caches of file and block data means that many recently accessed cache pages are likely to be counted as `active_file`. If enough of these kernel block buffers are on the active LRU list, the kubelet is liable to observe this as high resource use and taint the node as experiencing memory pressure - triggering pod eviction.

For more details, see https://github.com/kubernetes/kubernetes/issues/43916

You can work around that behavior by setting the memory limit and memory request the same for containers likely to perform intensive I/O activity. You will need to estimate or measure an optimal memory limit value for that container.

What's next

- Learn about API-initiated Eviction
- Learn about Pod Priority and Preemption

- Learn about PodDisruptionBudgets
- Learn about Quality of Service (QoS)
- Check out the Eviction API

17 - API-initiated Eviction

API-initiated eviction is the process by which you use the Eviction API to create an `Eviction` object that triggers graceful pod termination.

You can request eviction by calling the Eviction API directly, or programmatically using a client of the API server, like the `kubectl drain` command. This creates an `Eviction` object, which causes the API server to terminate the Pod.

API-initiated evictions respect your configured `PodDisruptionBudgets` and `terminationGracePeriodSeconds`.

Using the API to create an Eviction object for a Pod is like performing a policy-controlled `DELETE` operation on the Pod.

Calling the Eviction API

You can use a Kubernetes language client to access the Kubernetes API and create an `Eviction` object. To do this, you POST the attempted operation, similar to the following example:

policy/v1 | policy/v1beta1

> **Note:**
> `policy/v1` Eviction is available in v1.22+. Use `policy/v1beta1` with prior releases.

```
{
  "apiVersion": "policy/v1",
  "kind": "Eviction",
  "metadata": {
    "name": "quux",
    "namespace": "default"
  }
}
```

Alternatively, you can attempt an eviction operation by accessing the API using `curl` or `wget`, similar to the following example:

```
curl -v -H 'Content-type: application/json' https://your-cluster-api-endpoint.examp
```

How API-initiated eviction works

When you request an eviction using the API, the API server performs admission checks and responds in one of the following ways:

- `200 OK` : the eviction is allowed, the `Eviction` subresource is created, and the Pod is deleted, similar to sending a `DELETE` request to the Pod URL.
- `429 Too Many Requests` : the eviction is not currently allowed because of the configured PodDisruptionBudget. You may be able to attempt the eviction again later. You might also see this response because of API rate limiting.
- `500 Internal Server Error` : the eviction is not allowed because there is a misconfiguration, like if multiple PodDisruptionBudgets reference the same Pod.

If the Pod you want to evict isn't part of a workload that has a PodDisruptionBudget, the API server always returns `200 OK` and allows the eviction.

If the API server allows the eviction, the Pod is deleted as follows:

1. The `Pod` resource in the API server is updated with a deletion timestamp, after which the API server considers the `Pod` resource to be terminated. The `Pod` resource is also marked with the configured grace period.
2. The kubelet on the node where the local Pod is running notices that the `Pod` resource is marked for termination and starts to gracefully shut down the local Pod.
3. While the kubelet is shutting the Pod down, the control plane removes the Pod from EndpointSlice objects. As a result, controllers no longer consider the Pod as a valid object.
4. After the grace period for the Pod expires, the kubelet forcefully terminates the local Pod.
5. The kubelet tells the API server to remove the `Pod` resource.
6. The API server deletes the `Pod` resource.

Troubleshooting stuck evictions

In some cases, your applications may enter a broken state, where the Eviction API will only return `429` or `500` responses until you intervene. This can happen if, for example, a ReplicaSet creates pods for your application but new pods do not enter a `Ready`

state. You may also notice this behavior in cases where the last evicted Pod had a long termination grace period.

If you notice stuck evictions, try one of the following solutions:

- Abort or pause the automated operation causing the issue. Investigate the stuck application before you restart the operation.
- Wait a while, then directly delete the Pod from your cluster control plane instead of using the Eviction API.

What's next

- Learn how to protect your applications with a Pod Disruption Budget.
- Learn about Node-pressure Eviction.
- Learn about Pod Priority and Preemption.

18 - Node Declared Features

ⓘ **FEATURE STATE:** Kubernetes v1.36 [beta](enabled by default)

Kubernetes nodes use *declared features* to report the availability of specific features that are new or feature-gated. Control plane components utilize this information to make better decisions. The kube-scheduler, via the `NodeDeclaredFeatures` plugin, ensures pods are only placed on nodes that explicitly support the features the pod requires. Additionally, the `NodeDeclaredFeatureValidator` admission controller validates pod updates against a node's declared features.

This mechanism helps manage version skew and improve cluster stability, especially during cluster upgrades or in mixed-version environments where nodes might not all have the same features enabled. This is intended for Kubernetes feature developers introducing new node-level features and works in the background; application developers deploying Pods do not need to interact with this framework directly.

How it Works

1. **Kubelet Feature Reporting:** At startup, the kubelet on each node detects which managed Kubernetes features are currently enabled and reports them in the `.status.declaredFeatures` field of the Node. Only features under active development are included in this field.
2. **Scheduler Filtering:** The default kube-scheduler uses the `NodeDeclaredFeatures` plugin. This plugin:
   - In the `PreFilter` stage, checks the `PodSpec` to infer the set of node features required by the pod.
   - In the `Filter` stage, checks if the features listed in the node's `.status.declaredFeatures` satisfy the requirements inferred for the Pod. Pods will not be scheduled on nodes lacking the required features. Custom schedulers can also utilize the `.status.declaredFeatures` field to enforce similar constraints.
3. **Admission Control:** The `nodedeclaredfeaturevalidator` admission controller can reject Pods that require features not declared by the node they are bound to, preventing issues during pod updates.

Enabling node declared features

To use Node Declared Features, the `NodeDeclaredFeatures` feature gate must be enabled on the `kube-apiserver`, `kube-scheduler`, and `kubelet` components.

What's next

- Read the KEP for more details: KEP-5328: Node Declared Features
- Read about the `NodeDeclaredFeatureValidator` admission controller.